

The following two chapters will focus on the theory and application of *linear regression models*, which play a major role in statistics. We already studied some examples in Sect. 2.2. In addition to the direct application of linear regression models, they are also the basis of a variety of more complex regression methods. Examples are generalized linear models (Chap. 5), mixed models (Chap. 7), or semiparametric models (Chaps. 8 and 9).

In this chapter, we will focus exclusively on the classical linear model $y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + \varepsilon$, with independent and identically distributed errors, which were already introduced in Sect. 2.2.2. Section 3.1 defines the model, discusses model assumptions, and illustrates how covariate effects are modeled. The following Sects. 3.2 and 3.3 describe the theory of classical estimation and testing within the linear model framework. The method of least squares plays a major role for estimating the unknown regression coefficients $\beta_0, \dots, \beta_k$.

This chapter's last part (Sect. 3.4) discusses methods and strategies for model choice and diagnostics, as well as the application of the models and inference techniques presented so far. A careful and detailed case study on prices of used cars will show how linear models can be used in practice.

The concluding Sect. 3.5 gives suggestions for further reading and provides proofs of the theorems that have been omitted in the main text to improve readability.

In order to circumvent limitations of classical linear models, several extensions have been proposed in the literature. These extensions are devoted to the next chapter.

3.1 Model Definition

Suppose we are given a variable of primary interest y and we aim to model the relationship between this response variable y and a set of regressors or explanatory variables $x_1, \dots, x_k$. In general, we model the relationship between y and $x_1, \dots, x_k$ with a function $f(x_1, \dots, x_k)$. This relationship is not exact, as it is affected by random noise ε . In practice, we usually assume additive errors and thus obtain

$$y = f(x_1, \dots, x_k) + \varepsilon.$$

Our goal is to estimate the unknown function f , i.e., to separate the systematic component f from the random noise ε . Within the framework of linear models, the following specific assumptions regarding the unknown function f and the noise ε are made:

1. *The systematic component f is a linear combination of covariates*

The unknown function $f(x_1, \dots, x_k)$ is modeled as a linear combination of covariates, i.e.,

$$f(x_1, \dots, x_k) = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k.$$

The parameters $\beta_0, \beta_1, \dots, \beta_k$ are unknown and need to be estimated. The parameter β_0 represents the intercept. If we combine the covariates and the unknown parameters each into $p = (k+1)$ dimensional vectors, $\mathbf{x} = (1, x_1, \dots, x_k)'$ and $\boldsymbol{\beta} = (\beta_0, \dots, \beta_k)'$, then

$$f(\mathbf{x}) = \mathbf{x}'\boldsymbol{\beta}.$$

Note that the intercept β_0 in the model implies that the first element of the vector $\mathbf{x}$ equals one.

At first glance, the assumption of a linear function f , i.e., a linear relation between y and $\mathbf{x}$, appears to be very restrictive. As already demonstrated in Sect. 2.2 (p. 22), nonlinear relationships can also be modeled within the framework of linear models. Section 3.1.3 will elaborate on this aspect in even more detail.

2. *Additive Errors*

Another basic assumption of the linear model is additivity of errors, which implies

$$y = \mathbf{x}'\boldsymbol{\beta} + \varepsilon.$$

Even though this appears to be very restrictive, this assumption is reasonable for many practical applications. Moreover, problems, which at first do not show additive error structure, can be specified by models with additive errors after a transformation of the response variable y (refer to Sect. 3.1.2, p. 83).

In order to estimate the unknown parameters $\boldsymbol{\beta}$, we collect data y_i and $\mathbf{x}_i = (1, x_{i1}, \dots, x_{ik})'$, $i = 1, \dots, n$, and for every observation construct the equation

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik} + \varepsilon_i = \mathbf{x}_i' \boldsymbol{\beta} + \varepsilon_i. \quad (3.1)$$

If we define the vectors

$$\mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \quad \text{and} \quad \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix}$$

and the *design matrix* $\mathbf{X}$,

$$\mathbf{X} = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1k} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{pmatrix} = \begin{pmatrix} \mathbf{x}'_1 \\ \vdots \\ \mathbf{x}'_n \end{pmatrix},$$

then the n equations in Eq. (3.1) can be compactly summarized as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}.$$

For the remaining chapter we assume that $\mathbf{X}$ is of full column rank, i.e., $\text{rk}(\mathbf{X}) = k + 1 = p$, implying the columns of $\mathbf{X}$ are linearly independent. A necessary requirement is that the number of observations n must at least be equal to (or larger than) the number p of regression coefficients. The assumption is violated if one of the covariates is a linear transformation of another covariate, for example, $x_1 = a + b x_2$. For example, the variable x_1 could represent the height of a person in meters and x_2 the height in centimeters, implying $a = 0$ and $b = 1/100$. In general, the assumption of linear independence is always violated when at least one of the explanatory variables can be represented as a linear combination of the other covariates, implying redundancy of information. We will see that linear independence of the columns in the design matrix $\mathbf{X}$ is necessary in order to obtain *unique* estimators of the regression coefficients in $\boldsymbol{\beta}$.

Our model is finalized by appropriate assumptions on the error terms ε_i . Within the *classical linear model*, the following assumptions for the vector $\boldsymbol{\varepsilon}$ of errors are made:

1. *Expectation of the errors*

The errors have mean or expectation zero, i.e., $E(\varepsilon_i) = 0$ or in matrix notation $E(\boldsymbol{\varepsilon}) = \mathbf{0}$.

2. *Variances and correlation structure of the errors*

We assume a constant error variance σ^2 across observations, that is homoscedastic errors with $\text{Var}(\varepsilon_i) = \sigma^2$. The errors are called heteroscedastic when the variances vary among observations, i.e., $\text{Var}(\varepsilon_i) = \sigma_i^2$. In addition to homoscedastic variances, we assume that errors are uncorrelated, which means $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$. The assumption of homoscedastic and uncorrelated errors leads to the covariance matrix $\text{Cov}(\boldsymbol{\varepsilon}) = E(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}') = \sigma^2 \mathbf{I}$.

3. *Assumptions about the covariates and the design matrix*

We distinguish between two situations. On one hand, explanatory variables $x_1, \dots, x_k$ may be deterministic or non-stochastic, e.g., as is the case in designed experiments. In most cases, and also in almost every application presented in this book, both the response as well as the covariates are stochastic (observational data). For example, the rent data for apartments in Munich are a random sample of all apartments in the city of Munich. Hence, all characteristics of an apartment, for example, its area or year of construction, are realizations of random variables. In the case of stochastic regressors, the observations $(y_i, \mathbf{x}'_i)$, $i = 1, \dots, n$,

3.1 The Classical Linear Model

The model

$$y = X\beta + \varepsilon$$

is called the classical linear regression model, if the following assumptions are true:

1. $E(\varepsilon) = \mathbf{0}$.
2. $\text{Cov}(\varepsilon) = E(\varepsilon\varepsilon') = \sigma^2 I$.
3. The design matrix X has full column rank, i.e., $\text{rk}(X) = k + 1 = p$.

The classical normal regression model is obtained if additionally

4. $\varepsilon \sim N(\mathbf{0}, \sigma^2 I)$

holds. For stochastic covariates these assumptions are to be understood conditionally on X .

can be understood as realizations of a random vector $(y, \mathbf{x}')$, and all model assumptions are conditional on the design matrix, as, for example, $E(\varepsilon | X) = \mathbf{0}$ (instead of $E(\varepsilon) = \mathbf{0}$) or $\text{Cov}(\varepsilon | X) = \sigma^2 I$ (instead of $\text{Cov}(\varepsilon) = \sigma^2 I$). The latter implies that ε_i and $\mathbf{x}_i$ are (stochastically) independent. The assumption that errors and stochastic covariates are independent can be relaxed, for example, by allowing $\text{Var}(\varepsilon_i | \mathbf{x}_i)$ to depend on covariates, i.e., $\text{Var}(\varepsilon_i | \mathbf{x}_i) = \sigma^2(\mathbf{x}_i)$ (see in Chap. 4, Sect. 4.1.3).

In any case, we assume that the design matrix has full column rank, i.e. $\text{rk}(X) = k + 1 = p$.

4. Gaussian errors

To construct confidence intervals and hypothesis tests for the regression coefficients, we often assume a normal distribution for the errors (at least approximately). Together with assumptions 1 and 2, we obtain $\varepsilon_i \sim N(0, \sigma^2)$ or in matrix notation $\varepsilon \sim N(\mathbf{0}, \sigma^2 I)$. With stochastic covariates we have $\varepsilon_i | \mathbf{x}_i \sim N(0, \sigma^2)$ and $\varepsilon | X \sim N(\mathbf{0}, \sigma^2 I)$, implying that ε_i and $\mathbf{x}_i$ are independent.

For notational simplicity we usually suppress the dependence of terms on the design matrix X in case of stochastic covariates.

From our model assumptions it immediately follows that

$$E(y_i) = E(\mathbf{x}_i' \beta + \varepsilon_i) = \mathbf{x}_i' \beta = \beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik},$$

$$\text{Var}(y_i) = \text{Var}(\mathbf{x}_i' \beta + \varepsilon_i) = \text{Var}(\varepsilon_i) = \sigma^2,$$

$$\text{Cov}(y_i, y_j) = \text{Cov}(\varepsilon_i, \varepsilon_j) = 0,$$

for the mean and the variance of y_i , as well as the covariance between y_i and y_j , respectively. In matrix notation we obtain

$$E(\mathbf{y}) = X\beta \quad \text{and} \quad \text{Cov}(\mathbf{y}) = \sigma^2 I.$$

If we additionally assume normally distributed errors, we have

$$\mathbf{y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}).$$

Note that covariates do only affect the mean of $\mathbf{y}$. The variance σ^2 of y_i or the covariance matrix $\sigma^2 \mathbf{I}$ of $\mathbf{y}$ is independent of the covariates; however we relax this assumption in Chap. 4, Sect. 4.1.3.

3.1.1 Model Parameters, Estimation, and Residuals

Prior to discussing the model assumptions for the classical linear model, we introduce some notation and terminology. We distinguish the model parameters from their estimates by a “hat,” which means estimates of $\boldsymbol{\beta}$ and σ^2 are denoted by $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}^2$, respectively. This distinction is necessary, since it is more or less impossible to estimate the “true” parameter vector $\boldsymbol{\beta}$ without error; thus $\hat{\boldsymbol{\beta}} \neq \boldsymbol{\beta}$ in general. Regression parameters are usually estimated using the method of least squares, which we have begun to introduce in Sect. 2.2 for the simple linear model with just one regressor (see p. 24). Section 3.2 will outline estimation of $\boldsymbol{\beta}$ and σ^2 in full detail. Based on an estimator $\hat{\boldsymbol{\beta}}$ for $\boldsymbol{\beta}$, a straightforward estimator of the mean $E(y_i)$ of y_i is given by

$$\widehat{E(y_i)} = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \cdots + \hat{\beta}_k x_{ik} = \mathbf{x}_i' \hat{\boldsymbol{\beta}}.$$

We usually refer to this estimator as $\hat{y}_i$, i.e., $\hat{y}_i = \widehat{E(y_i)}$. The estimated error, i.e., the deviation $y_i - \mathbf{x}_i' \hat{\boldsymbol{\beta}}$ between the true value y_i and the estimated value $\hat{y}_i$, is called residual and denoted by $\hat{\varepsilon}_i$. We have

$$\hat{\varepsilon}_i = y_i - \hat{y}_i = y_i - \mathbf{x}_i' \hat{\boldsymbol{\beta}}.$$

Defining the vector of residuals $\hat{\boldsymbol{\varepsilon}} = (\hat{\varepsilon}_1, \dots, \hat{\varepsilon}_n)'$, we obtain

$$\hat{\boldsymbol{\varepsilon}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}.$$

It is important to understand that the residuals $\hat{\varepsilon}_i$ are not identical to the errors ε_i . These are, like the parameter vector $\boldsymbol{\beta}$, unknown. In fact the residuals $\hat{\varepsilon}_i$ can be seen as estimates (or more precisely predictions) of ε_i .

The residuals contain the variation in the data that could not be explained by covariates. In some situations, especially for model diagnostics, we compute *partial residuals*, which quantify the removal of the effect of most, but not of all, covariates. Partial residuals with respect to the j th covariate x_j are defined as follows:

$$\begin{aligned} \hat{\varepsilon}_{x_j, i} &= y_i - \hat{\beta}_0 - \cdots - \hat{\beta}_{j-1} x_{i, j-1} - \hat{\beta}_{j+1} x_{i, j+1} - \cdots - \hat{\beta}_k x_{ik} \\ &= y_i - \mathbf{x}_i' \hat{\boldsymbol{\beta}} + \hat{\beta}_j x_{ij} \\ &= \hat{\varepsilon}_i + \hat{\beta}_j x_{ij}. \end{aligned}$$

Here, the effect of all covariates with the exception of x_j is removed. We will see that the partial residuals are helpful for model choice, model diagnostics, and visualization.

3.1.2 Discussion of Model Assumptions

This section has two goals: First, the assumptions of the classical linear model are critically discussed. Secondly, we direct the reader to particular sections within this book, in which various restrictive assumptions of classical linear models will be relaxed to yield more flexible modeling approaches. For illustration, we usually use the simple regression model $y = \beta_0 + \beta_1 x + \varepsilon$ with one explanatory variable.

Linearity of Covariate Effects

At first, the assumption of linear covariate effects may appear quite restrictive. However, within the scope of linear models, nonlinear relations are possible as well. Consider, for instance, the following model, in which the effect of the explanatory variable z_i is logarithmic:

$$y_i = \beta_0 + \beta_1 \log(z_i) + \varepsilon_i.$$

Defining a new variable $x_i = \log(z_i)$ yields the linear model $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$. In general, nonlinear relationships can be connected to linear models provided that they are *linear in the parameters*. An example for a model, which is nonlinear in the parameter β_2 , is given by

$$y_i = \beta_0 + \beta_1 \sin(\beta_2 z_i) + \varepsilon_i.$$

This book discusses several techniques on how to model nonlinear relationships between a response variable y and an explanatory variable x . Section 3.1.3 will illustrate two simple techniques in detail: the modeling of nonlinear relationships through variable transformation and also through polynomials. Example 2.2 (p. 25) already presented an example for variable transformation. We also already used polynomial regression in Example 2.6 (p. 32). Both of these simple techniques do have the disadvantage that the functional form of the relationship must be known in advance. In Chaps. 8 and 9, we will discuss methods for estimating nonlinear covariate effects without special assumptions on the functional form.

Homoscedastic Error Variances

Homoscedastic error variances imply that the variance of ε_i does not systematically vary across individuals, e.g., by an increase or decrease with one or more covariates x_j . Figure 3.1 illustrates the difference between homo- and heteroscedastic variances with simulated data. Panel (a) displays the typical ideal setting of homoscedastic errors: All observations fluctuate with a stable variability around the

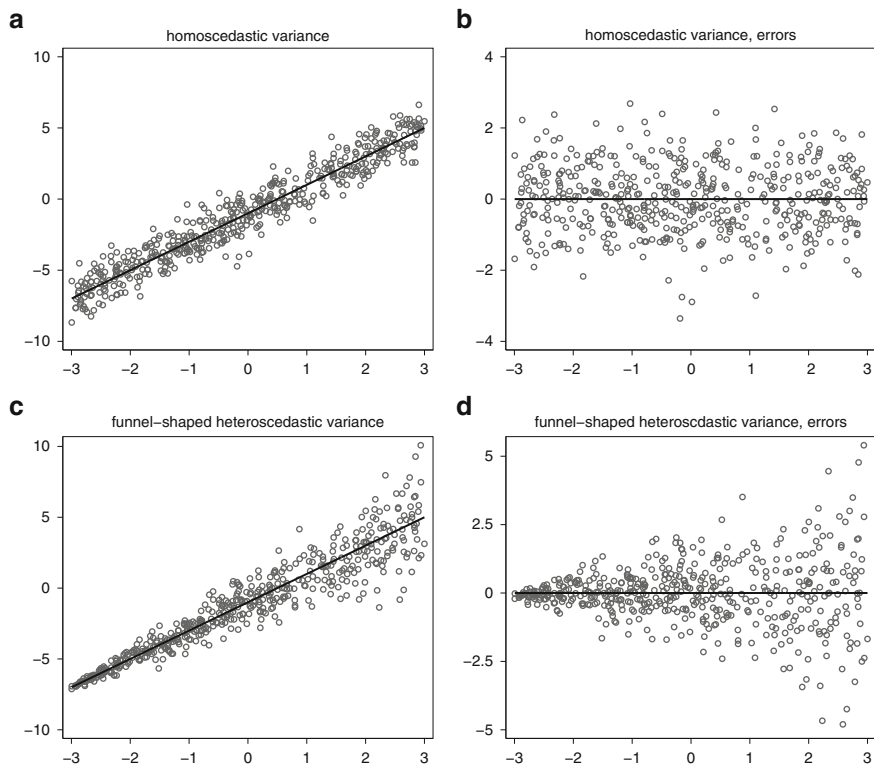


Fig. 3.1 Illustration for homo- and heteroscedastic variances: The graphs on the *left* show simulated data together with the true regression line. In the graphs on the *right* the corresponding errors are displayed. The data are based on the model $y_i \sim N(-1 + 2x_i, 1)$ [panels (a)—homoscedastic variance and (b)—homoscedastic variance, errors] and $y_i \sim N(-1 + 2x_i, (0.1 + 0.3(x_i + 3))^2)$ [panels (c)—funnel-shaped heteroscedastic variance and (d)—funnel-shaped heteroscedastic variance, errors]

regression line. Panel (b) shows the associated errors centered at zero. Since these data are simulated, we know the true regression coefficients and also the regression line and error values. In real data situations, we have to rely on estimators of the regression parameters. Since the errors $\varepsilon_i = y_i - \mathbf{x}_i' \boldsymbol{\beta}$ depend on the unknown regression parameters, we estimate the errors with the residuals $\hat{\varepsilon}_i = y_i - \mathbf{x}_i' \hat{\boldsymbol{\beta}}$. When verifying the assumption of homoscedastic errors, we rely on the examination of the residuals. Section 3.4.4 provides further details. Panels (c) and (d) in Fig. 3.1 provide a typical example for heteroscedastic variances. With an increasing x , the variation around the regression line also increases. The error values in panel (d) still fluctuate around zero, but the variances increase with increasing x .

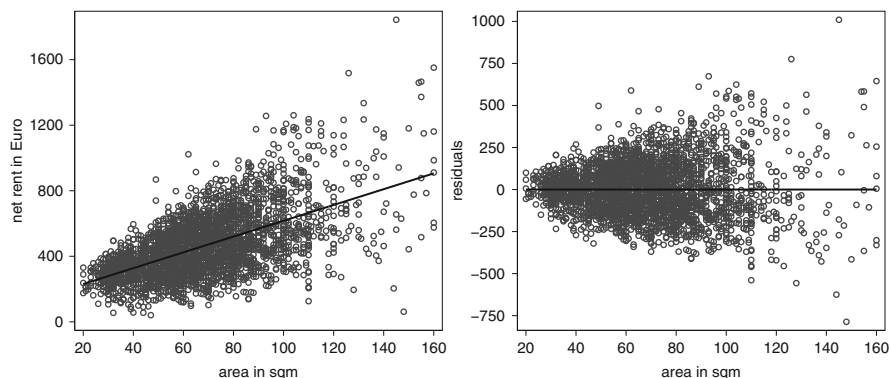


Fig. 3.2 Munich rent index: illustration of heteroscedastic variances. The *left panel* shows a scatter plot between net rent and area together with the estimated regression line. The *right panel* displays the corresponding residuals versus area

Example 3.1 Munich Rent Index—Heteroscedastic Variances

The funnel-shaped trend of the errors in Fig. 3.1c d is typical for many real data situations. As an example we take the Munich rent data; see Fig. 3.2 which shows the scatter plot between the net rent and living area together with the estimated regression line (left panel). The right panel of Fig. 3.2 shows a scatter plot of the corresponding residuals as a function of living area. The observed net rent scatters with an increasing variance around the plotted regression line. Clearly, a wider range of rent is found for larger living areas than for smaller ones. The scatter plot of the residuals shows the funnel-shaped trend mentioned above. We already encountered this phenomenon in Example 2.1 (p. 22), where we examined the relationship between net rent and area of apartments in average location that were built after 1966 (Fig. 2.1 on p. 23). We obtain a very similar pattern as presented in Fig. 3.2. The variability is, however, less pronounced, which is due to the fact that data for the apartments in average location are more homogeneous.

△

An obvious question arises: What are the consequences of ignoring heteroscedastic variances? In particular, we will see that the variances of estimated regression coefficients $\hat{\beta}$ are not properly estimated. This in turn may cause problems for hypothesis tests and confidence intervals for the regression coefficients.

Section 4.1 (p. 177) of the next chapter will discuss heteroscedastic variances in more detail. There we will generalize the classical linear model to allow for heteroscedastic variances of the errors.

Uncorrelated Errors

In some cases, especially in time series and longitudinal data, the assumption of uncorrelated errors is not realistic. Many applications show *autocorrelated errors*. For example, first-order autocorrelation reveals a linear relationship $\varepsilon_i = \rho \varepsilon_{i-1} + u_i$ between the errors ε_i at time i and the errors of the previous time period, ε_{i-1} . The u_i are independent and identically distributed random variables. Second-order or more generally l th-order autocorrelation is given if $\varepsilon_i = \rho_1 \varepsilon_{i-1} + \rho_2 \varepsilon_{i-2} + u_i$ or $\varepsilon_i = \rho_1 \varepsilon_{i-1} + \dots + \rho_l \varepsilon_{i-l} + u_i$, respectively.

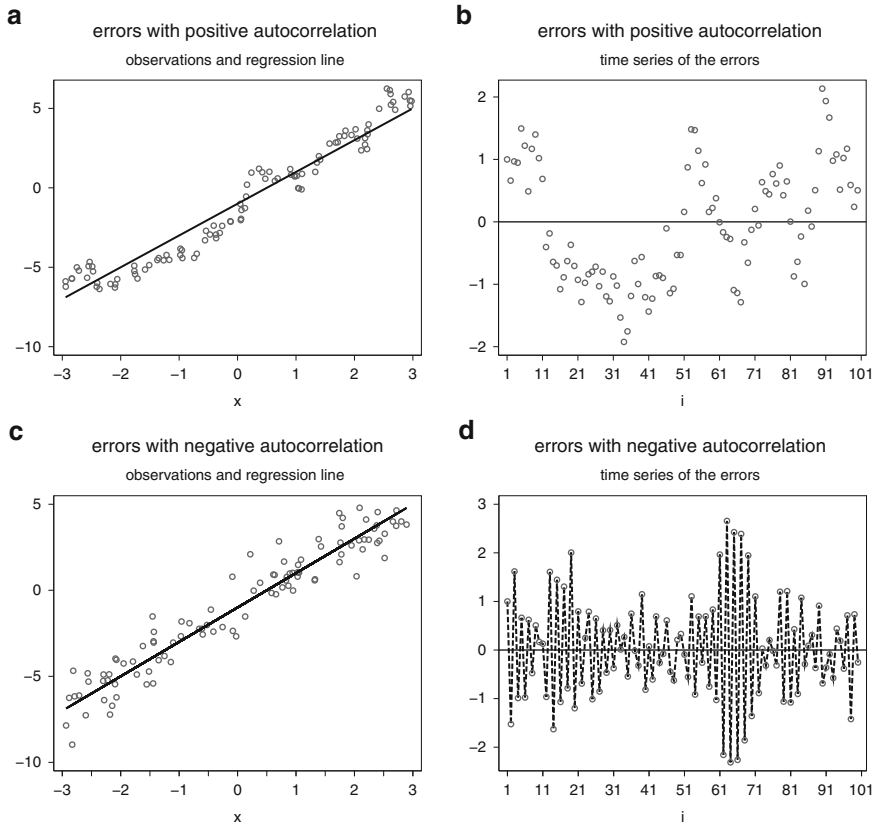


Fig. 3.3 Illustration for autocorrelated errors: Panels (a) and (b) show errors with positive autocorrelation and panels (c) and (d) correspond to negative autocorrelation. The respective graphs on the *left* show the (simulated) data including the (true) regression line. The graphs on the *right-hand side* display the corresponding errors. In case of negative autocorrelation, observations are connected in order to emphasize the changing algebraic sign. The data with positive correlation are simulated according to the model $y_i = -1 + 2x_i + \varepsilon_i$ where $\varepsilon_i = 0.9\varepsilon_{i-1} + u_i$ and $u_i \sim N(0, 0.5^2)$. The data with negative correlation in the errors are simulated according to $y_i = -1 + 2x_i + \varepsilon_i$ where $\varepsilon_i = -0.9\varepsilon_{i-1} + u_i$ and $u_i \sim N(0, 0.5^2)$

An example for autocorrelated errors is presented in Fig. 3.3. Panels (a) and (b) display simulated data with errors that have a first-order *positive autocorrelation*, which means that a positive (negative) error is likely to be followed by yet another positive (negative) error. Panels (c) and (d) illustrate the contrary, showing errors with negative autocorrelation. Positive (negative) errors are typically followed by negative (positive) errors, i.e., we frequently observe alternating signs of the errors.

Autocorrelated errors usually appear when the regression model is misspecified, e.g., a covariate is missing or the effect of a continuous covariate is nonlinear rather than linear. Figure 3.4 exemplifies such a situation. Panel (a) shows data

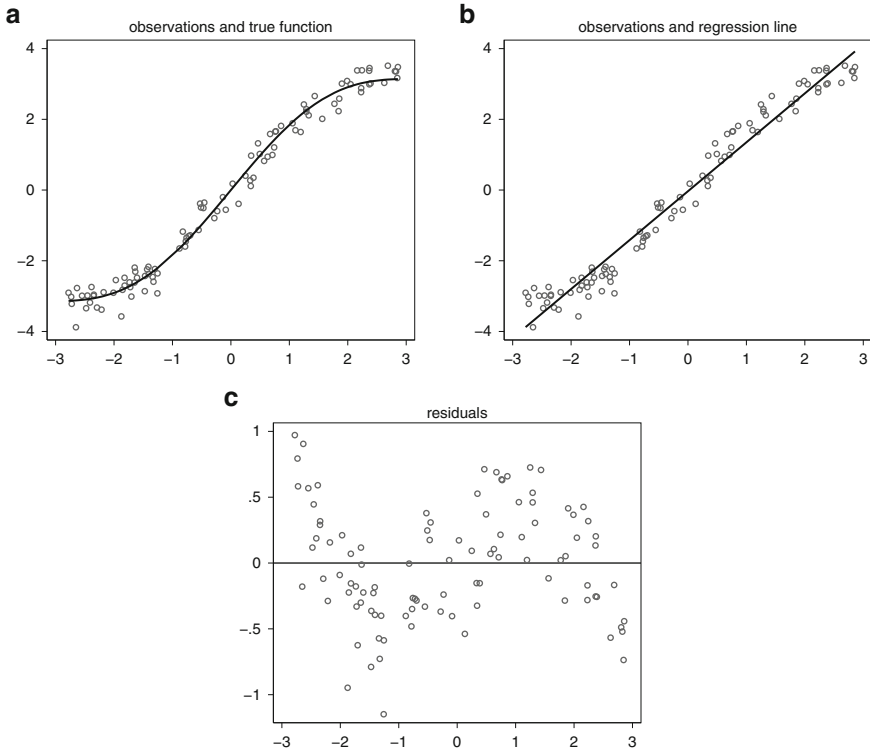


Fig. 3.4 Illustration for correlated residuals when the model is misspecified: Panel (a) displays (simulated) data based on the function $E(y_i | x_i) = \sin(x_i) + x_i$ and $\varepsilon_i \sim N(0, 0.3^2)$. Panel (b) shows the estimated regression line, i.e., the nonlinear relationship is ignored. The corresponding residuals can be found in panel (c)

simulated from the model $y_i = \sin(x_i) + x_i + \varepsilon_i$. The conditional mean of y_i is $E(y_i | x_i) = \sin(x_i) + x_i$, which is a nonlinear function of x ; see Fig. 3.4a. In panel (b), a simple linear regression line has been fitted, which means that the estimated model is misspecified. The corresponding residuals in panel (c) show positive autocorrelation.

Autocorrelated errors are most often encountered in time series or longitudinal data. In many cases, relevant covariates cannot be included in the model because they cannot be observed. If the unobserved but relevant covariates show a temporal or seasonal trend, correlated errors are induced. We again use simulated data for illustration. Panels (a) and (b) of Fig. 3.5 show a time series plot of variables x_1 and x_2 . Whereas x_1 is apparently subject to a clear temporal trend, x_2 fluctuates randomly around zero. Now consider the regression model $y_i = -1 + x_1 - 0.6x_2 + \varepsilon_i$, $i = 1, \dots, 100$, with independent and identically distributed normal errors $\varepsilon_i \sim N(0, 0.5^2)$. This is a classical linear regression model, which complies with the assumptions stated in Box 3.1 on p. 76. We obtain the estimates $\hat{y}_i =$

$-1.03 + 0.98x_{i1} - 0.61x_{i2}$. Figure 3.5c shows the corresponding residuals, which does not reveal any conspicuous behavior. Moreover, we do not see any evidence of autocorrelation. If we estimate a regression model that does not include covariate x_2 , we obtain $\hat{y}_i = -1.01 + 0.95 \cdot x_{i1}$. The corresponding time series of residuals is presented in Fig. 3.5d. Even though covariate x_2 is missing, there does not appear to be any anomalies in the fit. The estimated regression coefficients are close to their true values, and the residual variation over time appears to be random. In case that we neglect to model the variable x_1 fluctuating about some trend, we obtain the estimates $\hat{y}_i = -2.22 - 0.60 \cdot x_{i2}$. Figure 3.5e shows the residuals, which now suggest presence of autocorrelation. Why are the residuals rather inconspicuous when x_2 is not considered and why do the residuals show autocorrelation when neglecting x_1 ? The reason is that x_1 shows a distinct temporal trend, whereas x_2 does not. We can explain these relationships as follows: The omitted effects $\beta_1 x_1$ or $\beta_2 x_2$ can be absorbed into the error term. If x_1 (x_2) is omitted in the model, we can denote the corresponding errors as $\tilde{\varepsilon} = \beta_1 x_1 + \varepsilon$ ($\tilde{\varepsilon} = \beta_2 x_2 + \varepsilon$). Since x_2 essentially shows no time trend, the residuals $\tilde{\varepsilon}$ are still uncorrelated. However, if the effect of x_1 is absorbed into $\tilde{\varepsilon}$, the trend in x_1 is reflected in the errors and autocorrelated residuals are obtained.

Analogous to heteroscedastic errors, we also have to examine the consequences of ignoring correlated errors. Section 4.1 (p. 177) of the next chapter will discuss this in detail including estimation methods in the presence of correlated errors. Intuitively, it is clear that ignorance regarding correlation implies loss of information. Suppose we are interested in predicting the response based on a new observation $\mathbf{x}_{n+1}$ of the covariates at time $n + 1$. An obvious estimator in this situation is given by $\hat{y}_{n+1} = \mathbf{x}'_{n+1} \hat{\boldsymbol{\beta}}$. For correlated errors using this estimator results in a loss of information. If, for instance, errors are positively correlated, then positive residuals are more likely to be followed by other positive residuals. That is, if the residual ε_n at time n is positive, the residual ε_{n+1} in the next time period is likely positive again. The stronger the correlation, the less $\hat{\varepsilon}_n$ differs from $\hat{\varepsilon}_{n+1}$. Thus, it makes sense to predict a *higher* value for y_{n+1} than $\mathbf{x}'_{n+1} \hat{\boldsymbol{\beta}}$. On the other hand we should predict a *smaller* value than $\mathbf{x}'_{n+1} \hat{\boldsymbol{\beta}}$ if the residual $\hat{\varepsilon}_n$ is negative. Section 4.1 (p. 177) will show that $\hat{y}_{n+1} = \mathbf{x}'_{n+1} \hat{\boldsymbol{\beta}} + \hat{\rho} \hat{\varepsilon}_n$ represents an optimal prediction of y_{n+1} , where $\hat{\rho}$ is the (empirical) correlation coefficient between $\hat{\varepsilon}_i$ and $\hat{\varepsilon}_{i-1}$.

Additivity of Errors

In principle, many different models for the structure of the errors are conceivable. In the vast majority of cases, two alternative error structures are assumed: additive errors and multiplicative errors. An example for multiplicative errors is the exponential model

$$\begin{aligned} y_i &= \exp(\beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik} + \varepsilon_i) \\ &= \exp(\beta_0) \exp(\beta_1 x_{i1}) \cdot \dots \cdot \exp(\beta_k x_{ik}) \exp(\varepsilon_i), \end{aligned} \quad (3.2)$$

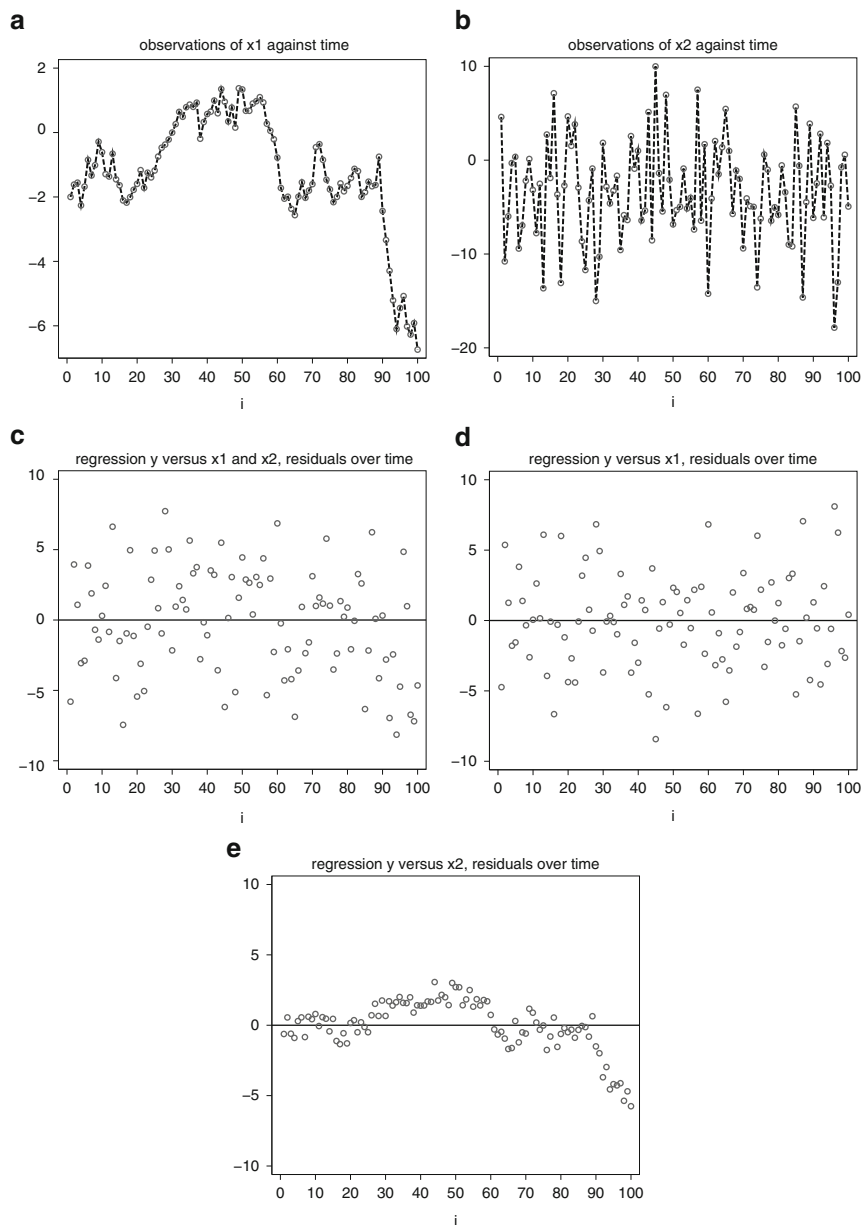


Fig. 3.5 Illustration for autocorrelated errors if relevant covariates showing a temporal trend are ignored. Panels (a) and (b) show the covariates over time. Panels (c–e) display the residuals for the regression models $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$ (correct model), $y_i = \beta_0 + \beta_1 x_{i1} + \varepsilon_i$ (x_2 ignored), and $y_i = \beta_0 + \beta_1 x_{i2} + \varepsilon_i$ (x_1 ignored)

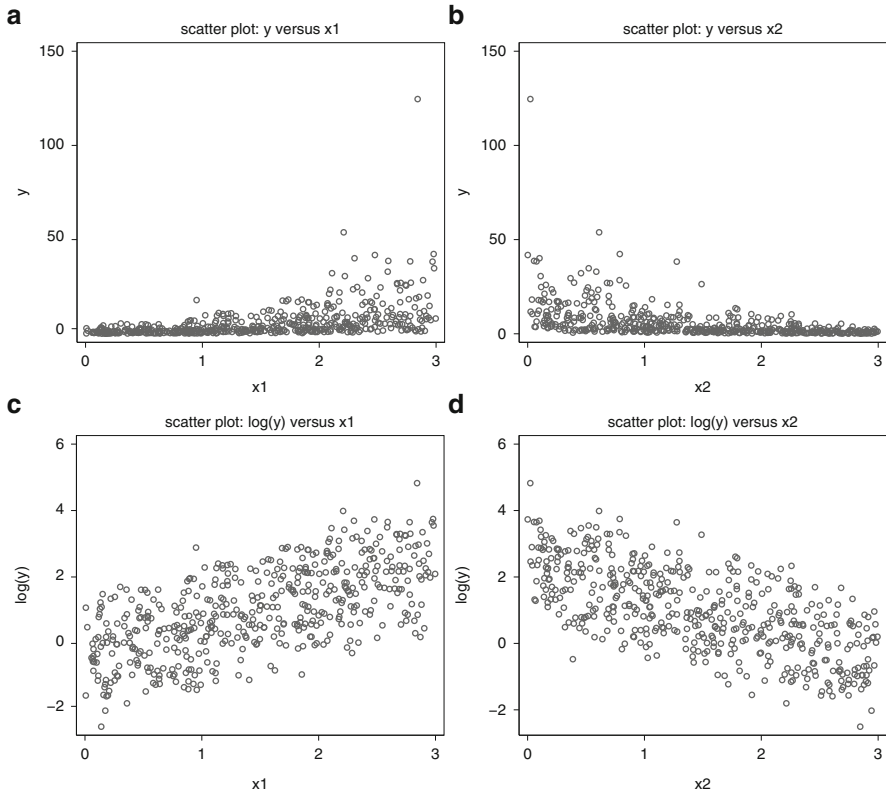


Fig. 3.6 Example for a multiplicative model: Panels (a) and (b) show scatter plots between simulated data y and x_1 , respectively, x_2 based on the model $y_i = \exp(1 + x_{i1} - x_{i2} + \varepsilon_i)$ with $\varepsilon_i \sim N(0, 0.4^2)$. Panels (c) and (d) display scatter plots of $\log(y)$ versus x_1 and x_2 , respectively

with multiplicative errors $\tilde{\varepsilon}_i = \exp(\varepsilon_i)$. Models with multiplicative error structure are more plausible for exponential relationships since the errors are proportional to the mean value of y . Figure 3.6 shows simulated data with multiplicative errors. Panels (a) and (b) display scatter plots of a response y with two explanatory variables x_1 and x_2 . The data are generated according to the model

$$y_i = \exp(1 + x_{i1} - x_{i2} + \varepsilon_i), \quad (3.3)$$

with $\varepsilon_i \sim N(0, 0.4^2)$. It is difficult to make a statement about the strength and type of the relationship between y and x_1 or x_2 on the basis of the scatter plots. Many popular models with multiplicative errors can be expressed as linear models with additive errors through a simple transformation of variables. A logarithmic transformation of (3.2) results in the linear model

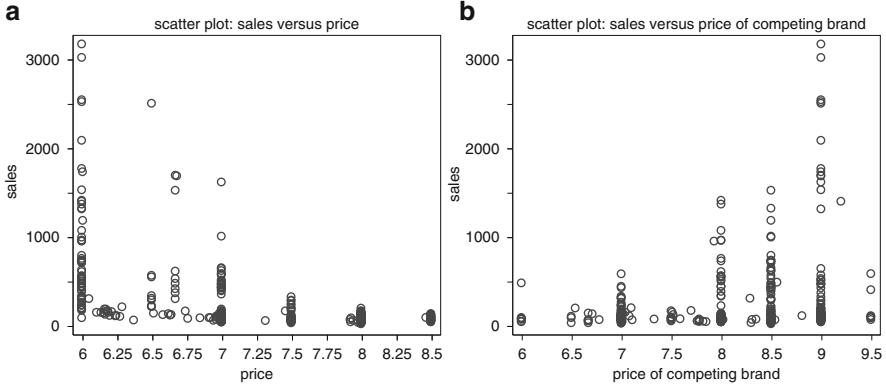


Fig. 3.7 Supermarket scanner data: scatter plot between the sales of a particular brand and its price [panel (a)] and the price of a competing brand [panel (b)], respectively

$$\log(y_i) = \beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik} + \varepsilon_i.$$

Hence, we can treat an exponential model within the scope of linear models by taking the logarithm of the response variable. Panels (c) and (d) in Fig. 3.6 show scatter plots between the logarithmic response value $\log(y)$ and the covariates x_1 and x_2 for the simulated model (3.3), which provides clear evidence of linear relationships under transformation. Provided that the errors are normally distributed, the response y is *log-normally distributed* (see Appendix B, p. 641) resulting in

$$E(y_i) = \exp(x_i' \beta + \sigma^2/2)$$

for the (conditional) mean of y_i .

Example 3.2 Supermarket Scanner Data

Figure 3.7 shows scatter plots for real data that look similar to the simulated data in Fig. 3.6. The graphs show scanner data, which are collected routinely during checkout at supermarkets. Here, we want to examine the dependence of weekly sales of a product (in our case a particular brand of coffee) on its own price and the price of a competing brand, respectively. Panel (a) shows the scatter plot between sales and the brand price; panel (b) shows a similar scatter plot between sales and the price for a competing brand of coffee. Based on the scatter plots alone, it is difficult to verify the hypothesis that a higher price leads to a decrease in sales and a higher price for the competing brand leads to an increase in sales.

△

3.1.3 Modeling the Effects of Covariates

Modeling the effects of covariates has already been explored in Sect. 2.2 (p. 22) and in connection with the discussion of model assumptions (last paragraph of Sect. 3.1.2). This section examines this aspect in more detail.

Continuous Covariates

As mentioned in our discussion of the model assumptions, we can fit nonlinear relationships within the scope of linear models. When dealing with continuous explanatory variables, a nonlinear specification is often necessary, and in this section we examine two simple methods to do so: simple *variable transformations* and *polynomial regression*. Chapter 8 will examine more flexible and automated methods.

Modeling with variable transformation or use of polynomials can be best demonstrated through examples. We use the Munich rent index data for illustration and start by modeling the relationship between net rent and area.

Example 3.3 Munich Rent Index—Variable Transformation

In connection with homo- and heteroscedastic variances in Sect. 3.1.2, a regression line between net rent and area has already been fit (see Fig. 3.2). Here, the assumption of a linear relationship seems to be justified. If we rather use the net rent per square meter as the response variable, we obtain the fit $\widehat{rentsqm}_i = 9.47 - 0.035 \cdot area_i$. Figure 3.8, panel (a) shows the scatter plot between net rent per square meter and living area, along with the fitted regression line. Panels (c) and (e) show the corresponding residuals and average residuals for every square meter.

We find that the residuals are mostly positive for smaller apartments ($area < 30$) indicating a nonlinear relationship between net rent per square meter and area. Variable transformations offer one possibility to specify such a nonlinear relationship. To do so, we consider the regression model

$$rentsqm_i = \beta_0 + \beta_1 \cdot f(area_i) + \varepsilon_i, \quad (3.4)$$

where f is an arbitrary function. The function f must be appropriately specified, i.e., f itself is fixed in advance and not estimated. Obviously, this model is a generalization of the linear model $rentsqm_i = \beta_0 + \beta_1 area_i + \varepsilon_i$. By defining the variable $x_i = f(area_i)$, we again obtain a linear model $rentsqm_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ as a special case. A transformation that seems to be appropriate for modeling the nonlinear relationship between $rentsqm$ and $area$ is given by $f(area_i) = 1/area_i$. We also chose this transformation in Example 2.2, but using only a subset of the data (apartments built after 1966 in average location). The least squares estimates for the entire data set result in

$$\widehat{rentsqm}_i = 4.73 + 140.18 \cdot f(area_i) = 4.73 + 140.18 \cdot 1/area_i.$$

The design matrix for this model is

$$X = \begin{pmatrix} 1 & 1/30 \\ 1 & 1/37 \\ \vdots & \vdots \\ 1 & 1/73 \\ 1 & 1/73 \end{pmatrix}.$$

The second column of the design matrix contains observations $x_1 = f(area_1) = 1/area_1 = 1/30$, $x_2 = 1/37$, ... of the variable x , which represents the transformed *area* variable.

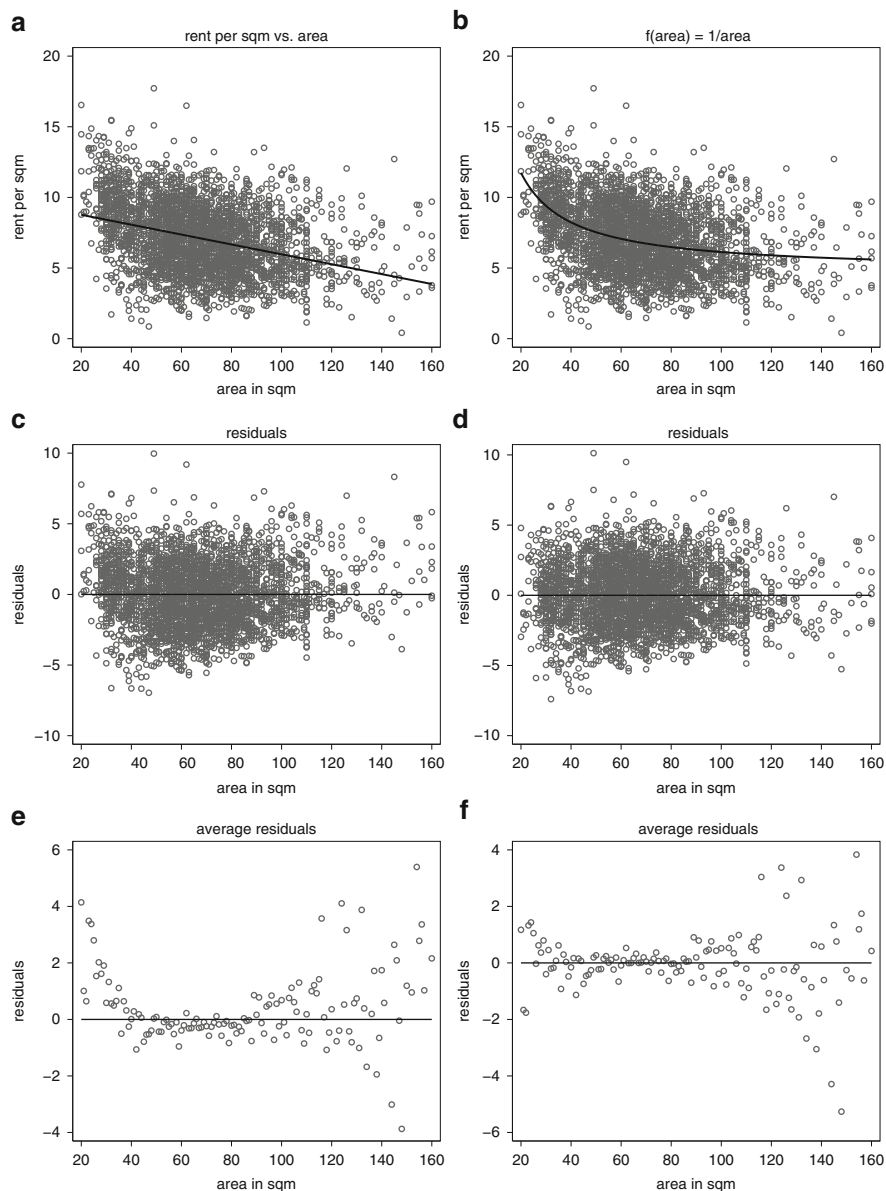


Fig. 3.8 Munich rent index: illustration for modeling nonlinear relationships via variable transformation. The *left* column shows the estimated regression line including observations [panel (a)], corresponding residuals [panel (b)], and average residuals for every distinct covariate value [panel (c)]. The *right* column displays the estimated nonlinear relationship $\widehat{\text{rentsqm}}_i = 4.73 + 140.18 \cdot 1/\text{area}_i$ [panel (d)] and the corresponding residual plots [panels (e) and (f)]

The interpretation of nonlinear relationships is best done by visualizing the estimated curves. Figure 3.8b plots the estimated relationship $\widehat{rentsqm}_i = 4.73 + 140.18 \cdot 1/area_i$ as a function of living area together with the observations $(rentsqm_i, area_i)$. Note that the curve represents the (conditional) mean of the response variable $rentsqm$ conditional on the living area. Overall the average net rent per square meter decreases with an increase of living area. The decrease of net rent is highest for small apartments up to about 40 to 45 square meters, then the curve levels off. Apartments of about 100 square meters or more have almost the same average net rent per square meter. When looking at the residuals $\hat{\varepsilon}_i = rentsqm_i - 4.73 - 140.18 \cdot 1/area_i$ in Fig. 3.8d or the average residuals in Fig. 3.8f, we can see that the chosen nonlinear curve fits the data well. Of course, transformations other than $f(area_i) = 1/area_i$ are possible, e.g., $f(area_i) = \log(area_i)$. In many cases different transformations lead to very similar fits. Hence, we do not have a panacea when choosing an appropriate transformation. We depend on the visual inspection of scatter plots or residuals in order to find reasonable and suitable variable transformations. Δ

Polynomial fitting offers another simple way to specify nonlinear relationships. In this case, we approximate the effect of a continuous covariate with a polynomial. We illustrate polynomial modeling again using the data of the Munich rent index.

Example 3.4 Munich Rent Index—Polynomial Regression

Assuming a second-order polynomial for the effect of area, we obtain the following regression model:

$$rentsqm_i = \beta_0 + \beta_1 \cdot area_i + \beta_2 \cdot area_i^2 + \varepsilon_i.$$

A polynomial of third degree leads to

$$rentsqm_i = \beta_0 + \beta_1 \cdot area_i + \beta_2 \cdot area_i^2 + \beta_3 \cdot area_i^3 + \varepsilon_i.$$

Typically, polynomials of a low degree are preferred. We rarely use polynomials of degree higher than three, as the resulting estimates become very unstable and exhibit high variability particularly at the borders of the covariate domain. Similar to using variable transformation, the nonlinear models can be connected to linear models. By defining new variables $x_{i1} = area_i$, $x_{i2} = area_i^2$ and $x_{i3} = area_i^3$, we obtain

$$rentsqm_i = \beta_0 + \beta_1 \cdot x_{i1} + \beta_2 \cdot x_{i2} + \beta_3 \cdot x_{i3} + \varepsilon_i,$$

when using a polynomial of degree 2 and

$$rentsqm_i = \beta_0 + \beta_1 \cdot x_{i1} + \beta_2 \cdot x_{i2} + \beta_3 \cdot x_{i3} + \varepsilon_i$$

when using a polynomial of degree 3. The corresponding design matrices are defined, respectively, by

$$X = \begin{pmatrix} 1 & 30 & 30^2 \\ 1 & 37 & 37^2 \\ \vdots & \vdots & \vdots \\ 1 & 73 & 73^2 \\ 1 & 73 & 73^2 \end{pmatrix} = \begin{pmatrix} 1 & 30 & 900 \\ 1 & 37 & 1369 \\ \vdots & \vdots & \vdots \\ 1 & 73 & 5329 \\ 1 & 73 & 5329 \end{pmatrix} \quad (3.5)$$

and

$$X = \begin{pmatrix} 1 & 30 & 30^2 & 30^3 \\ 1 & 37 & 37^2 & 37^3 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & 73 & 73^2 & 73^3 \\ 1 & 73 & 73^2 & 73^3 \end{pmatrix} = \begin{pmatrix} 1 & 30 & 900 & 27000 \\ 1 & 37 & 1369 & 50653 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & 73 & 5329 & 389017 \\ 1 & 73 & 5329 & 389017 \end{pmatrix}.$$

The estimated curves are given by

$$\begin{aligned} \widehat{rentsqm}_i &= 11.83 - 0.106 \cdot x_{i1} + 0.00047 \cdot x_{i2} \\ &= 11.83 - 0.106 \cdot area_i + 0.00047 \cdot area_i^2, \end{aligned}$$

when using a polynomial of degree 2 and

$$\begin{aligned} \widehat{rentsqm}_i &= 14.28 - 0.217 \cdot x_{i1} + 0.002 \cdot x_{i2} - 0.000006 \cdot x_{i3} \\ &= 14.28 - 0.217 \cdot area_i + 0.002 \cdot area_i^2 - 0.000006 \cdot area_i^3 \end{aligned}$$

for the polynomial of degree 3. To interpret results, we again visualize the estimated effects. Figure 3.9, panel (a) shows the estimated curve between net rent per square meter and living area for a second-degree polynomial. Panel (c) illustrates the corresponding residuals associated with this model. Panels (b) and (d) show the fit when using a third-degree polynomial. We obtain quite similar results compared to variable transformation $f(area_i) = 1/area_i$. Using visualization, we cannot decide as to which of the models is more appropriate. Here, we need more formal techniques for variable selection and model choice; see Sect. 3.4.

△

Thus far we have only considered one variable. The extension to more covariates is straightforward, as the next example demonstrates.

Example 3.5 Munich Rent Index—Additive Models



It is straightforward to include additional covariates with a linear effect into the model. For instance, we can extend the model (3.4) by adding the apartment's year of construction. We obtain

$$rentsqm_i = \beta_0 + \beta_1 \cdot f(area_i) + \beta_2 \cdot yearc_i + \varepsilon_i. \quad (3.6)$$

When using the transformation $f(area) = 1/area$, we obtain the fit

$$\widehat{rentsqm}_i = -65.41 + 119.36 \cdot 1/area_i + 0.036 \cdot yearc_i.$$

Alternatively, we can also estimate the effect of the year of construction nonlinearly. A model using a polynomial of degree 3 is given by

$$rentsqm_i = \beta_0 + \beta_1 \cdot 1/area_i + \beta_2 \cdot yearc_i + \beta_3 \cdot yearc_i^2 + \beta_4 \cdot yearc_i^3 + \varepsilon_i. \quad (3.7)$$

The corresponding fit is

$$\begin{aligned} \widehat{rentsqm}_i &= 29113.6 + 129.57 \cdot 1/area_i - 42.835 \cdot yearc_i + 0.020949 \cdot yearc_i^2 \\ &\quad - 0.00000340 \cdot yearc_i^3. \end{aligned}$$

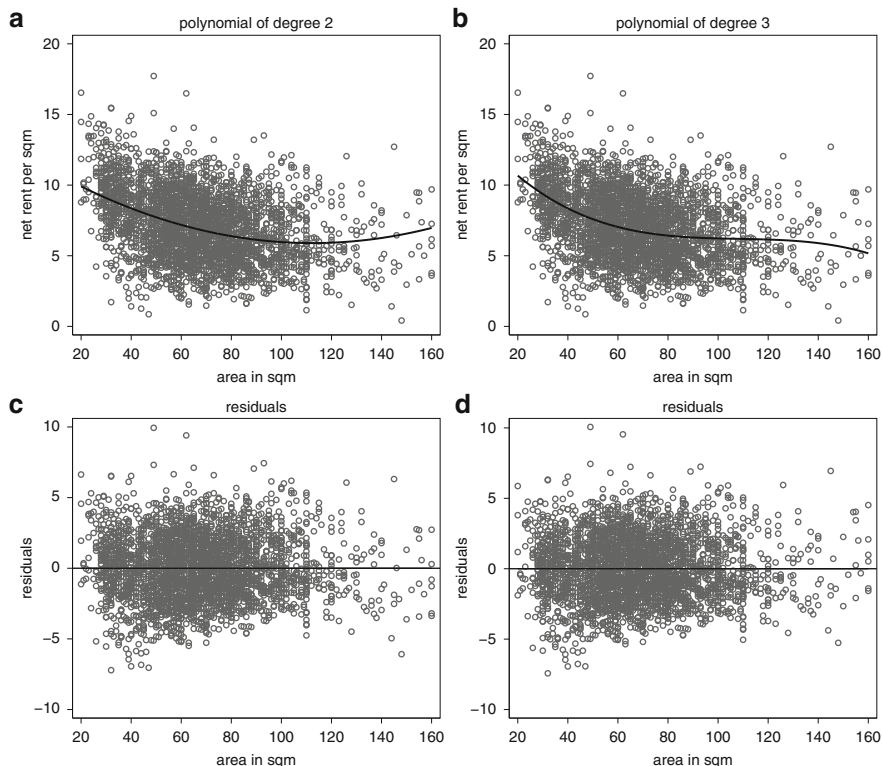


Fig. 3.9 Munich rent index: illustration for modeling nonlinear relationships using polynomials. The *upper panels* show fitted quadratic and cubic polynomials including observations. The *lower panels* display the corresponding residuals

The results are obtained with STATA. Note that with other statistics packages we sometimes obtained slightly different results due to rounding errors.

Combining the effects of the living area and the year of construction in the functions

$$f_1(\text{area}) = \beta_1 \cdot 1/\text{area}$$

and

$$f_2(\text{yearc}) = \beta_2 \cdot \text{yearc} + \beta_3 \cdot \text{yearc}^2 + \beta_4 \cdot \text{yearc}^3$$

we obtain the model

$$\text{rentsqm}_i = \beta_0 + f_1(\text{area}_i) + f_2(\text{yearc}_i) + \varepsilon_i.$$

The estimated functions are

$$\hat{f}_1(\text{area}) = 129.57 \cdot 1/\text{area}$$

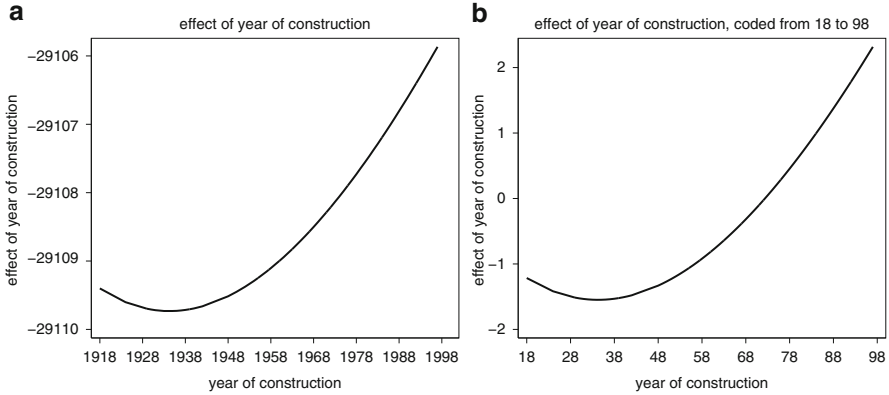


Fig. 3.10 Munich rent index: plots of the estimated nonlinear effect $\hat{f}_2(\text{yearc})$ of the year of construction

and

$$\hat{f}_2(\text{yearc}) = -42.835 \cdot \text{yearc} + 0.020949 \cdot \text{yearc}^2 - 0.00000340 \cdot \text{yearc}^3,$$

respectively. This is a special case of an additive model as in Chap. 2, Sect. 2.6.

The interpretation of the effects can again be achieved most successfully through visualization. We plot the estimated effects $\hat{f}_1(\text{area})$ and $\hat{f}_2(\text{yearc})$ for living area or year of construction. Figure 3.10a provides a visualization of the effect of year of construction. We first notice that the range of values on the vertical axis is approximately between $-29,110$ and $-29,106$. The reason for the unusual range is the scale of the year of construction, having values between 1918 and 1998. If we rather specify the year of construction with the values $18, \dots, 98$ (subtracting 1,900 of the original values), we obtain

$$\widehat{\text{rentsqm}}_i = 5.42 + 129.57 \cdot 1/\text{area}_i - 0.094 \cdot \text{yearc}_i + 0.0016 \cdot \text{yearc}_i^2 - 0.0000034 \cdot \text{yearc}_i^3.$$

Looking at the parameter estimates, it appears that the redefined effect of the year of construction differs clearly from the original effect. However, Fig. 3.10b shows that the effect is only shifted vertically, otherwise remaining the same. In fact, the level of the nonlinear function is not identifiable (as already pointed out in Sect. 2.6). If we add any constant a to $\hat{f}_2$ and subtract the same constant from $\hat{\beta}_0$, the estimated rent per square meter $\widehat{\text{rentsqm}}$ remains unchanged. This implies that the level of the nonlinear function can be arbitrarily changed, e.g., by transforming the explanatory variable.

For the sake of better interpretability, it is often useful to require that all functions “are centered around zero.” This condition is automatically fulfilled if all covariates are centered around zero. In case of the living area, we would replace the variable $\text{areainv} = 1/\text{area}$ by the centered version $\text{areainvc} = \text{areainv} - \overline{\text{areainv}}$, where $\overline{\text{areainv}}$ is the arithmetic mean of the variable areainv . We could do the same with the year of construction by replacing the variables yearc , $\text{yearc2} = \text{yearc}^2$ and $\text{yearc3} = \text{yearc}^3$ by $\text{yearcc} = \text{yearc} - \overline{\text{yearc}}$, $\text{yearc2c} = \text{yearc2} - \overline{\text{yearc2}}$ and $\text{yearc3c} = \text{yearc3} - \overline{\text{yearc3}}$.

In case of polynomial modeling, we can go a step further and use the so-called *orthogonal polynomials*, where we replace the usual basis $\text{yearc}, \text{yearc}^2, \text{yearc}^3$ for representing polynomials with an orthogonal polynomial basis. This implies that the columns of the design matrix corresponding to the variables yearc , yearc^2 , and yearc^3 are centered and

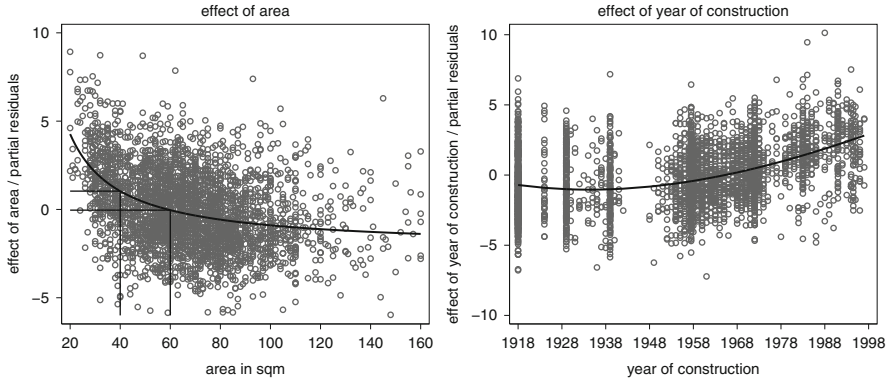


Fig. 3.11 Munich rent index: plots of the nonlinear and centered effects of living area and year of construction including partial residuals

orthogonal. Apart from the fact that the corresponding covariate effects are centered, orthogonal polynomials have other desirable properties, such as more stable computation of the least squares estimators. Typically, orthogonal polynomials are computed automatically using built-in functions provided by most statistical packages (e.g., in STATA the function `orthpol`). In Example 3.9 (p. 111), however, we show how to compute orthogonal polynomials using properties of the least squares method.

Using the orthogonal variables $yearco$, $yearco2$, $yearco3$, and the centered variable $areainvc$, we obtain the model fit

$$\widehat{rentsqm}_i = 7.11 + 129.57 \cdot areainvc_i + 0.79 \cdot yearco_i + 0.50 \cdot yearco2_i - 0.032 \cdot yearco3_i.$$

Figure 3.11 displays the corresponding effects $\hat{f}_1(area)$ and $\hat{f}_2(yearc)$ together with the partial residuals $\hat{\varepsilon}_{area,i}$ and $\hat{\varepsilon}_{yearc,i}$, which are defined by

$$\hat{\varepsilon}_{area,i} = rentsqm_i - \hat{\beta}_0 - \hat{f}_2(yearc_i) = \hat{\varepsilon}_i + \hat{f}_1(area_i)$$

and

$$\hat{\varepsilon}_{yearc,i} = rentsqm_i - \hat{\beta}_0 - \hat{f}_1(area_i) = \hat{\varepsilon}_i + \hat{f}_2(yearc_i),$$

respectively. The partial residuals for area, $\hat{\varepsilon}_{area,i}$, account for the remaining variation due to area, where all of the other effects are eliminated (in this case only the year of construction). Analogously, in the partial residuals for year of construction, $\hat{\varepsilon}_{yearc,i}$, the effect of the area, but not that of year of construction, is eliminated. By plotting the partial residuals in Fig. 3.11, we are able to visually assess whether or not the nonlinear fit is adequate.

We can interpret the effects of the living area and the year of construction accordingly: The plotted area effect specifies the influence of the living area on the average net rent per square meter, if the other covariates (in our case the year of construction) are held constant. For apartments with living area of 40 and 60 square meters, we obtain, for example, the effects $\hat{f}_1(40) \approx 1$ and $\hat{f}_1(60) \approx 0$. This implies that on average 60-square-meter-sized apartments are 1 Euro less expensive than apartments of size 40 square meters, assuming that both apartments were constructed in the same year. If we compare 60- and 80-square-meter-sized apartments, the difference in price is only 0.5 Euro. Even though in both examples we have a difference of 20 square meters of living space, the difference is cut in half in relation to the average net rent. The reason for this can be seen through the clear

3.2 Modeling Nonlinear Covariate Effects Through Variable Transformation

If the continuous covariate z has an approximately nonlinear effect $\beta_1 f(z)$ with known transformation f , then the model

$$y_i = \beta_0 + \beta_1 f(z_i) + \dots + \varepsilon_i$$

can be transformed into the linear regression model

$$y_i = \beta_0 + \beta_1 x_i + \dots + \varepsilon_i,$$

where $x_i = f(z_i) - \bar{f}$. By subtracting

$$\bar{f} = \frac{1}{n} \sum_{i=1}^n f(z_i),$$

the estimated effect $\hat{\beta}_1 x$ is automatically centered around zero. The estimated curve is best interpreted by plotting $\hat{\beta}_1 x$ against z (instead of x).

nonlinear effect, which becomes nearly constant for large values of the living area. We can interpret the effect of the year of construction in a similar fashion. Apartments that were constructed prior to World War II show roughly the same price (for the same living area). Apartments that were constructed after 1945 show an approximately linear price increase. $\triangle$

Categorical Covariates

To this point, we have discussed modeling the effect of continuous covariates. In this section, we will discuss categorical covariates and their characteristics. We will illustrate the methodology with the help of the Munich rent index data. More specifically, we will discuss appropriate modeling of the variable *location* with the three categories, 1 = average location, 2 = good location, and 3 = top location. In a first (naive) attempt we treat *location* as if it were continuous and obtain the model

$$\text{rentsqm}_i = \beta_0 + \beta_1 \cdot \text{location}_i + \varepsilon_i.$$

For simplicity, we have omitted all other covariates in this illustrative model. Using the least squares method, we obtain the fit

$$\widehat{\text{rentsqm}}_i = 6.54 + 0.39 \cdot \text{location}_i.$$

3.3 Modeling Nonlinear Covariate Effects Through Polynomials

If the continuous covariate z has an approximately polynomial effect $\beta_1 z + \beta_2 z^2 + \dots + \beta_l z^l$ of degree l , then the model

$$y_i = \beta_0 + \beta_1 z_i + \beta_2 z_i^2 + \dots + \beta_l z_i^l + \dots + \varepsilon_i$$

can be transformed into the linear regression model

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_l x_{il} + \dots + \varepsilon_i,$$

where $x_{i1} = z_i$, $x_{i2} = z_i^2$, $\dots$, $x_{il} = z_i^l$.

The centering (and possibly orthogonalization) of the vectors $\mathbf{x}^j = (x_{1j}, \dots, x_{nj})'$, $j = 1, \dots, l$, to $\mathbf{x}^1 - \bar{\mathbf{x}}_1, \dots, \mathbf{x}^l - \bar{\mathbf{x}}_l$ with the mean vector $\bar{\mathbf{x}}_j = (\bar{x}_j, \dots, \bar{x}_j)'$ facilitates interpretation of the estimated effects. A graphical illustration of the estimated polynomial is a useful way to interpret the estimated effect of z .

Due to the chosen coding, the effect of a good location would be twice as high as it would be for an average location (0.39 Euro versus $2 \cdot 0.39 = 0.78$ Euro). The effect for top location would be three times as high (0.39 Euro versus $3 \cdot 0.39 = 1.17$ Euro). If we coded the location with 2 = average location, 6 = good location, and 8 = top location, apartments in a good location would be three times as expensive as apartments in an average location; apartments in a top location would be four times as expensive as apartments in an average location. This shows that the results are dependent on the arbitrarily chosen coding of the categorical covariate. The problem is that we cannot interpret the distances between the categories in a reasonable way. A good location is not twice as good (or three times as good in the second coding) as an average location. A remedy is to define new covariates, so-called *dummy variables*, and estimate a separate effect for each category of the original covariate. In the case of location, we define the following three dummy variables:

$$alocation = \begin{cases} 1 & \text{location} = 1 \text{ (average location),} \\ 0 & \text{otherwise,} \end{cases}$$

$$glocation = \begin{cases} 1 & \text{location} = 2 \text{ (good location),} \\ 0 & \text{otherwise,} \end{cases}$$

$$tlocation = \begin{cases} 1 & \text{location} = 3 \text{ (top location).} \\ 0 & \text{otherwise,} \end{cases}$$

We obtain the model

$$\text{rentsqm}_i = \beta_0 + \beta_1 \cdot \text{allocation}_i + \beta_2 \cdot \text{glocation}_i + \beta_3 \cdot \text{tlocation}_i + \varepsilon_i.$$

If we look more closely, we find another difficulty when taking this approach: The regression parameters are not identifiable. To fully understand the identification problem, we inspect the definition of the location effects. For an average location ($\text{allocation}_i = 1, \text{glocation}_i = 0, \text{tlocation}_i = 0$), we obtain the effect $\beta_0 + \beta_1$. For good or top locations the effects result in $\beta_0 + \beta_2$ and $\beta_0 + \beta_3$, respectively. If we now add an arbitrary value a to the intercept β_0 and, at the same time, subtract a from the coefficients β_1, β_2 , and β_3 , we obtain the same total effects. Hence we cannot uniquely determine the regression parameters. Even with the help of the least squares method, we would not obtain unique estimators. This identification problem can be resolved in one of two ways: Either we use a model without intercept, or we remove one of the dummy variables *allocation*, *glocation*, or *tlocation* from the model. Since an intercept is usually included in the model, we restrict ourselves to the second option. If we do not consider the variable *glocation*, we obtain the model

$$\text{rentsqm}_i = \beta_0 + \beta_1 \cdot \text{allocation}_i + \beta_2 \cdot \text{tlocation}_i + \varepsilon_i.$$

The effects are now given by $\beta_0 + \beta_1$ for an average location, β_0 for good location, and $\beta_0 + \beta_2$ for top location. The regression parameters are uniquely determined and are thus identifiable. Using the method of least squares, we obtain the fit

$$\widehat{\text{rentsqm}}_i = 7.27 - 0.31 \cdot \text{allocation}_i + 0.90 \cdot \text{tlocation}_i.$$

Care must be taken when interpreting the estimated regression coefficients. They have to be interpreted with respect to the omitted category, in our example apartments in good location. We obtain an estimated effect of $\hat{\beta}_0 + \hat{\beta}_1 = 7.27 - 0.31 = 6.96$ for average location, $\hat{\beta}_0 = 7.27$ for good location, and $\hat{\beta}_0 + \hat{\beta}_2 = 7.27 + 0.90 = 8.17$ for the top location. Compared to apartments in good location, apartments in average location are 0.31 Euro less expensive and apartments in top location are 0.90 Euro more expensive. The category of good location omitted in the model and used for comparison is also called the *reference category*. Note that the interpretation of the estimated regression coefficients depends on the chosen reference category. However, all models are equivalent, which means that the estimated regression coefficients for a particular reference category can be computed from the regression coefficients of the original model. For example, if we use the average location as reference category, we obtain the fitted model

$$\widehat{\text{rentsqm}}_i = 6.96 + 0.31 \cdot \text{glocation}_i + 1.21 \cdot \text{tlocation}_i.$$

In comparison to apartments in average location, apartments in good location are 0.31 Euro more expensive and apartments in top location are 1.21 Euro more expensive. The results are fully consistent with the previous coding.

3.4 Dummy Coding for Categorical Covariates

For modeling the effect of a covariate $x \in \{1, \dots, c\}$ with c categories using dummy coding, we define the $c - 1$ dummy variables

$$x_{i1} = \begin{cases} 1 & x_i = 1, \\ 0 & \text{otherwise,} \end{cases} \quad \dots \quad x_{i,c-1} = \begin{cases} 1 & x_i = c - 1, \\ 0 & \text{otherwise,} \end{cases}$$

for $i = 1, \dots, n$, and include them as explanatory variables in the regression model

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_{i,c-1} x_{i,c-1} + \dots + \varepsilon_i.$$

For reasons of identifiability, we omit one of the dummy variables, in this case the dummy variable for category c . This category is called reference category. The estimated effects can be interpreted by direct comparison with the (omitted) reference category.

In principle, any of the categories of a categorical variable could be chosen as the reference. In practice, we usually choose the category which makes most sense for interpretation, for example, the most common category found in the data set. For an arbitrary categorical covariate x with c categories, dummy coding is summarized in Box 3.4.

Note that there is more than one coding scheme for categorical covariates. Another popular scheme is *effect coding*, which is defined by

$$x_{i1} = \begin{cases} 1 & x_i = 1, \\ -1 & x_i = c, \\ 0 & \text{otherwise,} \end{cases} \quad \dots \quad x_{i,c-1} = \begin{cases} 1 & x_i = c - 1, \\ -1 & x_i = c, \\ 0 & \text{otherwise.} \end{cases}$$

In contrast to dummy coding, effect coding produces new variables that are coded with -1 for the reference category yielding a sum to zero constraint as explained in the following example that illustrates the difference between the two coding schemes, using the Munich rent index data.

Example 3.6 Munich Rent Index—Effect Coding

If we choose an average location as the reference category, we obtain the two variables

$$glocation = \begin{cases} 1 & location = 2 \text{ (good location),} \\ -1 & location = 1 \text{ (average location),} \\ 0 & \text{otherwise,} \end{cases}$$

$$tlocation = \begin{cases} 1 & location = 3 \text{ (top location),} \\ -1 & location = 1 \text{ (average location),} \\ 0 & \text{otherwise,} \end{cases}$$

in effect coding. For the regression model

$$\text{rentsqm}_i = \beta_0 + \beta_1 \cdot \text{glocation}_i + \beta_2 \cdot \text{tlocation}_i + \varepsilon_i,$$

we obtain the fitted model

$$\widehat{\text{rentsqm}}_i = 7.47 - 0.19 \cdot \text{glocation}_i + 0.71 \cdot \text{tlocation}_i.$$

Due to the specific coding, we are able to compute an additional regression coefficient associated with the reference category. It is obtained as the negative sum of the other parameters, i.e., $\hat{\beta}_3 = -\hat{\beta}_1 - \hat{\beta}_2 = 0.19 - 0.71 = -0.52$. This results in estimated effects $\hat{\beta}_0 + \hat{\beta}_3 = 7.47 - 0.52 = 6.95$, $\hat{\beta}_0 + \hat{\beta}_1 = 7.47 - 0.19 = 7.28$, and $\hat{\beta}_0 + \hat{\beta}_2 = 7.47 + 0.71 = 8.18$ for average, good, and top locations, respectively. $\triangle$

Interactions Between Covariates

An interaction between covariates exists if the effect of a covariate depends on the value of at least one other covariate. To start with, consider the following simple model:

$$y = \beta_0 + \beta_1 x + \beta_2 z + \beta_3 x z + \varepsilon \quad (3.8)$$

between a response y and two other explanatory variables x and z . The term $\beta_3 x z$ is called an *interaction* between x and z . The terms $\beta_1 x$ and $\beta_2 z$ depend on only one variable and are called *main effects*. We can understand the impact of an interaction term when considering how $E(y)$ changes when one variable, e.g., x , changes by an amount d . We have

$$\begin{aligned} E(y | x + d, z) - E(y | x, z) &= \beta_0 + \beta_1 (x + d) + \beta_2 z + \beta_3 (x + d) z \\ &\quad - \beta_0 - \beta_1 x - \beta_2 z - \beta_3 x z \\ &= \beta_1 d + \beta_3 d z. \end{aligned}$$

Now we can distinguish between the two cases $\beta_3 = 0$ and $\beta_3 \neq 0$:

- In the case $\beta_3 = 0$, the interaction is dropped from the model and the expected change $\beta_1 d$ is *independent* from the specific value of the second covariate z .
- In the case $\beta_3 \neq 0$, the expected change $\beta_1 d + \beta_3 d z$ does not only depend on the amount d but also on the value of the covariate z .

Therefore, an interaction is always needed if the effect of the change of a variable is also dependent on the value of another variable. The specification of interactions is dependent on the type of the variables involved. In the sections to follow, we discuss interactions between two categorical variables, i.e. between a continuous and a categorical variable, as well as between two continuous variables.

Interactions Between Categorical Variables

We first consider the simplest case, the interaction between two binary variables x and z . In context of the rent index example, the value $x = 1$ or $z = 1$ could imply the existence of, e.g., a premium kitchen or bathroom. In this case, the coefficient β_1

in Eq. (3.8) measures the effect of a premium kitchen on the net rent, β_2 measures the effect of a premium bathroom, and β_3 measures the additional effect of having both a premium kitchen and a premium bathroom. The interpretation depends on the values of the coefficients. In our example, all three coefficients are most likely positive, leading to the following interpretation: The existence of either a premium kitchen ($x = 1$) or a premium bathroom ($z = 1$) will increase the average net rent. If the apartment has both a premium kitchen and a bathroom, there is an additional rent increase.

Our example shows that the interaction between two binary variables quantifies the effect of both characteristics associated with x and z occurring simultaneously. In the case of two arbitrary categorical variables x and z with c and m categories, respectively, modeling of interactions is more complicated. We next illustrate the situation of two categorical covariates, each with three categories: Define the dummy variables x_1, x_2 that correspond to x and the dummy variables z_1, z_2 corresponding to z . We choose the last category as the reference for both x and z . For modeling the interaction effect we have to consider all possible combinations of the values of x and z (with the exception of the reference categories), specifically x_1z_1, x_1z_2, x_2z_1 , and x_2z_2 . We now obtain

$$y = \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_3z_1 + \beta_4z_2 + \beta_5x_1z_1 + \beta_6x_1z_2 + \beta_7x_2z_1 + \beta_8x_2z_2 + \varepsilon.$$

The coefficients can be interpreted as follows:

β_0 :	effect if $x = 3$ (reference) and $z = 3$ (reference),
$\beta_0 + \beta_1$:	effect if $x = 1$ and $z = 3$ (reference),
$\beta_0 + \beta_2$:	effect if $x = 2$ and $z = 3$ (reference),
$\beta_0 + \beta_3$:	effect if $x = 3$ (reference) and $z = 1$,
$\beta_0 + \beta_4$:	effect if $x = 3$ (reference) and $z = 2$,
$\beta_0 + \beta_1 + \beta_3 + \beta_5$:	effect if $x = 1$ and $z = 1$,
$\beta_0 + \beta_1 + \beta_4 + \beta_6$:	effect if $x = 1$ and $z = 2$,
$\beta_0 + \beta_2 + \beta_3 + \beta_7$:	effect if $x = 2$ and $z = 1$,
$\beta_0 + \beta_2 + \beta_4 + \beta_8$:	effect if $x = 2$ and $z = 2$.

As is common with categorical covariates, the effects have to be interpreted as difference relative to the reference categories $x = 3$ and $z = 3$. The quantity $\beta_2 + \beta_4 + \beta_8$, e.g., measures the effect of the combination $x = 2, z = 2$ when compared to the combination $x = 3, z = 3$. Note that the interaction between x and z can also be modeled by defining the new categorical variable w , whose categories consist of all possible combinations of the values of x and z :

$$w = \begin{cases} 1 & x = 1, z = 1, \\ 2 & x = 1, z = 2, \\ 3 & x = 1, z = 3, \\ \vdots & \vdots \\ 9 & x = 3, z = 3. \end{cases}$$

Of course, one of the categories of w must be specified as the reference category when defining the eight dummy variables to include in the regression model. In many cases this approach is preferable because interpretation of the effects is more straightforward.

Example 3.7 Munich Rent Index—Interaction with Quality of Kitchen

The rent index is updated every two years, with the collection of new data. For financial reasons, the update during 2001 only consisted of data for 1,500 apartments. Due to the smaller sample size, a complete redesign of the rent index was not possible. Instead the following procedure was chosen: The same characteristics of the 1999 rent index were used, implying that the structure of the rent index did not change. For all characteristics possible changes of their effects compared to 1999 have been examined. To do so, both data sets of 1999 and 2001 were analyzed simultaneously, and changes in covariate effects have been investigated with the help of interactions. We will illustrate the approach using the quality of the kitchen (*kitchen*), with categories “kitchen below average” (reference category), “normal kitchen” (dummy variable *nkitchen*), and premium kitchen (dummy variable *pkitchen*). Note that we measure in this example the quality of the kitchen in three categories rather than two categories as in Table 1.2 (p. 6) and all examples so far. Our starting point is the model (3.7) which was developed in Example 3.5 (p. 90). Here, we modeled the effect of the living area with the help of the transformation $1/area$ and the effect of the year of construction through a third-degree polynomial. We now extend this model with an interaction term between the quality of the kitchen and the survey year. We obtain the estimate

$$\widehat{rentsqm} = \cdots - 0.26 \text{ year01} + 0.91 \text{ nkitchen} + 1.09 \text{ pkitchen} \\ + 0.41 \text{ nkitchen} \cdot \text{year01} + 0.74 \text{ pkitchen} \cdot \text{year01}.$$

The dummy variable *year01* specifies whether an observation has been taken from the year 2001 (*year01* = 1) or from the year 1999 (*year01* = 0). For interpretation purposes, we compare apartments with the same living area and year of construction. The results can be summarized as follows, on a per square meter basis:

- In 2001, apartments with a below average kitchen are approximately 0.26 Euro per square meter cheaper than apartments in 1999.
- Apartments with a normal kitchen are approximately $-0.26 + 0.41 = 0.15$ Euro more expensive in 2001 than in 1999.
- Apartments with a premium kitchen are approximately $-0.26 + 0.74 = 0.48$ Euro more expensive in 2001 than in 1999.

In comparison to the original rent index, the estimated surcharge is very high for average and above average kitchens. This results partly from the fact that the model in this example does not contain all relevant covariates. At this point it is impossible to determine if the interaction is necessary. The required inference techniques will be developed in the following sections; in particular, see Sects. 3.3 and 3.4. △

Interactions with Continuous and Categorical Variables

In the illustrative Example 2.5 (p. 30) of Chap. 2, we were already concerned with the modeling of an interaction between continuous and categorical variables. There, we used data from apartments that were built after 1966 and that were found in an average or good location. Our goal was to model the relationship of the net rent with the explanatory variables living area and location. Figure 2.3 (p. 27) gives the impression that the main effects model

$$rent = \beta_0 + \beta_1 \text{ area} + \beta_2 \text{ glocation} + \varepsilon$$

is not adequate, as the regression line between net rent and living area for apartments in a good location ($glocation = 1$) is very steep relative to the reference of average location. As a result, we replaced the main effect model by

$$rent = \beta_0 + \beta_1 area + \beta_2 glocation + \beta_3 inter + \varepsilon,$$

with the interaction variable $inter = area \cdot glocation$ and obtained the estimate

$$\widehat{rent} = 130.23 + 5.57 area + 5.20 glocation + 0.82 inter.$$

We can interpret the terms as follows:

- $\beta_1 area$: Effect of the living area for apartments in average location ($glocation = 0$).
- $\beta_1 area + \beta_2 + \beta_3 inter = \beta_2 + (\beta_1 + \beta_3)area$: Effect of the area for apartments in a good location ($glocation = 1$). The two linear area effects for normal and good location can be best interpreted through visualization; see Fig. 2.6 (p. 31).
- $\beta_2 + \beta_3 inter = \beta_2 + \beta_3 area$: This is the difference effect of apartments in good location in comparison to apartments in average location. This effect varies depending on the area. Due to the positive coefficients $\hat{\beta}_2 = 5.20$ and $\hat{\beta}_3 = 0.82$, apartments in good location are always more expensive than apartments in average location (when having the same living area). The difference is even more evident when the living area is larger. Every additional square meter of living area increases the difference to apartments in average location by 0.82 Euro.

Interactions for any continuous and categorical covariates x and z can be handled analogously. To keep things simple we first assume $z \in \{0, 1\}$ is binary. If the main effects and interaction effects are linear, we have the model

$$y = \beta_0 + \beta_1 z + \beta_2 x + \beta_3 xz + \dots + \varepsilon.$$

The dots indicate that the model may contain additional terms (of other covariates). The terms in the model can be interpreted both with respect to the continuous covariate x and the binary variable z . Referring to x , we obtain the following interpretation:

- $\beta_2 x$: Linear effect of the continuous covariate x , if $z = 0$.
- $\beta_1 + (\beta_2 + \beta_3) x$: Linear effect of the continuous covariate x , if $z = 1$.
On the contrary, we obtain the following interpretation with respect to z :
- $\beta_1 + \beta_3 x$: Difference effect for observations with $z = 1$ relative to $z = 0$. The difference effect varies depending on the value of x (unlike a constant difference effect in models without interaction).

To this point, we assumed that the main effect of the continuous variable x and the interaction effect are linear. Modeling nonlinear interaction effects within the scope of our current modeling framework is also possible. To do so, we express the model in terms of the functions f_1 and f_2 :

$$y = \beta_0 + \beta_1 z + f_1(x) + f_2(x)z + \dots + \varepsilon. \quad (3.9)$$

In the case of linear modeling, we obtain the special cases $f_1(x) = \beta_2 x$ and $f_2(x) = \beta_3 x$. Nonlinear relationships can be achieved using transformation of variables or through polynomials. For example, using polynomials of degree 2 yields the functions $f_1(x) = \beta_2 x + \beta_3 x^2$ and $f_2(x) = \beta_4 x + \beta_5 x^2$ and the model

$$\begin{aligned} y &= \beta_0 + \beta_1 z + \beta_2 x + \beta_3 x^2 + (\beta_4 x + \beta_5 x^2) z + \varepsilon \\ &= \beta_0 + \beta_1 z + \beta_2 x + \beta_3 x^2 + \beta_4 xz + \beta_5 x^2 z + \varepsilon. \end{aligned}$$

We can interpret the terms analogously to linear interaction effects:

- $f_1(x)$: Nonlinear effect of the continuous covariate x when $z = 0$.
- $\beta_1 + f_1(x) + f_2(x)$: Nonlinear effect of the continuous covariate x when $z = 1$. The two curves $f_1(x)$ and $\beta_1 + f_1(x) + f_2(x)$ show the strength of interaction. In an extreme case of no interaction both curves run parallel to each other, i.e., they only differ by the constant β_1 .
- $\beta_1 + f_2(x)$: Nonlinear difference effect for observations with $z = 1$ relative to $z = 0$. The effect varies nonlinearly, depending on x . We call this a varying coefficient term, since the effect of z varies in a way that depends on the value of the other variable x . The variable x “modifies” the effect of z and thus is also called an effect modifier. In Sect. 9.3.1, we will deal with methods for flexibly and automatically estimating nonlinear effects f_2 .

To facilitate interpretation, again it is convenient to center the main effect $f_1(x)$ around zero. Automatic centering can be achieved when using centered design vectors. We do not have to center the function f_2 , since it serves as the *difference effect* to the main effect. The estimated effects $\hat{f}_1$, $\hat{\beta}_1 + \hat{f}_1 + \hat{f}_2$ and $\hat{\beta}_1 + \hat{f}_2$ can be plotted together against x for interpretation.

Example 3.8 Munich Rent Index—Interaction Between Living Area and Location

For illustration, we use the data for the Munich rent index. For simplicity, we only consider apartments in average or top locations and model the relationship between net rent per square meter and the explanatory variables living area and location. In previous examples (e.g., Example 2.4, p. 28), we modeled the effect of the living area using the transformation $1/\text{area}$. If we assume a linear interaction effect between area and location, we have

$$\begin{aligned} \text{rentsqm} &= \beta_0 + \beta_1 \text{tlocation} + \beta_2 \text{areainvc} + \beta_3 \text{area} \cdot \text{tlocation} + \varepsilon \\ &= \beta_0 + \beta_1 \text{tlocation} + f_1(\text{area}) + f_2(\text{area}) \cdot \text{tlocation} + \varepsilon, \end{aligned}$$

where areainvc denotes the transformation $1/\text{area}$ centered around zero. The dummy variable tlocation is one for apartments in top location, zero otherwise. The function $f_1(\text{area}) = \beta_2 \text{areainvc}$ represents the nonlinear effect of living area and is automatically centered around zero. The function $f_2(\text{area}) = \beta_3 \text{area} \cdot \text{tlocation}$ represents the linear interaction effect. We obtain

$$\widehat{\text{rentsqm}} = 6.94 + 0.77 \text{tlocation} + 143.12 \text{areainvc} + 0.01 \text{area} \cdot \text{tlocation}.$$

To interpret the estimated results, we visualize:

- The estimated effect $\hat{f}_1(\text{area})$ of the living area for apartments in an average location (Fig. 3.12a, solid line)

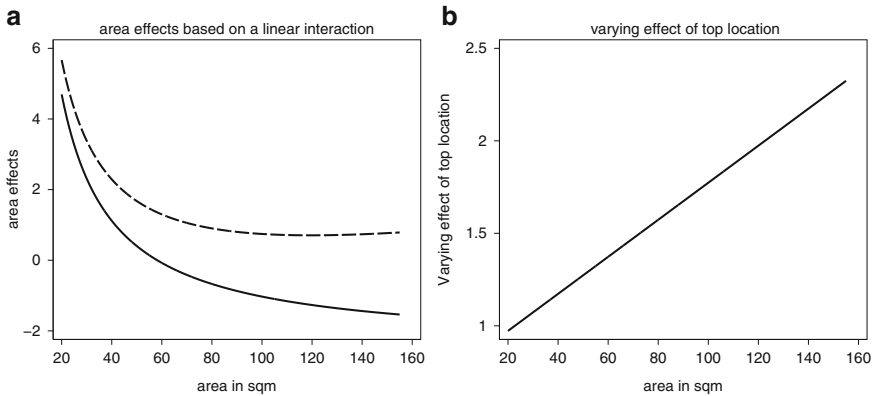


Fig. 3.12 Munich rent index: Panel (a) visualizes the area effect for average (solid line) and top location (dashed line). Panel (b) shows the effect of top location varying with respect to area

- The effect $\hat{\beta}_1 + \hat{f}_1(\text{area}) + \hat{f}_2(\text{area})$ of the living area for apartments in a top location (Fig. 3.12a, dotted line)
- The varying effect $\hat{\beta}_1 + \hat{f}_2(\text{area})$ of apartments in top location (relative to average location), depending on the living area (Fig. 3.12b)

These figures can be interpreted both with respect to the living area or the location:

- *Living area:* The effects of area in average or top location in Fig. 3.12a are similar. However, the fitted curves are not parallel to each other implying an interaction effect. If the living area increases, the average rent per square meter decreases. The decrease in rent is at first very steep, but the slope moderates for apartments of about 80 square meters or larger. Apartments in top location are always more expensive than apartments in average location, regardless of size. For smaller apartments this difference is much smaller than for larger apartments. This implies that with an increase in living area, we find an increase in the difference of the net rent per square meter for apartments in average relative to top location.
- *Location:* Regardless of living area, apartments in top location are more expensive than apartments in average location. The difference increases linearly with an increase in living area.

Thus far, we are unable to answer whether the inclusion of an interaction is significant and necessary. This question can be answered with inference techniques described in the sections to follow, specifically with statistical hypothesis testing (Sect. 3.3) and methods of model choice (Sect. 3.4).

△

Next, we briefly illustrate the case of an interaction involving multi-categorical rather than binary variables. To simplify notation, we limit ourselves to a variable with three categories 1, 2, and 3. Extensions to more than three categories are straightforward. Let z_1 and z_2 denote the dummy variables derived from z , with $z_1 = 1$ if $z = 1$, and $z_2 = 1$ if $z = 2$, and zero otherwise. The third level $z = 3$ serves as the reference category. Analogous to Eq. (3.9), an interaction model can be written as

$$y = \beta_0 + \beta_1 z_1 + \beta_2 z_2 + f_1(x) + f_2(x) z_1 + f_3(x) z_2 + \dots + \varepsilon,$$

where f_1 , f_2 , and f_3 are again linear or nonlinear functions. The model consists of the following effects:

- $f_1(x)$: Effect of the continuous variable x , if $z = 3$ (reference category).
- $\beta_1 + f_1(x) + f_2(x)$: Effect of the continuous variable x , if $z = 1$.
- $\beta_2 + f_1(x) + f_3(x)$: Effect of the continuous variable x , if $z = 2$.
- $\beta_1 + f_2(x)$: Varying effect of the level $z = 1$ depending on x . As always with categorical variables, the effect has to be interpreted as a difference relative to the reference category $z = 3$.
- $\beta_2 + f_3(x)$: Varying effect of the level $z = 2$, depending on x .

We can interpret the estimated results most easily if all the estimated effects $\hat{f}_1(x)$, $\hat{\beta}_1 + \hat{f}_1(x) + \hat{f}_2(x)$, and $\hat{\beta}_2 + \hat{f}_1(x) + \hat{f}_3(x)$ are visualized in one figure. The varying effects $\hat{\beta}_1 + \hat{f}_2(x)$ and $\hat{\beta}_2 + \hat{f}_3(x)$ can be plotted and analogously interpreted.

If we model all functions linearly, then we obtain the linear model

$$y = \beta_0 + \beta_1 z_1 + \beta_2 z_2 + \beta_3 x + \beta_4 x z_1 + \beta_5 x z_2 + \dots + \varepsilon.$$

Using quadratic polynomials results in

$$y = \beta_0 + \beta_1 z_1 + \beta_2 z_2 + \beta_3 x + \beta_4 x^2 + \beta_5 x z_1 + \beta_6 x^2 z_1 + \beta_7 x z_2 + \beta_8 x^2 z_2 + \dots + \varepsilon.$$

Interactions Between Continuous Variables

When estimating interactions between two continuous covariates, we need to model two-dimensional functions, e.g., by using two- or higher-dimensional polynomials. Here we reach the limit within the scope of linear models, since it is very difficult if not impossible to find adequate models through the inspection of (three-dimensional) scatter plots. In this case, automated methods are clearly superior, and we refer to Sects. 8.2 and 9.3.2.

3.2 Parameter Estimation

In this section we develop estimators for the unknown parameters β and σ^2 of the linear model and derive their statistical properties. Section 3.2.1 that follows addresses estimation of the regression coefficients β , and Sect. 3.2.2 discusses estimation of the variance σ^2 . In Sect. 3.2.3, we examine the statistical properties of these estimators.

3.2.1 Estimation of Regression Coefficients

The theory of estimation for the regression coefficients of linear models is closely connected to the method of least squares, first discovered by Legendre in 1806. In addition to the method of least squares, other principles for parameter estimation are possible, in particular the maximum likelihood estimator; see Appendix B.4 for

a general introduction to likelihood-based inference. We will see that the maximum likelihood estimator of the regression coefficients coincides with the least squares estimator with Gaussian errors. Alternatives include *robust methods*, which reduce the influence of outlying observations, compared to the least squares method.

The Method of Least Squares

According to the principle of least squares, the unknown regression coefficients β are estimated by minimizing the *sum of the squared deviations*

$$LS(\beta) = \sum_{i=1}^n (y_i - \mathbf{x}'_i \beta)^2 = \sum_{i=1}^n \varepsilon_i^2 = \boldsymbol{\varepsilon}' \boldsymbol{\varepsilon}, \quad (3.10)$$

with respect to $\beta \in \mathbb{R}^p$. Perhaps, at first thought, a more reasonable principle is minimizing the *sum of absolute deviations*

$$SM(\beta) = \sum_{i=1}^n |y_i - \mathbf{x}'_i \beta| = \sum_{i=1}^n |\varepsilon_i|.$$

In fact, this method is historically much older than the method of least squares. It was first proposed around 1760, by Rudjer Joseph Boscovich (1711–1787). In more modern terminology, we refer to this approach as *median regression*, since the minimizing solution yields an estimate for the median of the response variable conditional on the covariates. Median regression is a special case of *quantile regression*, which aims at estimating an arbitrary quantile q_τ , $\tau \in (0, 1)$ of the response variable conditional on the covariates. More on quantile regression can be found in Chap. 10.

Yet, the method of least squares remains the most common method for estimating the regression coefficients β . This is mainly due to the following reasons: On the one hand, the use of the least squares principle is relatively simple from a mathematical point of view. It is, for example, possible to differentiate $LS(\beta)$, in contrast to $SM(\beta)$, with respect to β . Additionally, estimators that rely on the least squares method have a number of desirable statistical properties; see Sect. 3.2.3. Figure 3.13 illustrates the difference between the least squares principle and the minimization of absolute differences by means of the simple model $y = \beta_0 + \beta_1 x + \varepsilon$. The illustration shows that observations with large deviations have relatively large impact as a matter of taking squares in the minimization criterion. Unlike median regression, least squares estimators have the disadvantage of being highly sensitive to outliers.

In order to determine the minimum of $LS(\beta)$, we rearrange Eq. (3.10) and obtain

$$\begin{aligned} LS(\beta) &= \boldsymbol{\varepsilon}' \boldsymbol{\varepsilon} \\ &= (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta) \\ &= \mathbf{y}'\mathbf{y} - \beta' \mathbf{X}'\mathbf{y} - \mathbf{y}'\mathbf{X}\beta + \beta' \mathbf{X}'\mathbf{X}\beta \\ &= \mathbf{y}'\mathbf{y} - 2\mathbf{y}'\mathbf{X}\beta + \beta' \mathbf{X}'\mathbf{X}\beta. \end{aligned} \quad (3.11)$$

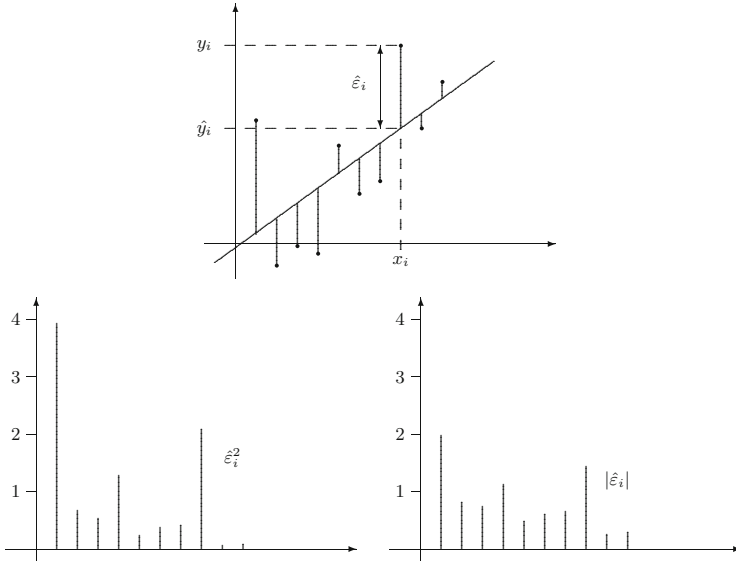


Fig. 3.13 Illustration of the least squares method: The *upper panel* shows some observations (y_i, x_i) , scattered around a straight line. According to the principle of least squares, the regression line is chosen such that the sum of squared differences in the *lower left panel* is minimized. As discussed before, squaring the residuals yields comparably large values for observations with larger deviations from the regression line. For comparison, the *lower right panel* displays absolute deviations where this effect is much smaller

We exploit the fact that the terms $\beta'X'y$ and $y'X\beta$ (as well as $y'y$ and $\beta'X'X\beta$) are scalars. Thus $\beta'X'y$ is equal to its transpose $(\beta'X'y)'$, and we have $\beta'X'y = (\beta'X'y)' = y'X\beta$.

We minimize $LS(\beta)$ by setting the vector of first derivatives zero and by showing that the matrix of second derivatives is positive definite. Readers unfamiliar with vector and matrix calculus should first inspect Appendix A.8 (p. 635). Applying two rules for the differentiation of vector functions [see Theorem A.33 (1) and (3)], we obtain

$$\frac{\partial LS(\beta)}{\partial \beta} = -2X'y + 2X'X\beta. \quad (3.12)$$

Taking second derivatives results in

$$\frac{\partial^2 LS(\beta)}{\partial \beta \partial \beta'} = \frac{\partial (-2X'y + 2X'X\beta)}{\partial \beta'} = 2X'X.$$

We thereby used Theorem A.33 (5). According to our assumptions, the columns of the design matrix X are linear independent, i.e., $\text{rk}(X) = k + 1 = p$. Thus, according to Theorem A.30 (p. 634), the matrix $X'X$ is positive definite.

We therefore obtain the least squares estimator $\hat{\beta}$ by setting Eq. (3.12) to zero or equivalently by solving the so-called *normal equations*:

$$X'X\hat{\beta} = X'y. \quad (3.13)$$

Since $X'X$ is positive definite and invertible (see Theorem A.28, p. 634), the normal equations have a unique solution given by the least squares estimator

$$\hat{\beta} = (X'X)^{-1}X'y. \quad (3.14)$$

Maximum Likelihood Estimation

We obtained the least squares estimator without any specific distributional assumptions regarding the error term ϵ . Assuming normally distributed errors, i.e., $\epsilon \sim N(0, \sigma^2 I)$, the maximum likelihood (ML) estimator for the unknown parameters can be computed (see Appendix B.4 for an introduction to the theory of maximum likelihood estimators). We next show that the ML estimator for β is equivalent to the least squares estimator.

Assuming normally distributed errors we have $y \sim N(X\beta, \sigma^2 I)$, which yields the likelihood

$$L(\beta, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2}(y - X\beta)'(y - X\beta)\right). \quad (3.15)$$

The log-likelihood is thus given by

$$l(\beta, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2}(y - X\beta)'(y - X\beta). \quad (3.16)$$

When maximizing the log-likelihood with respect to β , we can ignore the first two terms of the sum in Eq. (3.16) because they are independent of β . Maximizing $-\frac{1}{2\sigma^2}(y - X\beta)'(y - X\beta)$ is equivalent to minimizing $(y - X\beta)'(y - X\beta)$, which is the least squares criterion (3.10). The maximum likelihood estimator of β is therefore identical to the least squares estimator (3.14).

Predicted Values and Residuals

Based on the least squares estimator $\hat{\beta} = (X'X)^{-1}X'y$ for β , we can estimate the (conditional) mean of y by

$$\widehat{E(y)} = \hat{y} = X\hat{\beta}.$$

Substituting the least squares estimator further results in

$$\hat{y} = X(X'X)^{-1}X'y = Hy,$$

with the $n \times n$ -matrix

$$H = X(X'X)^{-1}X'.$$

3.5 Properties of the Hat Matrix

The hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ has the following properties:

1. $\mathbf{H}$ is symmetric.
2. $\mathbf{H}$ is idempotent. For the definition of idempotent matrices, see Appendix A, Definition A.12.
3. $\text{rk}(\mathbf{H}) = \text{tr}(\mathbf{H}) = p$, i.e., the trace is equal to the rank.
4. $\frac{1}{n} \leq h_{ii} \leq \frac{1}{r}$, where r represents the number of rows in $\mathbf{X}$ with different $\mathbf{x}_i$. If the rows are all different, then $\frac{1}{n} \leq h_{ii} \leq 1$.
5. The matrix $\mathbf{I} - \mathbf{H}$ is also symmetric and idempotent, with $\text{rk}(\mathbf{I} - \mathbf{H}) = n - p$.

The matrix $\mathbf{H}$ is also called the *prediction matrix* or *hat matrix*. Some easily verifiable properties of the hat matrix are summarized in Box 3.5.

With the help of the hat matrix $\mathbf{H}$, it is also possible to express the residuals $\hat{\epsilon}_i = y_i - \hat{y}_i$ in matrix notation. We have

$$\hat{\epsilon} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{H}\mathbf{y} = (\mathbf{I} - \mathbf{H})\mathbf{y}.$$

3.2.2 Estimation of the Error Variance

Maximum Likelihood Estimation

It is rather natural to estimate the variance σ^2 using maximum likelihood. We already determined the likelihood $L(\boldsymbol{\beta}, \sigma^2)$ and the log-likelihood $l(\boldsymbol{\beta}, \sigma^2)$ for the linear model; see Eqs. (3.15) and (3.16). Differentiation of the log-likelihood (3.16) with respect to σ^2 , and setting to zero, provides

$$\frac{\partial l(\boldsymbol{\beta}, \sigma^2)}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) \stackrel{!}{=} 0.$$

Substituting the ML or least squares estimator $\hat{\boldsymbol{\beta}}$ for $\boldsymbol{\beta}$ results in

$$-\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}(\mathbf{y} - \hat{\mathbf{y}})'(\mathbf{y} - \hat{\mathbf{y}}) = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\hat{\epsilon}'\hat{\epsilon} \stackrel{!}{=} 0,$$

which yields

$$\hat{\sigma}_{ML}^2 = \frac{\hat{\epsilon}'\hat{\epsilon}}{n}.$$

3.6 Parameter Estimation in the Classical Linear Model

Estimation of β

In the classical linear model, the estimator

$$\hat{\beta} = (X'X)^{-1}X'y$$

minimizes the least squares criterion

$$LS(\beta) = \sum_{i=1}^n (y_i - x_i'\beta)^2.$$

Under the assumption of normally distributed errors, the least squares estimator is also the ML estimator for β .

Estimation of σ^2

The estimator

$$\hat{\sigma}^2 = \frac{1}{n-p} \hat{\epsilon}'\hat{\epsilon}$$

is unbiased and can be characterized as the REML estimator for σ^2 .

However, this estimator for σ^2 is rarely used. The mean of the sum of squared residuals is

$$E(\hat{\epsilon}'\hat{\epsilon}) = (n-p) \cdot \sigma^2 \quad (3.17)$$

and thus

$$E(\hat{\sigma}_{ML}^2) = \frac{n-p}{n} \sigma^2.$$

This implies that the ML estimator for σ^2 is biased. A proof for Eq. (3.17) is provided in Sect. 3.5.2 on p. 168.

Restricted Maximum Likelihood Estimation

Using Eq. (3.17), we easily get an unbiased estimator $\hat{\sigma}^2$ for σ^2 . We obtain

$$\hat{\sigma}^2 = \frac{1}{n-p} \hat{\epsilon}'\hat{\epsilon}, \quad (3.18)$$

which is a commonly used estimator for σ^2 . Estimator (3.18) also possesses the interesting feature of being the *restricted maximum likelihood estimator* (REML). It can be shown that Eq. (3.18) maximizes the *marginal likelihood*

$$L(\sigma^2) = \int L(\boldsymbol{\beta}, \sigma^2) d\boldsymbol{\beta},$$

which is obtained by integration over the vector of regression coefficients $\boldsymbol{\beta}$. In general, the REML estimator for variance parameters is less biased than the ML estimator and thus is generally preferred. In the present case the REML estimator is even unbiased, in contrast to the ML estimator. A further application of the REML principle can be found when estimating linear mixed models; see Chap. 7.

3.2.3 Properties of the Estimators

We will now discuss the properties of the estimators in the classical linear model. First, we will discuss the geometric properties of the least squares estimator and then move to its statistical properties for finite and infinite samples. We will also discuss properties of the residuals.

Geometric Properties of Least Squares

From a geometric perspective, the (conditional) mean $\boldsymbol{\mu} = E(\mathbf{y}) = \mathbf{X}\boldsymbol{\beta}$ of the linear model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ is an n -dimensional vector in $\mathbb{R}^n$. Since $\boldsymbol{\mu}$ is a linear combination of the columns of the design matrix $\mathbf{X}$, $\boldsymbol{\mu}$ must be an element of the column space of $\mathbf{X}$. See Definition A.15 in Appendix A for the definition of the column space of a matrix. The column space of $\mathbf{X}$ is spanned by the columns $\mathbf{1}, \mathbf{x}^1, \dots, \mathbf{x}^k$, which are assumed to be linearly independent ($\text{rk}(\mathbf{X}) = k + 1 = p$). Thus the columns of $\mathbf{X}$ define a p -dimensional vector space.

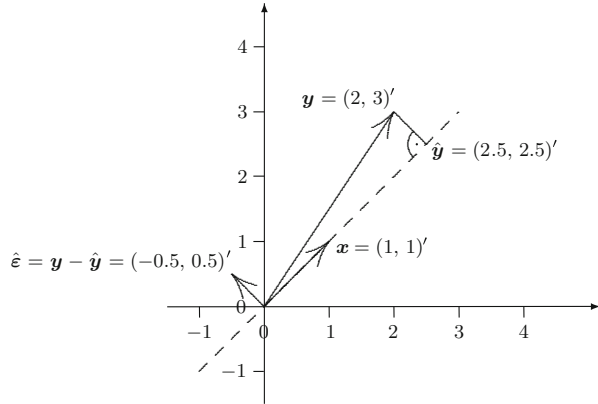
Geometrically, the response vector $\mathbf{y}$ is also a vector in $\mathbb{R}^n$. However, $\mathbf{y}$ is usually (similar to $\boldsymbol{\varepsilon}$) *not* an element of the column space of $\mathbf{X}$. We illustrate the scenario with the help of the very simple model

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \beta_0 + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \end{pmatrix},$$

having only two observations and one parameter β_0 . Since the design matrix only consists of the column vector $\mathbf{x}^0 = (1, 1)'$, the column space of $\mathbf{X}$ consists of all points in $\mathbb{R}^2$ that are on the line defined by the origin and the point $(1, 1)$; see Fig. 3.14.

For example, if we observe the vector $\mathbf{y} = (2, 3)'$, we obtain the value $\hat{\beta}_0 = 2.5$ as the least squares estimator for β_0 . The resulting predicted values are $\hat{\mathbf{y}} = (2.5, 2.5)'$ (see Fig. 3.14). Since the method of least squares minimizes $(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \boldsymbol{\varepsilon}'\boldsymbol{\varepsilon}$, the *Euclidean distance* between $\mathbf{y}$ and $\mathbf{X}\boldsymbol{\beta}$ is minimized. Figure 3.14 shows this distance is minimized if $\hat{\beta}_0$ is chosen such that the line connecting $\mathbf{y}$ with $\hat{\mathbf{y}}$ is orthogonal to $\hat{\mathbf{y}}$. The connecting line is in fact the residual vector $\hat{\boldsymbol{\varepsilon}}$. This in turn implies that the residuals $\hat{\boldsymbol{\varepsilon}}$ and the predicted values $\hat{\mathbf{y}}$ are orthogonal, i.e., $\hat{\boldsymbol{\varepsilon}}'\hat{\mathbf{y}} = 0$. Notice that $\mathbf{x}^0$ and $\hat{\boldsymbol{\varepsilon}}$ are also orthogonal to each other.

Fig. 3.14 Visualization of the geometric properties of the least squares estimator



We can generalize these observations for arbitrary linear models: The method of least squares yields parameter estimates $\hat{\beta}$ such that the residuals $\hat{\epsilon}$ and the predicted values $\hat{y}$ are orthogonal to each other. This can be easily proved using properties of the hat matrix H . Using $\hat{y} = Hy$ and $\hat{\epsilon} = (I - H)y$ (see p. 108), we obtain

$$\hat{y}'\hat{\epsilon} = y'H(I - H)y = y'H y - y'HH y = y'H y - y'H y = 0.$$

Moreover, all columns of the design matrix are orthogonal to the residuals, i.e., $(x^j)'\hat{\epsilon} = 0$, $j = 0, \dots, k$, or $X'\hat{\epsilon} = \mathbf{0}$. For a proof we take again advantage of properties of the hat matrix:

$$X'\hat{\epsilon} = X'(I - H)y = X'y - X'H y = X'y - X'X(X'X)^{-1}X'y = \mathbf{0}.$$

The fact that the columns of the design matrix and the residuals are orthogonal provides some more interesting implications; see properties 3–5 in Box 3.7. A proof can be found in Sect. 3.5.2, on p. 169.

Example 3.9 Orthogonal Design

The fact that the residuals and the columns of the design matrix are orthogonal can be used for constructing an orthogonal design matrix. An orthogonal design matrix implies uncorrelated covariates. Orthogonalizing the design matrix is useful, e.g., when constructing orthogonal polynomials to model nonlinear relationships (see Example 3.5, p. 90).

Let X be the design matrix with columns x^j . The goal is to transform the columns x^j such that the resulting columns $\tilde{x}^j$ of the transformed design matrix $\tilde{X}$ are orthogonal. To achieve this we use for $j = 1, \dots, k$ the transformations

$$\tilde{x}^j = x^j - \tilde{X}_j(\tilde{X}_j'\tilde{X}_j)^{-1}\tilde{X}_j'x^j,$$

where matrix $\tilde{X}_j$ contains the first $j - 1$ transformed vectors. We do not transform the first column of the design matrix X , associated with the intercept. The transformed vector $\tilde{x}^j$ can be viewed as the residual vector of a regression with x^j as the response variable and the j transformed vectors $\tilde{x}^0 = \mathbf{1}, \tilde{x}^1, \dots, \tilde{x}^{j-1}$ as the covariates. Due to the orthogonality

3.7 Geometric Properties of the Least Squares Estimator

The method of least squares has the following geometric properties:

1. The predicted values $\hat{y}$ are orthogonal to the residuals $\hat{e}$, i.e., $\hat{y}'\hat{e} = 0$.
2. The columns $\mathbf{x}^j$ of $\mathbf{X}$ are orthogonal to the residuals $\hat{e}$, i.e., $(\mathbf{x}^j)'\hat{e} = 0$ or $\mathbf{X}'\hat{e} = \mathbf{0}$.
3. The average of the residuals is zero, i.e.,

$$\sum_{i=1}^n \hat{e}_i = 0 \quad \text{or} \quad \frac{1}{n} \sum_{i=1}^n \hat{e}_i = 0.$$

4. The average of the predicted values $\hat{y}_i$ is equal to the average of the observed response y_i , i.e.,

$$\frac{1}{n} \sum_{i=1}^n \hat{y}_i = \bar{y}.$$

5. The regression hyperplane runs through the average of the data, i.e.,

$$\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}_1 + \cdots + \hat{\beta}_k \bar{x}_k.$$

of the residuals, $\tilde{\mathbf{x}}^j$ is orthogonal to the columns of $\tilde{\mathbf{X}}^j$. This implies that $\tilde{\mathbf{x}}^j$ is orthogonal to all $j - 1$ previously constructed variables $\tilde{\mathbf{x}}^1, \dots, \tilde{\mathbf{x}}^{j-1}$. Notice that the first transformed variable $\tilde{\mathbf{x}}^1$ results from a simple centering around the column mean value of $\mathbf{x}^1$. In linear algebra, this method is also known as Gram–Schmidt orthogonalization. $\triangle$

Analysis of Variance and Coefficient of Determination

Using the geometric properties of the least squares estimator, we can derive a fundamental analysis of variance formula for the empirical variance of observed responses y_i . This allows us to define the *coefficient of determination* or the proportion of total variance that is explained by the regression model. The coefficient of determination is closely related to the empirical correlation coefficient and can be used as a goodness-of-fit measure (among many others).

In Sect. 3.5.2 (p. 169), we prove the following decomposition formula:

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^n \hat{e}_i^2. \quad (3.19)$$

Division by n (or $n - 1$) on both sides leads to the analysis of variance formula:

$$s_y^2 = s_{\hat{y}}^2 + s_{\hat{\varepsilon}}^2,$$

where s_y^2 , $s_{\hat{y}}^2$, and $s_{\hat{\varepsilon}}^2$ are the empirical variances of the observed response, the predicted values, and the residuals. Thus we obtain an additive decomposition of the empirical variance of the response into the empirical variance of the predicted values and the residuals.

Based on the analysis of variance formula, we can define the coefficient of determination R^2 as a goodness-of-fit measure. It is defined as

$$R^2 = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = 1 - \frac{\sum_{i=1}^n \hat{\varepsilon}_i^2}{\sum_{i=1}^n (y_i - \bar{y})^2}.$$

As a consequence of Eq. (3.19), $0 \leq R^2 \leq 1$.

Using the analysis of variance formula, we can interpret the coefficient of determination in the following way: The closer R^2 is to 1, the smaller the residual sum of squares $\sum \hat{\varepsilon}_i^2$ and thus the better the fit to the data. The extreme case of $R^2 = 1$ implies $\sum \hat{\varepsilon}_i^2 = 0$, i.e., all residuals are zero and the fit to the data is perfect. If R^2 is close to 0, the sum of squared residuals is relatively large and the model fit is poor. The limit $R^2 = 0$ implies $\sum (\hat{y}_i - \bar{y})^2 = 0$, and hence $\hat{y}_i = \bar{y}$ for all i . That is the prediction of y_i is always equal to the mean $\bar{y}$ of the response variable and is thus *independent* of the explanatory variables. Consequently, the covariates do not have any explanatory power with regard to the mean of y . However, it is important to note that the model can be misspecified, e.g., there could be a nonlinear relationship for one of the covariates. In such a case, covariates can hold uncovered explanatory power, even if the coefficient of determination is close to zero.

In the special case of a simple regression model

$$y = \beta_0 + \beta_1 x + \varepsilon,$$

it can be shown that

$$R^2 = r_{xy}^2,$$

where r_{xy}^2 is the squared empirical correlation coefficient between x and y . This property of the coefficient of determination is responsible for the notation R^2 . In the simple regression model, as well as in the multiple regression model, we can interpret the coefficient of determination as the squared empirical correlation coefficient between the observations y and the predicted values $\hat{y}$. Formally we have

$$R^2 = r_{y\hat{y}}^2.$$

Finally, a warning is in order. It is quite common to choose models with preferably high coefficients of determination. However, we must be careful when

Table 3.1 Munich rent index: comparison of different models for the relationship between net rent per square meter (*rentsqm*) and area (*area*)

Model	Equation	R^2
M1	$\widehat{rentsqm} = 9.47 - 0.035 \cdot area$	0.116
M2	$\widehat{rentsqm} = 4.73 + 140.18 \cdot 1/area$	0.154
M3	$\widehat{rentsqm} = 11.83 - 0.106 \cdot area + 0.00047 \cdot area^2$	0.143
M4	$\widehat{rentsqm} = 14.28 - 0.22 \cdot area + 0.0020 \cdot area^2 - 0.000006 \cdot area^3$	0.150

comparing different models using the coefficients of determination. Model comparison using R^2 is only meaningful if the following three conditions are fulfilled:

- Every model must use the same response variable y , e.g., it is not possible to compare models with an original response y to that of a transformed response, e.g., $\log(y)$.
- Every model must have the same number of parameters.
- Every model should include an intercept β_0 .

Thus, in general, we *cannot* use the coefficient of determination for model comparison. It can be shown that the coefficient of determination cannot decrease if an additional explanatory variable is added to the model. The following example will illustrate the difficulty of using R^2 for model comparison.

Example 3.10 Munich Rent Index—Comparison of Models Using R^2

In Sect. 3.1.3 (p. 87), we examined several models to investigate the relationship between net rent per square meter and living area. Table 3.1 contains the estimated models and their corresponding coefficients of determination. First, we find that every coefficient of determination is relatively small. One reason is the strong variability in the data; see the scatter plot in Fig. 3.8 on p. 88. Another reason is that many important covariates of the original rent index are missing in the models of this illustrative example. The final model that was chosen for the official rent index for 1999 contained more than 20 explanatory variables, which led to a coefficient of determination of 0.49.

If we compare model M1 and M2, we find that M2 has a higher coefficient of determination. Since both models have the same number of parameters, a comparison based on the coefficient of determination is appropriate, and Model M2 should be preferred. Note, however, that we came to the same conclusion with residual analysis; see Fig. 3.8 (p. 88). The models M1, M3, and M4 are nested, which means that M4 contains both M3 and M1, and M3 contains M1 as a special case. As a consequence M1 *must* have the smallest R^2 value, M3 has the second smallest R^2 value, and M4 has the highest R^2 value. A comparison on the basis of the coefficient of determination does not make sense here, as the number of parameters differs across models. Nevertheless, we point out that a model with a higher number of regression parameters does not necessarily imply a higher coefficient of determination. This is the case, e.g., if the models have different explanatory variables (non-nested models). A comparison of models M2 and M3 demonstrates such a comparison: Even though M3 has three regression parameters and M2 only has two, M2 has a higher coefficient of determination. In such a case, M2 is preferred over M3, as it contains less parameters with a higher coefficient of determination. The coefficient of determination only increases automatically with an increase of parameters for *nested models*, i.e., if more complex models contain the other models as special cases, as is the case with the models M1, M3, and M4.

△

3.8 The Coefficient of Determination

Definition

$$R^2 = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = 1 - \frac{\sum_{i=1}^n \hat{\varepsilon}_i^2}{\sum_{i=1}^n (y_i - \bar{y})^2}.$$

Interpretation

The closer the coefficient of determination is to 1, the smaller is the residual sum of squares, and the better the fit is to the data. If $R^2 = 1$, all residuals are zero with perfect fit to the data.

Properties

1. In the simple linear regression model $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, the coefficient of determination corresponds to the squared correlation coefficient (according to Bravais–Pearson), implying $R^2 = r_{xy}^2$.
2. In the multiple regression model, the coefficient of determination can be understood as a squared correlation coefficient between the observations y and the predicted values $\hat{y}$, implying $R^2 = r_{y\hat{y}}^2$.
3. Let the vector x of the explanatory variables be partitioned into two subvectors x_1 and x_2 . Consider the full model M1 $y_i = \beta_0 + \beta'_1 x_{i1} + \beta'_2 x_{i2} + \varepsilon_i$ and the submodel M2 $y_i = \beta_0 + \beta'_1 x_{i1} + \varepsilon_i$. Then

$$R_{M1}^2 \geq R_{M2}^2,$$

i.e., for nested models, the coefficient of determination of the submodel is always less than or equal to the coefficient of determination of the full model.

4. Comparing the coefficient of determination across models is meaningful for a common response variable and if the models contain the same number of parameters and an intercept.

In Sect. 3.4, we will investigate other model choice criteria that are appropriate to compare models containing a varying number of parameters.

Statistical Properties Without Specific Distributional Assumptions

We will now investigate the statistical properties of the least squares estimator. Thereby we will admit an arbitrary distribution for the error term. In particular, we do not assume that the errors necessarily follow a normal distribution. Note,

however, that the following properties are only true if the model is otherwise *correctly specified*; see Sect. 3.4 for the consequences of model misspecification.

Expectation and Bias

For the expectation of the least squares estimator, we have

$$E(\hat{\beta}) = E((X'X)^{-1}X'y) = (X'X)^{-1}X'E(y) = (X'X)^{-1}X'X\beta = \beta.$$

Thus the least squares estimator is unbiased for β .

Covariance Matrix

Using Theorem B.2.5 (p. 647), the covariance matrix of the least squares estimator is given by

$$\begin{aligned} \text{Cov}(\hat{\beta}) &= \text{Cov}((X'X)^{-1}X'y) = (X'X)^{-1}X'\text{Cov}(y)((X'X)^{-1}X')' \\ &= \sigma^2 (X'X)^{-1}X'X(X'X)^{-1} = \sigma^2 (X'X)^{-1}. \end{aligned}$$

For the variances of the estimated regression coefficients $\hat{\beta}_j$, there exists a more easily interpretable formula (see, e.g., Wooldridge, 2006):

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{(1 - R_j^2) \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}.$$

We define R_j^2 here as the coefficient of determination for the regression between x_j , as the response variable, and all of the other explanatory variables (except x_j). We immediately see which components determine the precision of the estimates for the regression coefficients:

- The smaller the model variance σ^2 , the smaller the variance of $\hat{\beta}_j$ and thus the more accurate the estimation.
- The smaller the linear dependence between x_j and the other explanatory variables (measured through R_j^2), the smaller is the variance of $\hat{\beta}_j$. The variance, $\text{Var}(\hat{\beta}_j)$, is minimized for $R_j^2 = 0$, i.e., when the covariates are uncorrelated. *Orthogonal designs* guarantee uncorrelated regressors and are typically used in experimental design situations to maximize the precision of estimators. On the other hand, when some of the covariates are highly correlated, the estimators can be very imprecise. In the extreme case of $R_j^2 \rightarrow 1$ the variances explode towards infinity. We will explain this *collinearity* problem in detail in Sect. 3.4.
- The larger the variability of covariate x_j around its average, the smaller is the variance of $\hat{\beta}_j$.

The true covariance of $\hat{\beta}$ cannot be calculated in applications, as the error variance σ^2 is unknown. Rather, we have to estimate $\text{Cov}(\hat{\beta})$, by replacing σ^2 with

its estimate $\hat{\sigma}^2 = \hat{\mathbf{e}}'\hat{\mathbf{e}}/(n - p)$:

$$\widehat{\text{Cov}}(\hat{\boldsymbol{\beta}}) = \hat{\sigma}^2 (\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{n - p} \hat{\mathbf{e}}'\hat{\mathbf{e}}(\mathbf{X}'\mathbf{X})^{-1}.$$

The diagonal elements of this matrix represent the estimated variances of the least squares estimator $\hat{\beta}_j$. The square root of the diagonal elements are the estimated standard errors, which we abbreviate as se_j from this point forward:

$$\text{se}_j = \widehat{\text{Var}}(\hat{\beta}_j)^{1/2}, \quad j = 0, 1, \dots, k.$$

The estimated covariance matrix, especially the standard errors se_j , are used for statistical testing and confidence intervals for the regression coefficients; see Sect. 3.3.

Comparison with Linear Estimators

In the following, we want to compare the least squares estimator with the more general class of *linear estimators*. A linear estimator $\hat{\boldsymbol{\beta}}^L$ takes the form of

$$\hat{\boldsymbol{\beta}}^L = \mathbf{a} + \mathbf{A} \mathbf{y},$$

where $\mathbf{a} = (a_0, \dots, a_k)'$ is a $p \times 1$ -vector, and $\mathbf{A} = (a_{ij})$ is a matrix of dimension $p \times n$. Thus, the components β_j of $\boldsymbol{\beta}$ are estimated through a linear combination of the response observations y_i :

$$\hat{\beta}_j^L = a_j + a_{j1}y_1 + \dots + a_{jn}y_n \quad j = 0, \dots, k.$$

Obviously, the least squares estimator $\hat{\boldsymbol{\beta}}$ is a special case of a linear estimator with $\mathbf{a} = \mathbf{0}$ and $\mathbf{A} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$. We can derive the expectation and the covariance matrix of linear estimators in a similar way as we did with the least squares estimator. We have

$$\mathbf{E}(\hat{\boldsymbol{\beta}}^L) = \mathbf{a} + \mathbf{A} \mathbf{X} \boldsymbol{\beta} \quad \text{Cov}(\hat{\boldsymbol{\beta}}^L) = \sigma^2 \mathbf{A} \mathbf{A}'.$$

Linear estimators are, thus, not necessarily unbiased. A comparison of the least squares estimator with the subclass of *linear unbiased estimators* $\hat{\boldsymbol{\beta}}^L$ shows that the least squares estimator has minimal variances, i.e.,

$$\text{Var}(\hat{\beta}_j^L) \geq \text{Var}(\hat{\beta}_j), \quad j = 0, \dots, k.$$

This property also holds for any linear combination

$$b_0\beta_0 + b_1\beta_1 + \dots + b_k\beta_k = \mathbf{b}'\boldsymbol{\beta}$$

of β , i.e.,

$$\text{Var}(\mathbf{b}'\hat{\beta}^L) \geq \text{Var}(\mathbf{b}'\hat{\beta}).$$

These properties of the least squares estimator are known as Gauß–Markov Theorem. A proof can be found in this chapter’s appendix, on p. 170. The Gauß–Markov Theorem can also be used to obtain an “optimal” prediction for a new (future) observation y_0 with a given covariate vector $\mathbf{x}_0$. We use the conditional mean for prediction:

$$E(y_0 | \mathbf{x}_0) = \mathbf{x}_0' \beta.$$

An optimal estimator of the expectation (in terms of the Gauß–Markov Theorem) is then given by

$$\hat{y}_0 = \mathbf{x}_0' \hat{\beta}.$$

Statistical Properties Under Normality Assumption

Thus far, the derived statistical properties of the least squares estimator have been obtained without assuming any specific distribution for the error term ϵ . Assuming a normal distribution for the errors, i.e., $\epsilon \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$, additional properties of the least squares estimator can be derived. They will be especially useful when constructing hypothesis tests and confidence intervals for β .

Normality of the errors results in $\mathbf{y} \sim N(\mathbf{X}\beta, \sigma^2 \mathbf{I})$. It follows that the least squares estimator $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{y}$ will also follow a normal distribution, as $\hat{\beta}$ is a linear transformation of $\mathbf{y}$ (refer to Theorem B.5 on p. 649). Using the previously derived results for the expectation and the covariance matrix of the least squares estimator, we immediately have

$$\hat{\beta} \sim N(\beta, \sigma^2 (\mathbf{X}'\mathbf{X})^{-1}).$$

Moreover, Theorem B.8.1 (p. 651) shows that the distribution for the distance (weighted by the inverse covariance matrix) between the least squares estimator $\hat{\beta}$ and the true β follows a χ^2 -distribution with p degrees of freedom:

$$\frac{(\hat{\beta} - \beta)' (\mathbf{X}'\mathbf{X}) (\hat{\beta} - \beta)}{\sigma^2} \sim \chi_p^2.$$

This property is useful when deriving the distribution of a test statistic in hypothesis testing.

Asymptotic Properties of the Least Squares Estimator

The distributional properties of the least squares estimator are the basis for tests and confidence intervals that we will discuss in the next chapter. In order to have exact tests and intervals, the assumption of normally distributed errors is needed. However, some of the properties are still approximately valid, as long as the sample size n tends to infinity or is sufficiently large. For clarification, we index the model with the number of observations n :

3.9 Statistical Properties of the Least Squares Estimator

Without Specific Distributional Assumptions

1. *Expectation:* $E(\hat{\beta}) = \beta$, implying that the least squares estimator is unbiased.
2. *Covariance Matrix:* $\text{Cov}(\hat{\beta}) = \sigma^2 (X'X)^{-1}$. The diagonal elements can be expressed as

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{(1 - R_j^2) \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2},$$

where R_j^2 is the coefficient of determination of the regression between x_j as the response variable and the remaining columns of X as regressors. An estimator for the covariance matrix is given by

$$\widehat{\text{Cov}}(\hat{\beta}) = \hat{\sigma}^2 (X'X)^{-1} = \frac{1}{n-p} \hat{\mathbf{e}}' \hat{\mathbf{e}} (X'X)^{-1}.$$

3. *Gauß–Markov Theorem:* Among all linear and unbiased estimators $\hat{\beta}^L$, the least squares estimator has minimal variances, implying

$$\text{Var}(\hat{\beta}_j) \leq \text{Var}(\hat{\beta}_j^L), \quad j = 0, \dots, k.$$

For any linear combination $b'\beta$ it holds

$$\text{Var}(b'\hat{\beta}) \leq \text{Var}(b'\hat{\beta}^L).$$

With Normality Assumption

1. *Distribution of the response variable:*

$$\mathbf{y} \sim N(X\beta, \sigma^2 I).$$

2. *Distribution of the least squares estimator:*

$$\hat{\beta} \sim N(\beta, \sigma^2 (X'X)^{-1}).$$

3. *Distribution of weighted distance:*

$$\frac{(\hat{\beta} - \beta)'(X'X)(\hat{\beta} - \beta)}{\sigma^2} \sim \chi_p^2.$$

$$y_n = X_n \beta + \varepsilon_n, \quad E(\varepsilon_n) = \mathbf{0}, \quad \text{Cov}(\varepsilon_n) = \sigma^2 I_n.$$

Similarly, we index the least squares estimator $\hat{\beta}_n$ and the variance estimator $\hat{\sigma}_n^2$ with n . To obtain valid asymptotic results, we need to go beyond the assumptions 1–3 stated in Box 3.1 (p. 76). Further assumptions are needed regarding the limiting behavior of the design matrix X_n and with it the sequence $\mathbf{x}_1, \dots, \mathbf{x}_n, \dots$ of the design vectors. A standard assumption is that the matrix $X_n' X_n$ averaged over n converges to a limiting positive definite matrix V , i.e.,

$$\lim_{n \rightarrow \infty} \frac{1}{n} X_n' X_n = V, \quad V \text{ positive definite.} \quad (3.20)$$

In this case we have the following asymptotic results:

3.10 Asymptotic Properties of the Least Squares Estimator

1. The least squares estimator $\hat{\beta}_n$ for β and the ML or REML estimator $\hat{\sigma}_n^2$ for the variance σ^2 are consistent.
2. The least squares estimator asymptotically follows a normal distribution, specifically

$$\sqrt{n}(\hat{\beta}_n - \beta) \xrightarrow{d} N(\mathbf{0}, \sigma^2 V^{-1}).$$

That is the difference $\hat{\beta}_n - \beta$ normalized with $\sqrt{n}$ converges in distribution to the normal distribution on the right-hand side.

We use these asymptotic results for a sufficiently large sample size n as follows. First, $\hat{\beta}_n$ has an approximately normal distribution

$$\hat{\beta}_n \overset{a}{\sim} N(\beta, \sigma^2 V^{-1}/n).$$

If we replace σ^2 with the consistent estimator $\hat{\sigma}_n^2$ and V with the approximation $V \overset{a}{\sim} 1/n X_n' X_n$, we have

$$\hat{\beta}_n \overset{a}{\sim} N(\beta, \hat{\sigma}_n^2 (X_n' X_n)^{-1}).$$

This implies that, with sufficiently large sample size and provided that Eq. (3.20) holds, the least squares estimator has the same approximate normal distribution, regardless of the normal assumption for ε . Assumption (3.20) is particularly ensured if the observed covariate vectors $\mathbf{x}_i$, $i = 1, \dots, n$, are independent and identically distributed realizations of stochastic covariates $\mathbf{x} = (1, x_1, \dots, x_k)'$, i.e., if the observations $(y_i, \mathbf{x}_i)$ form a random sample from $(y, \mathbf{x})$. This condition is met for many empirical studies, e.g., in our applications on the Munich rent index and on malnutrition in developing countries. In such cases, the law of large numbers implies

$$\frac{1}{n} \mathbf{X}'_n \mathbf{X}_n = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}'_i \rightarrow E(\mathbf{x} \mathbf{x}') =: \mathbf{V}.$$

However, the condition is typically violated for regressors following a trend. A simple example is a linear trend $x_i = i$, i.e.,

$$y_i = \beta \cdot i + \varepsilon_i, \quad i = 1, \dots, n.$$

For simplicity the intercept β_0 is assumed to be zero. We have

$$\frac{1}{n} \mathbf{X}'_n \mathbf{X}_n = \frac{1}{n} \sum_{i=1}^n x_i^2 = \frac{1}{n} (1 + \dots + i^2 + \dots + n^2) \rightarrow \infty,$$

resulting in a violation of assumption (3.20).

In fact, consistency can be derived under the following more general condition:

$$(\mathbf{X}'_n \mathbf{X}_n)^{-1} \rightarrow \mathbf{0}. \quad (3.21)$$

Informally, this implies that the covariate information grows as the sample size increases. It can be shown that Eq. (3.21) is a necessary and sufficient condition for consistency of the least squares estimator and its variance estimator; see [Lai, Robins, and Wei \(1979\)](#) and [Drygas \(1976\)](#).

For asymptotic normality, we additionally need the condition

$$\max_{i=1, \dots, n} \mathbf{x}'_i (\mathbf{X}'_n \mathbf{X}_n)^{-1} \mathbf{x}_i \rightarrow 0 \quad \text{for } n \rightarrow \infty. \quad (3.22)$$

In short, this implies that the influence of each observation $\mathbf{x}_i$ is negligible relative to the entire information $\mathbf{X}'_n \mathbf{X}_n = \sum_{i=1}^n \mathbf{x}_i \mathbf{x}'_i$. The central limit theorem (specifically in the Lindeberg–Feller form) then applies. Provided that the conditions (3.21) and (3.22) are satisfied, the normal approximation still holds:

$$\hat{\beta}_n \stackrel{a}{\sim} N(\beta, \hat{\sigma}_n^2 (\mathbf{X}'_n \mathbf{X}_n)^{-1}),$$

which is important for practical use.

Example 3.11 Simple Linear Regression

For the simple linear regression model (excluding the intercept β_0)

$$y_i = \beta x_i + \varepsilon_i,$$

the following can be easily verified:

1. For a linear trend $x_i = i$, conditions (3.21) and (3.22) are satisfied.
2. For $x_i = 1/i$, both Eqs. (3.21) and (3.22) are violated, which implies that the least squares estimator is neither consistent nor asymptotically normal. The reason is that the



sequence of regressor values $x_i = 1/i$ converges too fast towards zero and, thus, does not provide increasing information with increasing sample size.

3. For $x_i = 1/\sqrt{i}$, conditions (3.21) and (3.22) are met. Despite that $x_i \rightarrow 0$, enough covariate information is provided as $n \rightarrow \infty$.

△



Statistical Properties of Residuals

We close this section with the examination of the statistical properties of the residuals $\hat{\varepsilon}_i = y_i - \mathbf{x}'_i \hat{\boldsymbol{\beta}}$. In terms of the hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$, the residuals can be expressed as

$$\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{H})\mathbf{y} = \mathbf{y} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y};$$

see p. 107.

Thus, we obtain the expectation of the residuals as

$$\mathbf{E}(\hat{\boldsymbol{\varepsilon}}) = \mathbf{E}(\mathbf{y}) - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{E}(\mathbf{y}) = \mathbf{X}\boldsymbol{\beta} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\boldsymbol{\beta} = \mathbf{0},$$

as well as their covariance matrix

$$\text{Cov}(\hat{\boldsymbol{\varepsilon}}) = \text{Cov}((\mathbf{I} - \mathbf{H})\mathbf{y}) = (\mathbf{I} - \mathbf{H})\sigma^2\mathbf{I}(\mathbf{I} - \mathbf{H})' = \sigma^2(\mathbf{I} - \mathbf{H}).$$

In deriving the covariance matrix, we used Theorem B.2.5 (p. 647) in Appendix B in combination with the fact that the matrix $\mathbf{I} - \mathbf{H}$ is symmetric and idempotent. Specifically for the variances of the residuals, we have

$$\text{Var}(\hat{\varepsilon}_i) = \sigma^2(1 - h_{ii}),$$

where h_{ii} is the i th diagonal element of the hat matrix. We state the following:

- Similar to the error terms, the residuals have mean zero.
- In contrast to the error terms, the residuals are *not* uncorrelated.
- In contrast to the error terms, the residuals have *heteroscedastic variances*. Since $\frac{1}{n} \leq h_{ii} \leq 1$ (see p. 108), the variance of the i th residual approaches zero as h_{ii} approaches one.

If we additionally assume normally distributed errors, we are able to derive the distribution of the residuals. We then have

$$\hat{\boldsymbol{\varepsilon}} \sim \mathbf{N}(\mathbf{0}, \sigma^2(\mathbf{I} - \mathbf{H})). \quad (3.23)$$

Due to $\text{rk}(\mathbf{H}) = p \leq n$, this is a singular normal distribution; see also Sect. B.3.2 (p. 650) in Appendix B.

Using Eq. (3.23), we can also make statements about the residual sum of squares. In Sect. 3.5.2 (p. 171), we show

$$\frac{\hat{\boldsymbol{\varepsilon}}'\hat{\boldsymbol{\varepsilon}}}{\sigma^2} \sim \chi^2_{n-p},$$

3.11 Statistical Properties of Residuals

Without Specific Distributional Assumptions

1. *Expectation:* $E(\hat{\boldsymbol{\varepsilon}}) = \mathbf{0}$, i.e., the residuals have mean zero.
2. *Variance:*

$$\text{Var}(\hat{\varepsilon}_i) = \sigma^2(1 - h_{ii}),$$

which means that the residuals (in contrast to the errors ε_i) have heteroscedastic variances.

3. *Covariance Matrix:*

$$\text{Cov}(\hat{\boldsymbol{\varepsilon}}) = \sigma^2(\mathbf{I} - \mathbf{H}) = \sigma^2(\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'),$$

which implies that the residuals (in contrast to the errors) are not uncorrelated.

With Normal Assumption

4. *Distribution of Residuals:*

$$\hat{\boldsymbol{\varepsilon}} \sim N(\mathbf{0}, \sigma^2(\mathbf{I} - \mathbf{H})).$$

5. *Distribution of the Residual Sum of Squares:*

$$\frac{\hat{\boldsymbol{\varepsilon}}'\hat{\boldsymbol{\varepsilon}}}{\sigma^2} = (n - p) \frac{\hat{\sigma}^2}{\sigma^2} \sim \chi_{n-p}^2.$$

6. *Independence:* The residual sum of squares $\hat{\boldsymbol{\varepsilon}}'\hat{\boldsymbol{\varepsilon}}$ and the least squares estimator $\hat{\boldsymbol{\beta}}$ are independent.

which is equivalent to

$$(n - p) \frac{\hat{\sigma}^2}{\sigma^2} \sim \chi_{n-p}^2.$$

Moreover, it can be shown that the residual sum of squares and the least squares estimator are independent (again see Sect. 3.5.2 p. 171 for a proof). Both statements are necessary for the derivation of hypothesis tests regarding the regression coefficients.

Standardized and Studentized Residuals

In practice, the residuals are often used to confirm model assumptions in a linear model. However, the residuals are not always adequate for this purpose. As we have seen, the residuals are neither homoscedastic nor uncorrelated. In most cases,

the correlation is negligible, but this is not the case with the heteroscedasticity of the residuals. Thus, the verification of the assumption of homoscedastic errors is problematic, as heteroscedastic residuals are in general not an indicator for heteroscedastic errors. An obvious solution of the heteroscedasticity problem is standardization. Dividing through the estimated standard deviation of the residuals, we obtain the *standardized residuals*

$$r_i = \frac{\hat{\varepsilon}_i}{\hat{\sigma} \sqrt{1 - h_{ii}}}. \quad (3.24)$$

Provided that the model assumptions are correct, the standardized residuals are homoscedastic. Thus, the analysis of standardized residuals helps to assess whether or not the assumption of homoscedastic variances is violated. We often plot the standardized residuals against the predicted values or the values of covariates; see Sect. 3.4.4 on model diagnosis for examples.

Since the residuals are normally distributed and $(n - p)\hat{\sigma}^2/\sigma^2$ follows a χ^2_{n-p} -distribution, it is tempting to assume the standardized residuals follow a t-distribution; see Definition B.13 in Appendix B. This conclusion is not correct, as $\hat{\varepsilon}_i$ is part of $\hat{\sigma}$, thus numerator and denominator in Eq. (3.24) are not stochastically independent. However, it is possible to bypass the problem of dependence. To do so, we define “leave-one-out” estimators, which are based on all observations excluding the i th observation. We can then define residuals, which are based on these “leave-one-out” estimators, and show that they follow a valid t-distribution.

Define $\mathbf{X}_{(i)}$ and $\mathbf{y}_{(i)}$ as the design matrix and response vector, respectively, from which the i th row is removed. The corresponding least squares estimator $\hat{\boldsymbol{\beta}}_{(i)}$, which is based on all observations except the i th one, is then given by

$$\hat{\boldsymbol{\beta}}_{(i)} = (\mathbf{X}'_{(i)}\mathbf{X}_{(i)})^{-1}\mathbf{X}'_{(i)}\mathbf{y}_{(i)}.$$

Using $\hat{\boldsymbol{\beta}}_{(i)}$, we obtain predicted values $\hat{y}_{(i)} = \mathbf{x}'_i\hat{\boldsymbol{\beta}}_{(i)}$ and thus residuals

$$\hat{\varepsilon}_{(i)} = y_i - \hat{y}_{(i)} = y_i - \mathbf{x}'_i(\mathbf{X}'_{(i)}\mathbf{X}_{(i)})^{-1}\mathbf{X}'_{(i)}\mathbf{y}_{(i)}$$

for the i th observation. Simple calculations show

$$\hat{\varepsilon}_{(i)} \sim N(0, \sigma^2(1 + \mathbf{x}'_i(\mathbf{X}'_{(i)}\mathbf{X}_{(i)})^{-1}\mathbf{x}_i))$$

or

$$\frac{\hat{\varepsilon}_{(i)}}{\sigma(1 + \mathbf{x}'_i(\mathbf{X}'_{(i)}\mathbf{X}_{(i)})^{-1}\mathbf{x}_i)^{1/2}} \sim N(0, 1).$$

According to property 5 of the preceding Box 3.11, it follows

$$(n - p - 1) \frac{\hat{\sigma}_{(i)}^2}{\sigma^2} \sim \chi^2_{n-p-1},$$

where

$$\hat{\sigma}_{(i)}^2 = \frac{1}{n-p-1} \left[(y_1 - \mathbf{x}'_1 \hat{\boldsymbol{\beta}}_{(i)})^2 + \dots + (y_{i-1} - \mathbf{x}'_{i-1} \hat{\boldsymbol{\beta}}_{(i)})^2 + (y_{i+1} - \mathbf{x}'_{i+1} \hat{\boldsymbol{\beta}}_{(i)})^2 + \dots + (y_n - \mathbf{x}'_n \hat{\boldsymbol{\beta}}_{(i)})^2 \right]$$

is an estimator for σ^2 , which is not based on the i th observation. Now we can use Definition B.13 of the t-distribution (p. 644) and obtain the *studentized residuals*

$$r_i^* = \frac{\hat{\varepsilon}_{(i)}}{\hat{\sigma}_{(i)}(1 + \mathbf{x}'_i(\mathbf{X}'_{(i)}\mathbf{X}_{(i)})^{-1}\mathbf{x}_i)^{1/2}} \sim t_{n-p-1}.$$

A crucial requirement for this distributional statement is that $\hat{\varepsilon}_{(i)}$ and $\hat{\sigma}_{(i)}$ are independent, which holds because we did not use the i th observation y_i when calculating $\hat{\sigma}_{(i)}$. Another requirement is that the model is correctly specified. Knowledge of the distribution of the studentized residuals can be used for model diagnosis to identify observations which are not in agreement with the estimated model; see Sect. 3.4.4 for details.

The standardized and studentized residuals help to circumvent the problem of heteroscedastic residuals. However there remains the problem of correlation among the residuals. The literature does offer suggestions on how to define uncorrelated residuals. As mentioned, in practice the correlation of standardized and studentized residuals in correctly specified models can be neglected. Thus we do not pursue this issue further.

Box 3.12 summarizes the various definitions of residuals. The box also shows (without proof) alternative definitions of studentized residuals. In particular, we find that the studentized residuals can be computed from the standardized residuals. The beauty of this result is that it is unnecessary to recompute the least squares estimator every time we delete one observation: all that we need is the full model least squares estimator and the diagonal elements of the hat matrix. Finally, we recap the definition of partial residuals as used in Example 3.5 (p. 90).

3.3 Hypothesis Testing and Confidence Intervals

This section describes statistical tests for hypotheses regarding the unknown regression parameters $\boldsymbol{\beta}$. Due to the duality between statistical tests and confidence intervals, we are also able to construct confidence intervals for the regression parameters $\boldsymbol{\beta}$. A requirement for the construction of (exact) tests and confidence intervals is the assumption of normally distributed errors. Therefore, we first assume independently and identically normally distributed errors, i.e., $\varepsilon_i \sim N(0, \sigma^2)$. However, tests and confidence intervals are relatively robust to mild departures from the normality assumption. Moreover, in Sect. 3.3.1 we will show that the tests and confidence intervals, derived under the assumption of normality, remain valid for large sample size even with non-normal errors.

3.12 Overview of Residuals

Ordinary Residuals

The residuals are given by

$$\hat{\varepsilon}_i = y_i - \hat{y}_i = y_i - \mathbf{x}_i' \hat{\boldsymbol{\beta}} \quad i = 1, \dots, n.$$

Standardized Residuals

The standardized residuals are defined by

$$r_i = \frac{\hat{\varepsilon}_i}{\hat{\sigma} \sqrt{1 - h_{ii}}},$$

where h_{ii} is the i th diagonal element of the hat matrix.

Studentized Residuals

The studentized residuals are defined by

$$r_i^* = \frac{\hat{\varepsilon}_{(i)}}{\hat{\sigma}_{(i)}(1 + \mathbf{x}_i'(\mathbf{X}_{(i)}' \mathbf{X}_{(i)})^{-1} \mathbf{x}_i)^{1/2}} = \frac{\hat{\varepsilon}_i}{\hat{\sigma}_{(i)} \sqrt{1 - h_{ii}}} = r_i \left(\frac{n - p - 1}{n - p - r_i^2} \right)^{1/2}.$$

The studentized residuals are used to verify model assumptions and to discover outliers (see Sect. 3.4.4).

Partial Residuals

The partial residuals regarding covariate x_j are defined by

$$\hat{\varepsilon}_{x_j, i} = y_i - \hat{\beta}_0 - \dots - \hat{\beta}_{j-1} x_{i, j-1} - \hat{\beta}_{j+1} x_{i, j+1} - \dots - \hat{\beta}_k x_{ik} = \hat{\varepsilon}_i + \hat{\beta}_j x_{ij}.$$

In the partial residuals $\hat{\varepsilon}_{x_j, i}$, all covariate effects with the exception of the one associated with x_j are removed. Hence, they are very useful for exploring whether the influence of x_j is modeled correctly (see Sect. 3.4.4).

Example 3.12 Munich Rent Index—Hypothesis Testing

We revisit the data from the Munich rent index to illustrate hypothesis testing. We use the data for the 1999 rent index, in combination with the follow-up data from 2001; see Example 3.7 (p. 100). Consider the regression model

$$\begin{aligned} \text{rentsqm}_i = & \beta_0 + \beta_1 \text{areainvc}_i + \beta_2 \text{yearco}_i + \beta_3 \text{yearco2}_i \\ & + \beta_4 \text{yearco3}_i + \beta_5 \text{nkitchen}_i + \beta_6 \text{pkitchen}_i + \beta_7 \text{year01}_i + \varepsilon_i, \end{aligned} \quad (3.25)$$

where $areainvc$ is the transformation $1/area$ centered around zero and $year0$, $year02$, $year03$ constitute third-order orthogonal polynomials (see Example 3.5 on p. 90) for the year of construction. The dummy variable $year01$ is an indicator for the follow-up period, i.e., $year01 = 1$ if an observation is from the follow-up and $year01 = 0$ otherwise. The following estimated model results:

$$\widehat{rentsqm_i} = 6.94 + 124.34 areainvc_i + 0.73 yearco_i + 0.44 yearco2_i \\ - 0.01 yearco3_i + 1.04 nkitchen + 1.31 pkitchen - 0.19 year01.$$

We find that the average net rent per square meter in the year 2001 decreases about 0.19 Euro, in comparison to the net rent in 1999. First of all, this change is relatively small. Moreover a decrease from 1999 to 2001 is surprising. It raises the question if such a net rent decrease can be extrapolated to the entire population or if it is merely a random phenomenon of the observed random sample. Thus we wish to test if the regression parameter β_7 is significantly different from zero. This can be achieved by testing the hypothesis

$$H_0 : \beta_7 = 0 \quad \text{against} \quad H_1 : \beta_7 \neq 0,$$

using an appropriate statistical test. Of course, we also want to test whether or not other variables should be included in the regression model. The test for significance of the variable $kitchen$ with the three categories, “substandard kitchen” (reference category), “standard kitchen” (dummy variable $nkitchen$), and “premium kitchen” (dummy variable $pkitchen$), is more complicated since it involves a categorical variable at three levels. In this case, we have to test the hypothesis

$$H_0 : \begin{pmatrix} \beta_5 \\ \beta_6 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \text{against} \quad H_1 : \begin{pmatrix} \beta_5 \\ \beta_6 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Generally, a rent index should be as simple as possible. In statistical terms, this means that the model should be sparse and avoid overparameterization. Having this in mind, we may ask whether it is really necessary to differentiate between standard and premium kitchens. The estimated regression coefficients for standard kitchens are not very different from premium ones. The corresponding statistical hypotheses are given by

$$H_0 : \beta_5 = \beta_6 \quad \text{against} \quad H_1 : \beta_5 \neq \beta_6$$

or equivalently

$$H_0 : \beta_5 - \beta_6 = 0 \quad \text{against} \quad H_1 : \beta_5 - \beta_6 \neq 0.$$

△

The discussed problems are exemplary for the most common statistical hypotheses in linear models:

1. *Test of significance:*

$$H_0 : \beta_j = 0 \quad \text{against} \quad H_1 : \beta_j \neq 0.$$

2. *Composite test of a subvector $\beta_1 = (\beta_1, \dots, \beta_r)'$:*

$$H_0 : \beta_1 = \mathbf{0} \quad \text{against} \quad H_1 : \beta_1 \neq \mathbf{0}.$$

3. Test of equality:

$$H_0 : \beta_j - \beta_r = 0 \quad \text{against} \quad H_1 : \beta_j - \beta_r \neq 0.$$

We can treat these three test problems as special cases of tests for *general linear hypotheses*

$$H_0 : \mathbf{C}\boldsymbol{\beta} = \mathbf{d} \quad \text{against} \quad H_1 : \mathbf{C}\boldsymbol{\beta} \neq \mathbf{d}, \quad (3.26)$$

where $\mathbf{C}$ is an $r \times p$ -matrix with $\text{rk}(\mathbf{C}) = r \leq p$. This means that under H_0 a total of r linear-independent conditions apply. When testing the significance of an influential variable, $\mathbf{d} = 0$ is a scalar and $\mathbf{C}$ is a $1 \times p$ -matrix given by $\mathbf{C} = (0, \dots, 0, 1, 0, \dots, 0)$, where the one is at the $(j + 1)$ -th column of the matrix. In the case of testing the first r components, we obtain the r -dimensional vector $\mathbf{d} = \mathbf{0}$ and the $r \times p$ -matrix

$$\mathbf{C} = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & & & \ddots & & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 1 & 0 & \cdots & 0 \end{pmatrix}.$$

Lastly, when testing equality of two coefficients, we obtain the scalar $\mathbf{d} = 0$ and the $1 \times p$ -matrix $\mathbf{C} = (0, \dots, 1, \dots, -1, \dots, 0)$, where the one is at the $(j + 1)$ -th position and the minus one at the $(r + 1)$ -th position.

In the next section, we will develop a test for general linear hypotheses (3.26), which will contain the test problems 1–3 given above as special cases.

3.3.1 Exact F-Test

We first assume Gaussian errors to derive an exact test. In order to develop an appropriate test statistic for the general test problem (3.26), we proceed as follows:

1. Compute the residual sum of squares $\text{SSE} = \hat{\boldsymbol{\varepsilon}}' \hat{\boldsymbol{\varepsilon}}$ for the full model.
2. Compute the residual sum of squares $\text{SSE}_{H_0} = \hat{\boldsymbol{\varepsilon}}'_{H_0} \hat{\boldsymbol{\varepsilon}}_{H_0}$ for the model under the null hypothesis, i.e., under the restriction $\mathbf{C}\boldsymbol{\beta} = \mathbf{d}$.
3. Calculate the statistic

$$\frac{\Delta \text{SSE}}{\text{SSE}} = \frac{\text{SSE}_{H_0} - \text{SSE}}{\text{SSE}}, \quad (3.27)$$

which is the relative distance in residual sum of squares between the restricted model and the full model.

To interpret the test statistic, we first note that the fit to the data under the restriction is at most equal to the unrestricted fit. Figure 3.15 gives an illustration in the case of a linear model $y = \beta x + \varepsilon$, with only one parameter β . This figure shows the residual sum of squares $\text{LS}(\beta)$ depending on β . The least squares estimator $\hat{\beta} = 1.78$ is additionally marked. If the restriction $0 \leq \beta \leq 1$ applies, the parameter estimates can only be chosen within the two vertical lines. In this limited parameter

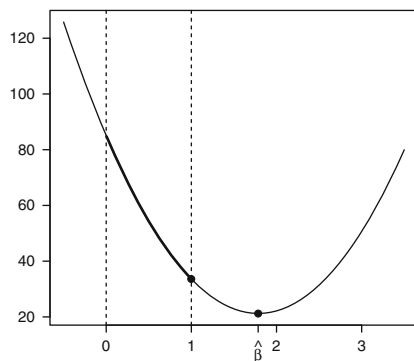


Fig. 3.15 Illustration of the difference in goodness of fit between the unconstrained least squares estimator and the estimator under the constraint $0 \leq \beta \leq 1$. The (unconstrained) least squares estimator is labeled as $\hat{\beta}$. For the constrained solution, we have $\hat{\beta} = 1$

space, the residual sum of squares reaches its minimum for $\hat{\beta} = 1$. In summary, the difference $SSE_{H_0} - SSE$ is always greater or equal to zero, since the fit to the data under the restriction $C\beta = d$ can be, at best, as good as with the unconstrained least squares estimator. A formal proof for $\Delta SSE \geq 0$ will be given in the appendix of this chapter on p. 172.

The above illustration also shows the main idea behind the statistic (3.27). The smaller the difference between SSE_{H_0} and SSE , the closer the two minima are, shown in Fig. 3.15, and the more likely it is that we will retain the null hypothesis. On the other hand, the larger the difference, the more likely it is that we will reject the null hypotheses. The test statistic actually used is

$$F = \frac{\frac{1}{r} \Delta SSE}{\frac{1}{n-p} SSE} = \frac{n-p}{r} \frac{\Delta SSE}{SSE}, \quad (3.28)$$

where r represents the number of (linear independent) restrictions, or the number of rows in C . The additional constant factor $\frac{n-p}{r}$ is not important for interpretation. It ensures that the distribution of the test statistic under the null hypothesis is a known distribution.

In order to derive the distribution of the test statistic under H_0 , we proceed as follows:

1. Determine the least squares estimator under H_0

In Sect. 3.5.2 (p. 172), we derive the least squares estimator $\hat{\beta}^R$ under H_0 , i.e., under the restriction $C\beta = d$. We obtain

$$\hat{\beta}^R = \hat{\beta} - (X'X)^{-1}C'(C(X'X)^{-1}C')^{-1}(C\hat{\beta} - d).$$

2. Determine the difference in residual sum of squares

In Sect. 3.5.2, we derive the difference ΔSSE in the residual sum of squares, given by

$$\Delta \text{SSE} = (\mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d})' (\mathbf{C}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{C}')^{-1} (\mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d}).$$

3. Stochastic properties of the difference in residual sum of squares

When determining the distribution of the test statistic under H_0 , we need the following stochastic properties of ΔSSE :

- (a) $E(\Delta \text{SSE}) = r\sigma^2 + (\mathbf{C}\boldsymbol{\beta} - \mathbf{d})' (\mathbf{C}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{C}')^{-1} (\mathbf{C}\boldsymbol{\beta} - \mathbf{d})$ (whether the restriction is fulfilled or not).
- (b) Under H_0 , we have $\frac{1}{\sigma^2} \cdot \Delta \text{SSE} \sim \chi_r^2$.
- (c) ΔSSE and SSE are stochastically independent.

Proofs of these statements are again found in Sect. 3.5.2.

4. Distribution of the test statistic

Using the stochastic properties in 3, we can now derive the distribution of the test statistic under the null hypothesis: According to property 3(b), we obtain under H_0 :

$$\frac{1}{\sigma^2} \Delta \text{SSE} \sim \chi_r^2.$$

Furthermore, we have

$$\frac{1}{\sigma^2} \text{SSE} \sim \chi_{n-p}^2;$$

refer to Box 3.11 on p. 123. Finally, ΔSSE and SSE are stochastically independent according to property 3(c). Thus, under H_0 , the test statistic (3.28) follows an F-distribution with r and $n - p$ degrees of freedom, i.e., $F \sim F_{r, n-p}$. This result follows from the definition of the F-distribution; see Definition B.14 (p. 645) in Appendix B.1.

This leads us to the following test: Let α be the significance level. We then reject the null hypothesis if the test statistic is larger than the $(1 - \alpha)$ -quantile of the corresponding F-distribution, i.e.,

$$F > F_{r, n-p}(1 - \alpha).$$

Connection to the Wald Test

We next demonstrate an interesting connection to the Wald test. See Sect. B.4.4 (p. 662) in Appendix B for the general background and the idea of the Wald test. The steps to derive the distribution of the test statistic for the F-test (see above) yield the relationship

$$\Delta \text{SSE} = (\mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d})' (\mathbf{C}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{C}')^{-1} (\mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d}).$$

With $\text{SSE} = \hat{\boldsymbol{\varepsilon}}'\hat{\boldsymbol{\varepsilon}} = (n - p)\hat{\sigma}^2$, we have

$$\begin{aligned} F &= \frac{(\mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d})' (\hat{\sigma}^2 \mathbf{C}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{C}')^{-1} (\mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d})}{r} \\ &= \frac{(\mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d})' \widehat{\text{Cov}}(\mathbf{C}\hat{\boldsymbol{\beta}})^{-1} (\mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d})}{r}. \end{aligned}$$

This representation offers another interesting interpretation of the test statistic. Clearly, F compares the difference between the estimator $C\hat{\beta}$ and the hypothesized value d , weighted by the inverse of the estimated covariance matrix

$$\widehat{\text{Cov}}(C\hat{\beta}) = \hat{\sigma}^2 C(X'X)^{-1}C'$$

of $C\hat{\beta}$. Note that $\hat{\sigma}^2$ denotes the estimator of σ^2 based on the unconstrained model. The Wald test has an analogous construction, and we have the relationship

$$W = rF$$

between the Wald W and the F statistics. Lastly, we note that the F -test can also be equivalently derived as a likelihood ratio test.

F-Tests for Some Specific Test Problems

In the following, we examine the F -test statistic in detail for some specific test problems:

1. *Test of significance (t-test):*

$$H_0 : \beta_j = 0 \quad \text{against} \quad H_1 : \beta_j \neq 0 \quad j = 1, \dots, k.$$

In this special case, it can be shown that

$$F = \frac{\hat{\beta}_j^2}{\widehat{\text{Var}}(\hat{\beta}_j)} \sim F_{1, n-p}.$$

Equivalently, the test can be based on the “t-statistic,”

$$t_j = \frac{\hat{\beta}_j}{\text{se}_j}, \quad (3.29)$$

where $\text{se}_j = \widehat{\text{Var}}(\hat{\beta}_j)^{1/2}$ denotes the estimated standard deviation or standard error of $\hat{\beta}_j$. We can view the original F-test statistic as the square of t_j . According to Definition B.14 in Appendix B.1 t_j is t-distributed with $n - p$ degrees of freedom. We obtain the critical value for the rejection region of the null hypothesis as the $(1 - \alpha/2)$ -quantile of the t-distribution with $n - p$ degrees of freedom. Thus, we reject the null hypothesis, if

$$|t_j| > t_{1-\alpha/2}(n - p).$$

We can test the slightly more general hypotheses

$$H_0 : \beta_j = d_j \quad \text{against} \quad H_1 : \beta_j \neq d_j \quad j = 1, \dots, k$$

in a similar way, by using the modified test statistic

$$t_j = \frac{\beta_j - d_j}{\text{se}_j}.$$

2. *Composite test of an r -dimensional subvector $\boldsymbol{\beta}_1 = (\beta_1, \dots, \beta_r)'$:*

$$H_0 : \boldsymbol{\beta}_1 = \mathbf{0} \quad \text{against} \quad H_1 : \boldsymbol{\beta}_1 \neq \mathbf{0}.$$

In this case, we obtain the test statistic

$$F = \frac{1}{r} \hat{\boldsymbol{\beta}}_1' \widehat{\text{Cov}(\hat{\boldsymbol{\beta}}_1)}^{-1} \hat{\boldsymbol{\beta}}_1 \sim F_{r, n-p}. \quad (3.30)$$

The estimated covariance matrix of the subvector $\hat{\boldsymbol{\beta}}_1$ consists of the corresponding elements of the full covariance matrix $\hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}$.

3. *Test for significance of regression:*

We want to test if there is a linear relationship between the response and *any* of the regressors, i.e.,

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0.$$

Note, that the alternative hypothesis does not imply that *all* variables are influential. Rather it simply states that at least one of the k covariates is influential. Under H_0 the least squares estimator simplifies to $\hat{\beta}_0 = \bar{y}$. Consequentially, we obtain

$$\text{SSE}_{H_0} = \sum_{i=1}^n (y_i - \bar{y})^2,$$

for the residual sum of squares under the null hypothesis. For the difference in the residual sum of squares between the null model and the full model we then have

$$\Delta \text{SSE} = \text{SSE}_{H_0} - \text{SSE} = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2,$$

where we applied the analysis of variance formula (3.19) on p. 112. This yields

$$\begin{aligned} F &= \frac{n-p}{k} \frac{\sum (\hat{y}_i - \bar{y})^2}{\sum \hat{\varepsilon}_i^2} \\ &= \frac{n-p}{k} \frac{\sum (\hat{y}_i - \bar{y})^2}{\sum (y_i - \bar{y})^2 - \sum (\hat{y}_i - \bar{y})^2} \end{aligned}$$

Table 3.2 Munich rent index: output for the regression model (3.25) on p. 126

Variable	Coefficient	Standard error	t-statistic	p-value	95 % Confidence-interval	
<i>intercept</i>	6.936	0.037	184.50	<0.001	6.862	7.009
<i>areainv</i>	124.341	4.445	27.97	<0.001	115.625	133.057
<i>yearco</i>	0.729	0.030	24.29	<0.001	0.670	0.788
<i>yearco2</i>	0.436	0.030	14.53	<0.001	0.377	0.495
<i>yearco3</i>	−0.011	0.029	−0.38	0.701	−0.068	0.046
<i>nkitchen</i>	1.044	0.101	10.28	<0.001	0.845	1.243
<i>pkitchen</i>	1.306	0.152	8.57	<0.001	1.007	1.604
<i>year01</i>	−0.192	0.062	−3.08	0.002	−0.314	−0.069

$$\begin{aligned}
&= \frac{n-p}{k} \frac{\sum (\hat{y}_i - \bar{y})^2 / \sum (y_i - \bar{y})^2}{1 - \sum (\hat{y}_i - \bar{y})^2 / \sum (y_i - \bar{y})^2} \\
&= \frac{n-p}{k} \frac{R^2}{1 - R^2}.
\end{aligned}$$

This result can be interpreted as follows: For a small coefficient of determination R^2 it is more likely to retain the null hypothesis “no linear relationship” (because F is small) than when the coefficient of determination is close to one (as F is then comparably large).

Example 3.13 Munich Rent Index—Standard Output and Hypothesis Tests

At this point, we are able to understand standard output for regression problems provided by software packages. Table 3.2 shows estimation results for the regression model (3.25) of Example 3.12 on p. 126. The table consists of six columns. From the left to the right we find the variable names, the estimated regression coefficients $\hat{\beta}_j$, the standard errors of the coefficients se_j , the test statistics t_j of the tests for $H_0 : \beta_j = 0$ against $H_1 : \beta_j \neq 0$, the corresponding p-values, as well as the respective 95% confidence intervals (see Sect. 3.3.2 on p. 136). According to Eq. (3.29), the t -statistic is the ratio between the estimated regression coefficient (second column) and the standard error (third column). As usual, the p-value states the minimal significance level α such that the null hypothesis $H_0 : \beta_j = 0$ can be rejected.

We first notice that the decrease of the average net rent by 0.19 Euro in the year 2001 is significant when comparing it to the year of 1999. We can reject the hypothesis $H_0 : \beta_7 = 0$ for every significance level $\alpha > 0.002$. As a matter of fact, there appeared to be a small market break in the housing market during the survey period of the rent index recording. Thus, the estimated decline is explainable.

In order to analyze the significance of the kitchen effect, we test the hypotheses

$$H_0 : \begin{pmatrix} \beta_5 \\ \beta_6 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \text{versus} \quad H_1 : \begin{pmatrix} \beta_5 \\ \beta_6 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

To compute the test statistic (3.30), we need the estimated covariance matrix $\widehat{\text{Cov}}(\hat{\beta})$ of $\hat{\beta}$, which is given by



$$\begin{pmatrix} 0.00141321 & & & & & & & & \\ 0.00732468 & 19.765577 & & & & & & & \\ 0.00007468 & -0.01912079 & 0.00090096 & & & & & & \\ 0.000076 & 0.02088758 & 0.00001069 & 0.00090025 & & & & & \\ -0.00003344 & 0.00206132 & 0.00000174 & 0.00000641 & 0.00085407 & & & & \\ -0.00109763 & -0.09199374 & -0.00038591 & -0.00039643 & 0.00009397 & 0.01031718 & & & \\ -0.00113005 & 0.0237961 & -0.00062755 & -0.00072908 & -0.00023632 & 0.0014074 & 0.02321696 & & \\ -0.00127122 & 0.0025953 & -0.00003465 & -0.00002294 & 0.00010375 & 0.00005632 & 0.00018081 & 0.0038842 \end{pmatrix}.$$

Due to symmetry in the covariance matrix, only the elements below the main diagonal are provided. The test statistic can now be computed as

$$F = \frac{1}{2} \begin{pmatrix} 1.044 \\ 1.306 \end{pmatrix}' \begin{pmatrix} 0.01031718 & 0.0014074 \\ 0.0014074 & 0.02321696 \end{pmatrix}^{-1} \begin{pmatrix} 1.044 \\ 1.306 \end{pmatrix} = 82.22.$$

With a significance level of $\alpha = 0.05$, the corresponding quantile of the F-distribution is given by

$$F_{2,4559-8}(0.95) = 3.00.$$

Since $F = 82.22 > 3.00 = F_{2,4551}(0.95)$, we can reject the null hypothesis at this level. The quality of the kitchen, thus, has a significant influence on the average net rent. Note that this test does not necessarily imply that both regression coefficients are different from zero for the two *kitchen* dummy variables. The null hypothesis is rejected when at least one coefficient significantly differs from zero. However, taking a look at the p-values in Table 3.2 for the variables *nkitchen* and *pkitchen*, we do find that both regression coefficients are significant. The practical implementation of the test is not as intricate as demonstrated here. Most of the statistics software packages provide simple and fast routines for linear tests. In STATA, for example, the command `test nkitchen pkitchen` would have produced the desired result.

We conclude this example by answering the question whether a distinction between a standard and a premium kitchen is necessary. We want to test the hypothesis

$$H_0 : \beta_5 - \beta_6 = 0 \quad \text{against} \quad H_1 : \beta_5 - \beta_6 \neq 0.$$

We obtain the test statistic $F = 2.23$. For $\alpha = 0.05$, the corresponding quantile of the F-distribution is given by $F_{1,4551}(0.95) = 3.84$. Since $F = 2.23 < 3.84 = F_{1,4551}(0.95)$, we cannot reject the null hypothesis. Hence, the difference between a standard and premium kitchen is not significant. This test has a p-value of $p = 0.13$. $\triangle$

Asymptotic Properties of the F-Test

The applicability of the F-test, considered thus far, requires normally distributed errors. Respecting the connection to the Wald test, we can show that the test can even be applied if the errors are not normally distributed. As already shown, we have the relationship $W = rF$. Moreover, $W \stackrel{a}{\sim} \chi_r^2$ holds; see Sect. B.4.4 in Appendix B. Thus the $F_{r,n-p}$ -distribution converges in distribution with $n \rightarrow \infty$ to a χ_r^2/r -distribution, and we are able to use critical values or p-values of the F-test (for large sample size) even in the case of non-normal errors.

3.13 Testing Linear Hypotheses

Hypotheses

1. General linear hypothesis:

$$H_0 : C\beta = d \quad \text{against} \quad H_1 : C\beta \neq d$$

where C is a $r \times p$ -matrix with $\text{rk}(C) = r \leq p$ (r linear independent restrictions).

2. Test of significance (t -test):

$$H_0 : \beta_j = 0 \quad \text{against} \quad H_1 : \beta_j \neq 0$$

3. Composite test of a subvector:

$$H_0 : \beta_1 = \mathbf{0} \quad \text{against} \quad H_1 : \beta_1 \neq \mathbf{0}$$

4. Test for significance of regression:

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0 \quad \text{against}$$

$$H_1 : \beta_j \neq 0 \text{ for at least one } j \in \{1, \dots, k\}$$

Test Statistics

Assuming normal errors we obtain under H_0 :

$$1. F = 1/r (C\hat{\beta} - d)' (\hat{\sigma}^2 C(X'X)^{-1}C')^{-1} (C\hat{\beta} - d) \sim F_{r,n-p}$$

$$2. t_j = \frac{\hat{\beta}_j}{\text{se}_j} \sim t_{n-p}$$

$$3. F = \frac{1}{r} (\hat{\beta}_1)' \widehat{\text{Cov}}(\hat{\beta}_1)^{-1} (\hat{\beta}_1) \sim F_{r,n-p}$$

$$4. F = \frac{n-p}{k} \frac{R^2}{1-R^2} \sim F_{k,n-p}$$

Critical Values

Reject H_0 in the case of:

$$1. F > F_{r,n-p}(1-\alpha)$$

$$3. F > F_{r,n-p}(1-\alpha)$$

$$2. |t| > t_{n-p}(1-\alpha/2)$$

$$4. F > F_{k,n-p}(1-\alpha)$$

The tests are relatively robust against moderate departures from normality. In addition, the tests can be applied for large sample size, even with non-normal errors.

3.3.2 Confidence Regions and Prediction Intervals

Confidence Intervals and Ellipsoids for Regression Coefficients

The duality between two-sided tests and confidence intervals or confidence regions allows us to construct a confidence interval for a single parameter β_j , $j = 0, \dots, k$ or a confidence ellipsoid for a subvector β_1 of β . To construct a confidence interval for β_j under normality, we use the test statistic $t_j = (\hat{\beta}_j - d_j)/\text{se}_j$ corresponding to the test $H_0 : \beta_j = d_j$. Recall that we reject the null hypothesis when $|t_j| > t_{n-p}(1 - \alpha/2)$. The test is constructed such that the probability of rejecting H_0 when H_0 is true equals α . Therefore, under H_0 we have

$$P(|t_j| > t_{n-p}(1 - \alpha/2)) = \alpha.$$

Thus the probability of not rejecting H_0 (given H_0 is true) is provided by

$$P(|t_j| < t_{n-p}(1 - \alpha/2)) = P(|(\hat{\beta}_j - \beta_j)/\text{se}_j| < t_{n-p}(1 - \alpha/2)) = 1 - \alpha.$$

This is equivalent to

$$P(\hat{\beta}_j - t_{n-p}(1 - \alpha/2) \cdot \text{se}_j < \beta_j < \hat{\beta}_j + t_{n-p}(1 - \alpha/2) \cdot \text{se}_j) = 1 - \alpha,$$

and we obtain

$$[\hat{\beta}_j - t_{n-p}(1 - \alpha/2) \cdot \text{se}_j, \hat{\beta}_j + t_{n-p}(1 - \alpha/2) \cdot \text{se}_j]$$

as a $(1 - \alpha)$ -confidence interval for β_j .

In a similar way, we can create a $(1 - \alpha)$ -confidence region for a r -dimensional subvector β_1 of β ; see Box 3.14.

Example 3.14 Munich Rent Index—Confidence Intervals

We illustrate the construction of confidence intervals using the regression model taken from the last example; see p. 133. A 95% confidence interval for the regression coefficient β_7 , associated with the time dummy variable, can be calculated as

$$-0.192 \pm 1.96 \cdot 0.062 = [-0.313, -0.070].$$

Here we used $\text{se}_7 = 0.062$ (see Table 3.2) and $t_{n-p}(1 - \alpha/2) = t_{4551}(0.975) = 1.96$. The minor differences found between this confidence interval and the one found in Table 3.2 result from rounding error.

△

Prediction Intervals

In Sect. 3.2.3 on p. 118, we derived an optimal prediction $\hat{y}_0 = \mathbf{x}'_0 \hat{\beta}$ for a new (future) observation at location $\mathbf{x}_0$. Specifically, $\hat{y}_0$ is an estimator for the (conditional) mean $E(y_0) = \mathbf{x}'_0 \beta = \mu_0$ of the future observation y_0 . In addition to the point estimate, interval estimation for μ_0 is often of interest and is easy to construct. Since $\hat{\beta} \sim N(\beta, \sigma^2(X'X)^{-1})$, it follows for the linear combination

3.14 Confidence Regions and Prediction Intervals

Provided that we have (at least approximately) normally distributed errors or a large sample size, we obtain the following confidence intervals or regions and prediction intervals:

Confidence Interval for β_j

A confidence interval for β_j with level $1 - \alpha$ is given by

$$[\hat{\beta}_j - t_{n-p}(1 - \alpha/2) \cdot \text{se}_j, \hat{\beta}_j + t_{n-p}(1 - \alpha/2) \cdot \text{se}_j].$$

Confidence Ellipsoid for Subvector β_1

A confidence ellipsoid for $\beta_1 = (\beta_1, \dots, \beta_r)'$ with level $1 - \alpha$ is given by

$$\left\{ \beta_1 : \frac{1}{r} (\hat{\beta}_1 - \beta_1)' \widehat{\text{Cov}}(\hat{\beta}_1)^{-1} (\hat{\beta}_1 - \beta_1) \leq F_{r, n-p}(1 - \alpha) \right\}.$$

Confidence Interval for μ_0

A confidence interval for $\mu_0 = E(y_0)$ of a future observation y_0 at location $\mathbf{x}_0$ with level $1 - \alpha$ is given by

$$\mathbf{x}_0' \hat{\beta} \pm t_{n-p}(1 - \alpha/2) \hat{\sigma} (\mathbf{x}_0' (\mathbf{X}' \mathbf{X})^{-1} \mathbf{x}_0)^{1/2}.$$

Prediction Interval

A prediction interval for a future observation y_0 at location $\mathbf{x}_0$ with level $1 - \alpha$ is given by

$$\mathbf{x}_0' \hat{\beta} \pm t_{n-p}(1 - \alpha/2) \hat{\sigma} (1 + \mathbf{x}_0' (\mathbf{X}' \mathbf{X})^{-1} \mathbf{x}_0)^{1/2}.$$

$$\mathbf{x}_0' \hat{\beta} \sim N(\mathbf{x}_0' \beta, \sigma^2 \mathbf{x}_0' (\mathbf{X}' \mathbf{X})^{-1} \mathbf{x}_0).$$

Standardizing yields

$$\frac{\mathbf{x}_0' \hat{\beta} - \mu_0}{\sigma (\mathbf{x}_0' (\mathbf{X}' \mathbf{X})^{-1} \mathbf{x}_0)^{1/2}} \sim N(0, 1).$$

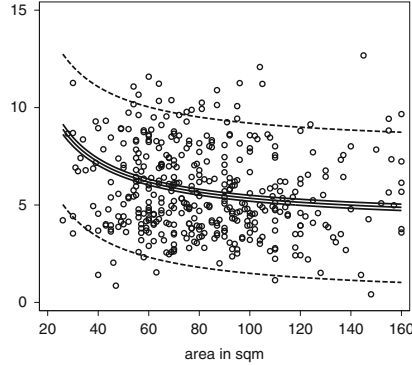


Fig. 3.16 Munich rent index: estimated rent per square meter depending on the living area including 95 % confidence interval (solid lines) and 95 % prediction interval (dashed lines). The values of the remaining covariates have been set to $yearc = 1918$, $nkitchen = 0$, $gkitchen = 0$, and $year01 = 0$. Additionally included are the observations available for this covariate pattern

If we substitute σ^2 with the estimator $\hat{\sigma}^2$, the resulting expression follows a t-distribution with $n - p$ degrees of freedom, and we have

$$P\left(-t_{n-p}(1 - \alpha/2) \leq \frac{\mathbf{x}'_0 \hat{\boldsymbol{\beta}} - \mu_0}{\hat{\sigma}(\mathbf{x}'_0(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0)^{1/2}} \leq t_{n-p}(1 - \alpha/2)\right) = 1 - \alpha.$$

Consequently,

$$[\mathbf{x}'_0 \hat{\boldsymbol{\beta}} - t_{n-p}(1 - \alpha/2)\hat{\sigma}(\mathbf{x}'_0(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0)^{1/2}, \mathbf{x}'_0 \hat{\boldsymbol{\beta}} + t_{n-p}(1 - \alpha/2)\hat{\sigma}(\mathbf{x}'_0(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0)^{1/2}]$$

is a $(1 - \alpha)$ -confidence interval for μ_0 .

In many cases, one is also interested in determining an interval, which contains the future observation y_0 with high probability. In relation to the rent index, this means that a prospective tenant wants to know the range of reasonable rent values for a specific apartment. Thus, we are looking for a prediction interval for the future observation y_0 . For this purpose we look at the prediction error $\hat{\varepsilon}_0 = y_0 - \mathbf{x}'_0 \hat{\boldsymbol{\beta}}$. We have

$$\hat{\varepsilon}_0 \sim N(0, \sigma^2 + \sigma^2 \mathbf{x}'_0(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0).$$

Standardizing and replacing $\hat{\sigma}^2$ for σ^2 yields

$$\frac{y_0 - \mathbf{x}'_0 \hat{\boldsymbol{\beta}}}{\hat{\sigma}(1 + \mathbf{x}'_0(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0)^{1/2}} \sim t_{n-p}$$

and therefore

$$P\left(-t_{n-p}(1 - \alpha/2) \leq \frac{y_0 - \mathbf{x}'_0 \hat{\boldsymbol{\beta}}}{\hat{\sigma}(1 + \mathbf{x}'_0(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0)^{1/2}} \leq t_{n-p}(1 - \alpha/2)\right) = 1 - \alpha.$$

Thus, with $(1 - \alpha)$ -confidence, we find the future observation at $\mathbf{x}_0$ to be within the prediction interval

$$\mathbf{x}'_0 \hat{\boldsymbol{\beta}} \pm t_{n-p}(1 - \alpha/2) \hat{\sigma} (1 + \mathbf{x}'_0 (\mathbf{X}'\mathbf{X})^{-1} \mathbf{x}_0)^{1/2}.$$

Per construction, the prediction interval is always wider than the corresponding confidence interval for μ_0 . In applications with high error variance, the interval can be considerably wider; see the following Example 3.15 on the rent index.

Even though at first the two intervals appear to be similar, they are in fact very different. In the first case, we constructed a confidence interval for $E(y_0) = \mu_0$. This implies that the random interval overlaps the (fixed and constant) mean $E(y_0)$ with probability $1 - \alpha$. In the second case, we rather constructed an interval which is very likely (more precisely with probability $1 - \alpha$) to contain the (random) future observation y_0 .

Example 3.15 Munich Rent Index—Prediction Intervals

We again use the model from Example 3.13 (p. 133). Figure 3.16 shows the estimated rent per square meter for apartments constructed in the year 1918, with a standard kitchen ($nkitchen = 0$, $pkitchen = 0$) and time of assessment in 1999, depending on the living area. The 95 % confidence and prediction intervals are also included: individual confidence intervals were connected with lines leading to the bands shown. Due to the high variability of the rent index data, prediction intervals are much wider than confidence intervals for $E(y_0)$.

△

3.4 Model Choice and Variable Selection

In many applications, a large (potentially enormous) number of candidate predictor variables are available, and we face the challenge and decision as to which of these variables to include in the regression model. As a typical example, take the Munich rent index data with approximately 200 variables that are collected through a questionnaire. The following are two naive (but often practiced) approaches to the model selection problem:

- Strategy 1: Estimate the most complex model which includes all available covariates.
- Strategy 2: First, estimate a model with all variables. Then, remove all insignificant variables from the model.

We discuss and illustrate each of these approaches using two simulated data sets.

Example 3.16 Polynomial Regression

We investigate the simulated data illustrated in Fig. 3.17a. The scatter plot suggests polynomial modeling of the relationship between y and x resulting in the regression model

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_l x_i^l + \varepsilon_i.$$

As with any polynomial regression, we have to choose the polynomial degree l . Panels (c–e) in Fig. 3.17 show the estimated relationship for $l = 1$ (regression line), $l = 2$, and $l = 5$. Apparently, a simple regression line, i.e., a first-order polynomial, does not fit the

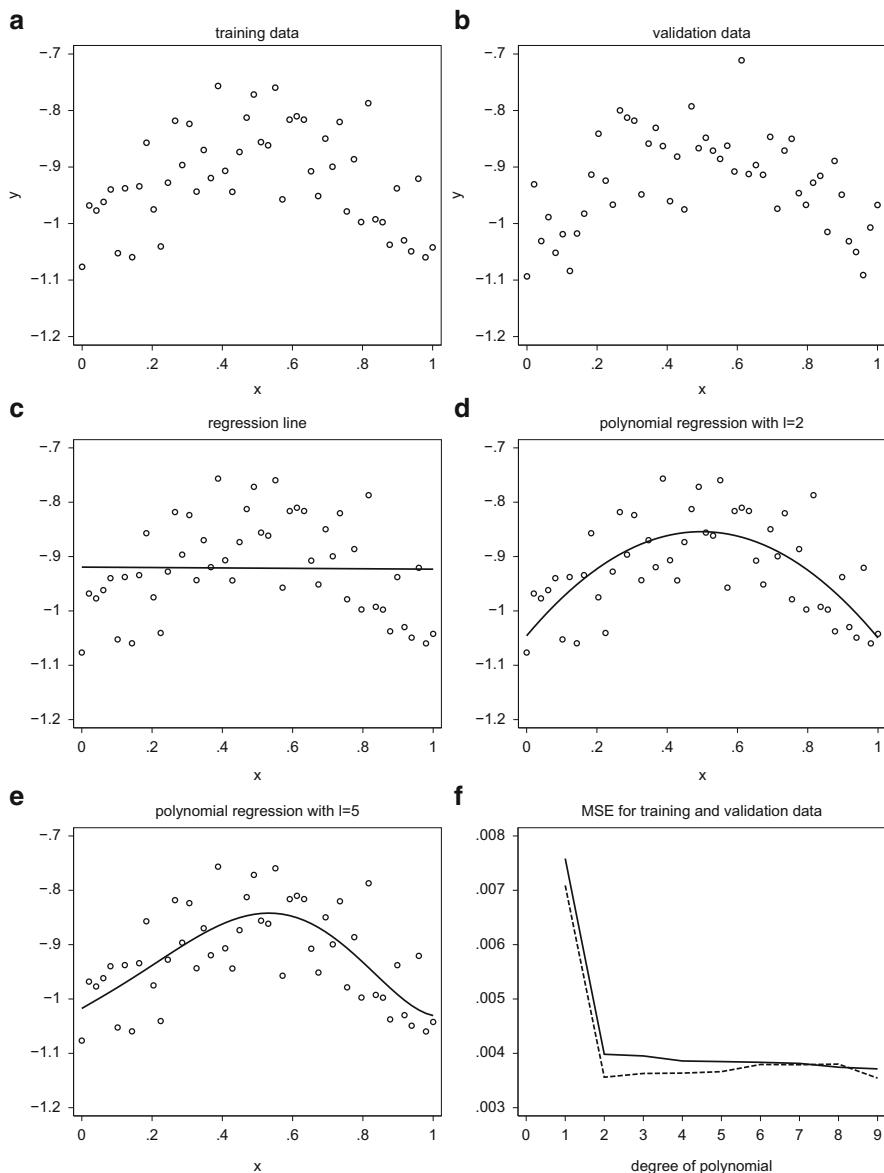


Fig. 3.17 Simulated training data y_i [panel (a)] and validation data y_i^* [panel (b)] based on 50 design points x_i , $i = 1, \dots, 50$. The true model used for simulation is $y_i = -1 + 0.3x_i + 0.4x_i^2 - 0.8x_i^3 + \varepsilon_i$ with $\varepsilon_i \sim N(0, 0.07^2)$. Panels (c–e) show estimated polynomials of degree $l = 1, 2, 5$ based on the training set. Panel (f) displays the mean squared error $MSE(l)$ of the fitted values in relation to the polynomial degree (*solid line*). The *dashed line* shows $MSE(l)$, if the estimated polynomials are used to predict the validation data y_i^*

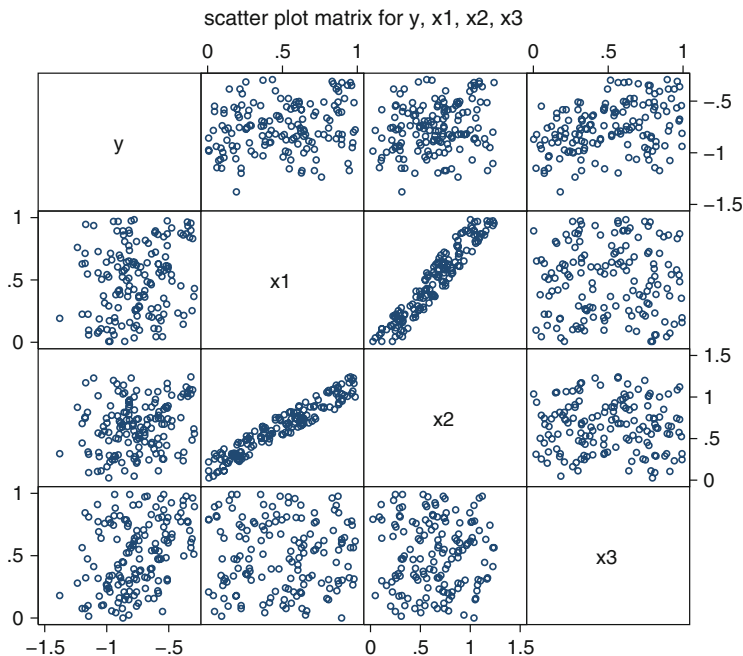


Fig. 3.18 Scatter plot matrix for the variables y , x_1 , x_2 , and x_3

data well. With polynomial degrees $l > 2$ onward, a satisfactory fit to the data appears to be guaranteed. Figure 3.17f additionally displays the mean squared error

$$\text{MSE}(l) = \frac{1}{50} \sum_{i=1}^{50} (y_i - \hat{y}_i(l))^2$$

of the fitted models depending on the order of the polynomial (continuous line). Clearly, $\text{MSE}(l)$ decreases monotonically with increased l . This suggests that the fit to the data is better with larger polynomial order. This finding appears to confirm the first strategy described above, namely to include as many regressors as possible into the model.

In a next step, we investigate how well the fitted models predict new observations that have been simulated according to the same model. Figure 3.17b shows additionally simulated observations for every design point x_i , $i = 1, \dots, 50$. We refer to this data set as the validation sample, whereas we refer to the first data set (used for estimation) as the training set. Figure 3.17f shows the mean squared error of $\hat{y}_i^*$ for the data y_i^* (dashed line) in the validation set. Apparently, the fit to the new data is initially getting better with an increase of the polynomial order. However, from the polynomial order $l = 3$ onward, the fit is getting worse. We recognize the following: The more complex the model, the better is the fit to the data that were used for estimation. However, with new data resulting from the same data generating process, models that are too complex can cause a poorer fit. $\triangle$

Example 3.17 Correlated Covariates

Consider the $n = 150$ observations $(y_i, x_{i1}, x_{i2}, x_{i3})$, $i = 1, \dots, 150$, in the scatter plot matrix in Fig. 3.18. The data were generated as follows: The variables x_1 and x_3 are

Table 3.3 Results for the model based on covariates x_1 , x_2 , and x_3

Variable	Coefficient	Standard error	t-value	p-value	95 % Confidence interval	
<i>intercept</i>	−0.970	0.047	−20.46	<0.001	−1.064	−0.877
x_1	0.146	0.187	0.78	0.436	−0.224	0.516
x_2	0.027	0.177	0.15	0.880	−0.323	0.377
x_3	0.227	0.052	4.32	<0.001	0.123	0.331

Table 3.4 Results for the correctly specified model based on covariates x_1 and x_3

Variable	Coefficient	Standard error	t-value	p-value	95 % Confidence interval	
<i>intercept</i>	−0.967	0.039	−24.91	<0.001	−1.042	−0.889
x_1	0.173	0.055	3.17	0.002	0.065	0.281
x_3	0.226	0.052	4.33	<0.001	0.123	0.330

independent and uniformly distributed on $[0,1]$. The variable x_2 is defined as $x_2 = x_1 + u$, where u is also uniformly distributed on $[0,1]$. Thus, the variables x_1 and x_2 are highly correlated. Finally, the response variable y is simulated according to the model

$$y \mid x_1, x_2, x_3 \sim N(-1 + 0.3x_1 + 0.2x_3, 0.2^2).$$

The conditional mean of y is thus dependent on x_1 and x_3 , but not on x_2 . In the following, we assume, however, that we do not know the true model (as is typically the case in practice). At first, we estimate a regression model with all available covariates x_1 , x_2 , and x_3 , and we obtain the results provided in Table 3.3. Clearly, x_1 and x_2 are nonsignificant. If we followed strategy 2, i.e., if we eliminate the nonsignificant variables from the model, we would eliminate not only the nonrelevant covariate x_2 , but also the relevant variable x_1 . If we instead estimate a correctly specified model with true predictor variables x_1 and x_3 , we obtain the results shown in Table 3.4. When having a correct model specification, not only is x_3 significant but so is the previously insignificant variable x_1 . We conclude: If we first consider all variables and then eliminate the insignificant variables from the model, it is possible that also important variables will be eliminated. The main reason for such unfortunate model estimation circumstances is the existing correlation among covariates. $\triangle$

3.4.1 Effect of Model Specification on Bias, Variance, and Prediction Quality

We now strengthen the new insights of the previous examples with more theoretical considerations. In particular, we focus on the following questions:

1. *Irrelevant Variables*: What can be said about the bias and the variance of the least squares estimator, in the case that we include irrelevant variables in the model?
2. *Missing Variables*: What can be said about the bias and the variance of the least squares estimator, if we omit relevant variables in the model?
3. *Prediction Quality*: What effect does the model specification, more specifically the selected variables in the model, have on prediction?

Effect of Model Specification on Bias and Variance of the Least Squares Estimator

Consider a partition of the available explanatory variables $\mathbf{x} = (x_0, x_1, \dots, x_k)'$ with $x_0 \equiv 1$ into the subsets $\mathbf{x}_1 = (x_0, x_1, \dots, x_{k_1})'$ and $\mathbf{x}_2 = (x_{k_1+1}, \dots, x_k)'$. We look at the two models

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$$

and

$$\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{u}.$$

The first model uses all available variables. The second model uses only the subset $\mathbf{x}_1$. We obtain the least squares estimators

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \quad \text{and} \quad \tilde{\boldsymbol{\beta}}_1 = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{y}$$

respectively. For the estimator $\tilde{\boldsymbol{\beta}}_1$ of the submodel, we obtain

$$\begin{aligned} E(\tilde{\boldsymbol{\beta}}_1) &= E((\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{y}) \\ &= (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'E(\mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}) \\ &= (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'(\mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2) \\ &= \boldsymbol{\beta}_1 + (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2\boldsymbol{\beta}_2 \end{aligned}$$

and

$$\begin{aligned} \text{Cov}(\tilde{\boldsymbol{\beta}}_1) &= \text{Cov}((\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{y}) = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\sigma^2\mathbf{I}\mathbf{X}_1(\mathbf{X}_1'\mathbf{X}_1)^{-1} \\ &= \sigma^2(\mathbf{X}_1'\mathbf{X}_1)^{-1}. \end{aligned}$$

for the mean and covariance matrix. The statistical properties of the estimator $\hat{\boldsymbol{\beta}}$ in the full model have already been derived in Sect. 3.2.3; see Box 3.9 on p. 119.

We now investigate the following two situations:

- *Missing Variables:* Even though the complete model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ is correct, we mistakenly estimate the reduced model $\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{u}$. In this case we neglect the relevant variables $\mathbf{x}_2$.
- *Irrelevant Variables:* Even though the reduced model $\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{u}$ is correct, we mistakenly estimate the full model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$. In this case, we included irrelevant variables in the model. The variables in $\mathbf{x}_2$ are redundant.

In the first case of missing variables the following applies:

- $\tilde{\boldsymbol{\beta}}_1$ is biased. An exception is the case when $\mathbf{X}_1'\mathbf{X}_2 = \mathbf{0}$, i.e., every variable in $\mathbf{X}_1$ is uncorrelated to every variable in $\mathbf{X}_2$.
- It can be shown that the difference $\text{Cov}(\hat{\boldsymbol{\beta}}_1) - \text{Cov}(\tilde{\boldsymbol{\beta}}_1)$ of covariance matrices is positive semi-definite. This implies that the components of the estimator $\tilde{\boldsymbol{\beta}}_1$ based on the submodel $\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{u}$ show a smaller variance than the corresponding

components of the estimator $\hat{\beta}_1$ based on the correct model $y = X\beta + \epsilon$. Thus we have $\text{Var}(\hat{\beta}_j) \geq \text{Var}(\tilde{\beta}_j)$.

- Moreover, it can be shown that situations exist, in which the components in $\tilde{\beta}_1$ based on the misspecified submodel actually show a smaller MSE than the components in $\hat{\beta}_1$, which are based on the full model, i.e., $\text{MSE}(\hat{\beta}_j) \geq \text{MSE}(\tilde{\beta}_j)$. Hence, it is possible that a sparse model with unconsidered covariates has better statistical properties than the correctly specified full model.

In the second case of irrelevant variables, we have:

- Even though irrelevant variables were considered, $\hat{\beta}$ is unbiased. Of course, the estimator $\tilde{\beta}_1$ based on the true model is also unbiased.
- It can be shown that the estimators for the components in β_1 based on $\hat{\beta}$ have larger variance than based on $\tilde{\beta}_1$. Thus, we have $\text{Var}(\hat{\beta}_j) \geq \text{Var}(\tilde{\beta}_j)$. If the estimated model contains irrelevant variables, then the precision of the estimators decreases.

Analogous statements can be made about $\hat{y}$.

We can reach the following conclusion: Preferably, the specified model should not contain irrelevant covariates. Moreover, we should aim for a sparse model so that bias and variance, and thus MSE, are small.

Consequences of the Model Specification on Prediction Quality

Next we take a look at prediction quality in linear models. Thereby, we do not necessarily assume that the model is correctly specified. Our considerations will become useful when constructing model choice criteria for the comparison of different models.

We assume independent observations $y_i, i = 1, \dots, n$, with expectation $E(y_i) = \mu_i$ and variance $\text{Var}(y_i) = \sigma^2$. The variables $x_0 = 1, x_1, \dots, x_k$ are available as potential regressors. In the following we assume that a subset of the available variables will be used for estimation. The specified model is defined by the subset $M \subset \{0, 1, 2, \dots, k\}$ of included covariates with corresponding design matrix X_M . For the least squares estimator we obtain

$$\hat{\beta}_M = (X'_M X_M)^{-1} X'_M y.$$

An estimator $\hat{y}_M$ for the vector μ of means $\mu_i = E(y_i)$ is given by

$$\hat{y}_M = X_M \hat{\beta}_M.$$

We can view the estimator $\hat{y}_{iM}$ also as a prediction for future observations $y_{n+i} = \mu_i + \epsilon_{n+i}, i = 1, \dots, n$, with given covariates $x_{i1}, \dots, x_{ik}$. In the following, we derive a formula for the sum of the expected squared prediction errors $\sum E(y_{n+i} - \hat{y}_{iM})^2$. To do so, we need the following, easily verifiable, properties of $\hat{y}_M$:

1. Expectation:

$$E(\hat{y}_M) = X_M (X'_M X_M)^{-1} X'_M E(y).$$

2. Covariance matrix:

$$\text{Cov}(\hat{\mathbf{y}}_M) = \sigma^2 \mathbf{X}_M (\mathbf{X}_M' \mathbf{X}_M)^{-1} \mathbf{X}_M'.$$

3. Sum of the variances:

$$\sum_{i=1}^n \text{Var}(\hat{y}_{iM}) = \sigma^2 \text{tr}(\mathbf{X}_M (\mathbf{X}_M' \mathbf{X}_M)^{-1} \mathbf{X}_M') = |M| \sigma^2,$$

where $|M|$ represents the cardinal number of M , i.e., the number of the covariates included in the model. The sum of the variances increases as more covariates are included in the model.

4. Sum of the mean squared errors (SMSE):

$$\begin{aligned} \text{SMSE} &= \sum_{i=1}^n \text{E}(\hat{y}_{iM} - \mu_i)^2 \\ &= \sum_{i=1}^n \text{E}((\hat{y}_{iM} - \mu_{iM}) + (\mu_{iM} - \mu_i))^2 \\ &= \sum_{i=1}^n \text{Var}(\hat{y}_{iM}) + 2 \sum_{i=1}^n \text{E}((\hat{y}_{iM} - \mu_{iM})(\mu_{iM} - \mu_i)) + \sum_{i=1}^n (\mu_{iM} - \mu_i)^2 \\ &= |M| \sigma^2 + \sum_{i=1}^n (\mu_{iM} - \mu_i)^2. \end{aligned}$$

Here we used $\mu_{iM} = \text{E}(\hat{y}_{iM})$ as an abbreviation for the expectation of the estimator $\hat{y}_{iM}$.

These properties provide us with the expected squared prediction error:

$$\begin{aligned} \text{SPSE} &= \sum_{i=1}^n \text{E}(y_{n+i} - \hat{y}_{iM})^2 \\ &= \sum_{i=1}^n \text{E}((y_{n+i} - \mu_i) - (\hat{y}_{iM} - \mu_i))^2 \\ &= \sum_{i=1}^n (\text{E}(y_{n+i} - \mu_i)^2 - 2\text{E}((y_{n+i} - \mu_i)(\hat{y}_{iM} - \mu_i)) + \text{E}(\hat{y}_{iM} - \mu_i)^2) \\ &= \sum_{i=1}^n \text{E}(y_{n+i} - \mu_i)^2 + \sum_{i=1}^n \text{E}(\hat{y}_{iM} - \mu_i)^2 \\ &= n \sigma^2 + \text{SMSE} \\ &= n \sigma^2 + |M| \sigma^2 + \sum_{i=1}^n (\mu_{iM} - \mu_i)^2. \end{aligned}$$

Note that in line 3 of the above derivation, the expectation for the cross product term can be written as the product of expectations due to the independence of $\hat{y}_{iM}$ and y_{n+i} . This way the entire term becomes zero. Thus, the expected squared prediction error can be decomposed into three additive terms:

- *Irreducible Prediction Error:* The first term $n\sigma^2$ depends on the error variance. Hence, it cannot be reduced, even by sophisticated inference techniques. This term is therefore referred to as the irreducible prediction error.
- *Variance:* The second term consists of the sum of variances $\text{Var}(\hat{y}_{iM})$ of the estimators $\hat{y}_{iM}$. This term can be manipulated through model choice. It becomes smaller as fewer variables are included in the model.
- *Squared Bias:* The last term $\sum(\mu_{iM} - \mu_i)^2$ can be seen as a bias term. It consists of the squared bias of the estimator $\hat{y}_{iM}$ for the expectation μ_i . This term can also be manipulated through model choice and becomes smaller as more variables are included in the model.

The decomposition of the expected prediction error into an irreducible error, a variance term, and a squared bias term is not limited to linear models but rather a *fundamental property of prediction* in all statistical models.

The formula for SPSE shows a classical bias–variance trade-off. The more complex the model, the smaller the squared bias and the greater the variance. On the contrary, simpler models show a greater squared bias and in return for that a smaller variance. This bias–variance trade-off is not only characteristic for linear models, but for all statistical models; see for example the discussion of the bias–variance trade-off with regard to smoothing techniques in Sect. 8.1.8.

3.4.2 Model Choice Criteria

The theoretical considerations regarding prediction quality in the last section, especially the fundamental formula for the expected squared prediction error, help to develop tools for model choice, in particular variable selection. A possible approach is to *minimize* the expected squared prediction error: We include those covariates in the model, which minimize SPSE. Unfortunately, SPSE is not directly accessible, since μ_i and σ^2 are unknown. Thus, we need to estimate SPSE. Two strategies are possible:

1. Estimate SPSE using new and independent data

If in fact additional observations y_{n+i} are available, we are able to estimate $\text{SPSE} = \sum E(y_{n+i} - \hat{y}_{iM})^2$ simply by

$$\widehat{\text{SPSE}} = \sum_{i=1}^n (y_{n+i} - \hat{y}_{iM})^2.$$

In practice, it is usually not possible to use this approach, as it rarely happens that additional observations are collected. An alternative procedure is the following:

- Randomly split the data into two parts, i.e., a test and a validation sample.

- Use the test data set to estimate the specified model, i.e., for the estimation of regression coefficients β and the expectations μ_i .
- Use the validation set to assess the goodness of fit, i.e., for the estimation of SPSE.

This approach is possible if the available data set is large enough. If the data set is too small, partitioning the data would cause loss of accuracy in estimation of the regression coefficients. How large the sample size has to be in order to proceed in a reasonable manner depends on the problem. As far as we are aware, there is not any rule of thumb or general advice in the literature. This is the reason why we often use the second strategy.

2. Estimate SPSE using existing data

A naive estimator for SPSE would be the use of the squared sum of residuals $\sum (y_i - \hat{y}_{iM})^2$. Note that this sum *underestimates* on average the expected prediction error, as it can be shown that

$$E\left(\sum_{i=1}^n (y_i - \hat{y}_{iM})^2\right) = \text{SPSE} - 2|M|\sigma^2.$$

Thus a better estimate for SPSE is given by

$$\widehat{\text{SPSE}} = \sum_{i=1}^n (y_i - \hat{y}_{iM})^2 + 2|M|\hat{\sigma}^2.$$

Accordingly, we choose a model that minimizes $\widehat{\text{SPSE}}$. In doing so we have to keep in mind that we always use the same estimator for $\hat{\sigma}^2$. Preferably, this estimator should be based on the full model with all available variables, in order to keep the bias in $\hat{\sigma}^2$ small. The criterion $\widehat{\text{SPSE}}$ has the typical structure of many model choice criteria. It consists of two terms: The first term, the sum of squared residuals, measures the goodness of fit and becomes smaller the more complex the model becomes. The second term $2|M|\hat{\sigma}^2$ measures model complexity and becomes smaller as models become simpler.

We next present some of the most widely used criteria for model choice in linear models. Their derivation follows ideas similar to that of $\widehat{\text{SPSE}}$.

The Corrected Coefficient of Determination

In Sect. 3.2.3, we defined the coefficient of determination R^2 as a measure for the goodness of fit to the data. When comparing different models, the use of the coefficient of determination is limited, since the coefficient of determination will always increase (never decrease) with the addition of a new covariate into the model. The *corrected coefficient of determination* adjusts for this problem, by including a correction term for the number of parameters. The corrected coefficient of determination is defined by

$$\bar{R}^2 = 1 - \frac{n-1}{n-p}(1 - R^2).$$

The corrected coefficient of determination is very popular and is provided by default with all statistical program packages. At this point, we advise against its usage, since the “penalty” for newly included covariates appears to be too small. It can be shown that $\bar{R}^2$ already increases when a variable with a t-value greater than 1 is included in the model, implying we would include variables with a p-value of about 0.3.

Mallow's C_p

Mallow's C_p (“complexity parameter”) relies directly on the ideas presented above and is defined by

$$C_p = \frac{\sum_{i=1}^n (y_i - \hat{y}_{iM})^2}{\hat{\sigma}^2} - n + 2|M|.$$

C_p can be understood as an estimate of SMSE/σ^2 . Thus minimizing C_p produces the same optimal model as minimizing $\widehat{\text{SPSE}}$.

Akaike Information Criterion

The Akaike information criterion (AIC) is one of the most widely used criteria for model choice within the scope of likelihood-based inference; see Appendix B, p. 664. In general, AIC is defined by

$$\text{AIC} = -2 \cdot l(\hat{\beta}_M, \hat{\sigma}^2) + 2(|M| + 1),$$

where $l(\hat{\beta}_M, \hat{\sigma}^2)$ is the maximum value of the log-likelihood, i.e., when the ML estimators $\hat{\beta}_M$ and $\hat{\sigma}^2$ are inserted into the log-likelihood. Smaller values of the AIC correspond to a better model fit. Note that the total number of parameter is $|M| + 1$ since the error variance σ^2 is also counted as a parameter. In a linear model with Gaussian errors, we obtain

$$\begin{aligned} -2l(\hat{\beta}_M, \hat{\sigma}^2) &= n \log(\hat{\sigma}^2) + \frac{1}{\hat{\sigma}^2}(\mathbf{y} - \mathbf{X}_M \hat{\beta}_M)'(\mathbf{y} - \mathbf{X}_M \hat{\beta}_M) \\ &= n \log(\hat{\sigma}^2) + \frac{n\hat{\sigma}^2}{\hat{\sigma}^2} \\ &= n \log(\hat{\sigma}^2) + n, \end{aligned}$$

and thus

$$\text{AIC} = n \cdot \log(\hat{\sigma}^2) + 2(|M| + 1),$$

when ignoring the constant n .

Note that the ML estimator $\hat{\sigma}^2 = \hat{\mathbf{e}}'\hat{\mathbf{e}}/n$ is considered in AIC and not the usual unbiased variance estimator.

Cross Validation

Cross validation imitates partitioning of the data into a test set for parameter estimation and a validation set to assess predictive quality. Cross validation is based on the following general principle:

- Partition the data set into r subsets $1, \dots, r$, of similar size.
- Start with the first data set as validation set and use the combined remaining $r - 1$ data sets for parameter estimation. Based on the estimates, obtain predictions for the validation set and determine the sum of the squared prediction errors.
- Cycle through the partitions using the second, third, up to the r th data set as validation sample and all other data sets for estimation. Determine the sum of squared prediction errors.
- Use the model with the smallest sum of squared prediction errors, where the final parameter estimates reflect all data. The partition into test and validation samples serves only to estimate the expected squared prediction error.

An important special case is the so-called “leave-one-out” cross validation, which uses all observations with the exception of *one* for the estimation of model parameters. We use this “leave-one-out” estimator to predict the deleted observation and to determine the squared prediction error. If we denote the “leave-one-out” estimator with $\hat{y}_{iM}^-$, we obtain the cross validation score

$$CV = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_{iM}^-)^2.$$

At first glance, computation of the cross validation score CV appears to be quite expensive, since the least squares estimator has to be calculated n times. It can be shown, however, that the cross validation score can be computed with the help of the original estimators $\hat{y}_{iM}$ that are based on all data. We obtain

$$CV = \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{y}_{iM}}{1 - h_{iiM}} \right)^2,$$

where h_{iiM} are the diagonal elements of the hat matrix $\mathbf{H}_M = \mathbf{X}_M (\mathbf{X}_M' \mathbf{X}_M)^{-1} \mathbf{X}_M'$.

Bayesian Information Criterion

The Bayesian information criterion (BIC) is generally defined by

$$BIC = -2 \cdot l(\hat{\boldsymbol{\beta}}_M, \hat{\sigma}^2) + \log(n) (|M| + 1);$$

see in Appendix B, p. 676. The BIC multiplied by 1/2 is also known as Schwartz criterion. Assuming Gaussian errors, we obtain

$$BIC = n \cdot \log(\hat{\sigma}^2) + \log(n) (|M| + 1).$$

The form of the BIC is similar to that of the AIC, and again smaller values indicate a better model fit. Note, however, that the BIC and AIC are motivated in a very

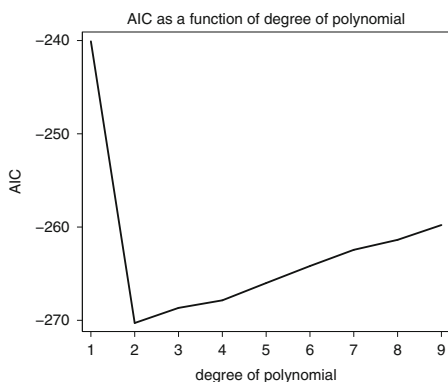


Fig. 3.19 AIC as a function of the polynomial degree for the simulated data of Example 3.16 (p. 139)

different way. From a practical point of view, the main difference is that the BIC penalizes complex models much more than the AIC. Thus, the resulting “best” models are typically more parsimonious when using the BIC rather than the AIC.

Example 3.18 Polynomial Regression—Model Choice with AIC

Figure 3.19 plots AIC for the simulated data from Example 3.16 (p. 139) as a function of the polynomial degree. AIC obtains a minimum for $l = 2$ resulting in a reasonable model, even though we do not obtain the polynomial order of the true model with $l = 3$. △

3.4.3 Practical Use of Model Choice Criteria

We can use the various model choice criteria to select the most promising models from candidate models. We recommend the following approach:

- On the basis of scientific knowledge, perhaps gained from previous research, we obtain a preselection of potential models. The models can differ in the number of variables but also in model type (e.g., linear versus nonlinear). The total number of potential models should be as small as possible.
- All potential models can now be assessed with the aid of one of the various model choice criteria. The summary of the results should not be restricted to the “best” model. As a rule, there are a number of competitive models having approximately equal model fit, differing only in small aspects from each other. These differences cause some uncertainty regarding the conclusions.

Unfortunately, this method is not always practical, since the number of regressor variables and modeling variants can be very large in many applications. In short, the calculation of all models is often impossible. In this case, we can use the following partially heuristic methods:

- *Complete Model Selection (All-Subset-Selection)*: In case that the number of covariates is smaller than about 40, we can determine the best model (in the sense of a model choice criterion) with the “leaps and bounds” algorithm introduced

by [Furnival and Wilson \(1974\)](#). The algorithm returns the optimal model thereby avoiding the computation of all models. An implementation can be found in the software SAS and also in the R package `leaps`.

- *Forward Selection*: Based on a starting model, forward selection includes one additional variable in every iteration of the algorithm. The variable which offers the greatest reduction of a preselected model choice criteria (C_p , AIC, CV, BIC) is chosen. The algorithm terminates if no further reduction is possible.
- *Backward Elimination*: Backward elimination starts with the full model containing all potential covariates. Subsequently, in every iteration, the covariate which provides the greatest reduction of the model choice criteria (C_p , AIC, CV, BIC) is eliminated from the model. The algorithm terminates if no further reduction is possible.
- *Stepwise Selection*: Stepwise selection is a combination of forward selection and backward elimination. In every iteration of the algorithm, the inclusion and the deletion of a variable are both possible.

Generally speaking, forward, backward, and stepwise selection do not offer the best model in the sense of the model choice criteria. However, typically a model that is close to the best model is selected.

Unfortunately, the available software for model choice is often problematic. A main problem is that dummy variables of multi-categorical variables are, in general, neither jointly included nor removed from the model. Instead it can happen that only a subset of the dummies of a categorical variable appears in the model. This causes an indirect change of the reference category and the interpretation of the regression coefficients. Moreover, other difficulties can occur, e.g., methods for hierarchical terms are frequently missing. For example, if we use polynomials for modeling nonlinear covariate effects, it can happen that the selected model has a squared and cubic term, but not a linear one. Often it is impossible to exclude certain terms from selection and force them into the model.

The listed procedures should not be confounded with an algorithm proposed by Efroymson in the 1960s, even though the approach is similar. In contrast to what has been proposed above, the Efroymson algorithm includes or excludes those variables in/from the model, which have the highest or lowest t -value.

The procedure terminates when no variable that potentially needs to be included has a p -value of less than a previously fixed maximal p -value (e.g., 0.05) and when no variable that needs to be excluded has a p -value greater than a minimal p -value (e.g., 0.1). This automatic procedure, which is implemented in all major statistical software packages, is often viewed as obscure among statisticians due to the following two reasons:

- Forward, backward, and stepwise selection usually provide different results. This also happens when using a global model choice criterion such as AIC. We can, however, compare the different selected models with the help of the global model choice criterion. When using the Efroymson approach, discrimination between the different models is impossible.
- The repetitive use of the t -test statistic, to assess whether or not a regression coefficient is different from zero, suggests exact tests. However, the t -test statistic

does not follow a t-distribution under the null hypothesis, since during the selection process we do not test an arbitrary variable but rather the variable with the *maximal* t-value.

Example 3.19 Prices of Used Cars—Model Choice



We illustrate the approaches for model choice using data from the sales price of pre-owned VW Golf automobiles. Our goal is to model the relationship between the sales price in 1,000 Euro (variable *price*) and the five explanatory variables “age of the car in months” (*age*), “kilometer reading in 1,000 km” (*kilometer*), “number of months until the next appointment with the Technical Inspection Agency” (*TIA*), “ABS brake yes/no” (*extras1*), and “sunroof yes/no” (*extras2*). Examinations of this kind play an important role in the context of hedonic price indices. In contrast to common price indices, hedonic price indices take quality differences of a product into consideration. This is especially of importance when dealing with product groups that undergo fast technological changes.

Figure 3.20 provides scatter plots and box plots to get a first impression about the relationships between the response variable and the various explanatory variables. The plots suggest the following effects: We can assume a linear or monotonically decreasing nonlinear effect for the variables *age* and *kilometer*, which could be appropriately modeled using (orthogonal) polynomials of degree three or less. The variable *TIA* appears to either have no effect or a very weak linear effect on the average sales price. Cars with ABS (*extras1*) seem to be slightly more expensive than cars without ABS; the effect, however, remains arguable. We can attest to no difference in the average sales price for models with or without sunroof (*extras2*). All in all, there seems to be a relationship with *age* and the kilometer reading. The effects of the remaining variables appear doubtful.

Based on these considerations, we first examine eight regression models (see Table 3.5), which do not differ in the modeling of the variables *age* and *kilometer*. For the remaining three regressors, all possible model combinations will be tested. Using the AIC criterion, we obtain the first model in Table 3.5 as the preliminary best model. Figure 3.21 displays the AIC values for the eight models under consideration. In addition, the AIC for a ninth model based on automatic variable selection is provided; see below.

Since only five explanatory variables are available, we can even determine the AIC best model with the help of the “leaps and bounds” algorithm. This model attains an AIC value of 389.35. It differs from the current “best model,” in that it only makes use of polynomials of second degree (not third) for the variables *age* and *kilometer*. Figure 3.21 shows that the obtained AIC for this ninth model is considerably smaller than the best AIC value of all the models that we examined so far. Table 3.6 shows the estimation output for model 9. Figure 3.22 provides additional visualization of the estimated effects for *age* and *kilometer* reading. Table 3.6 reveals that all regression coefficients differ significantly from zero at the 5 % level, and only one quadratic term is not significant at the 1 % level. Note that a model, which is selected according to a global goodness-of-fit criterion, can contain nonsignificant coefficients. This is only remarkable at first glance. All selection criteria are global measures, which, for example, minimize the prediction error. Statistical tests can rather be seen as local measures, which in addition are constructed asymmetrically. This means that the type I error is controlled and kept small so that we favor the null hypothesis over the alternative hypothesis.

△

We finally point out that there are alternative ways for variable selection than the optimization of a model choice criteria such as AIC or the cross validation score. In fact there is vital current research in variable selection for linear models. In this book we will present some of the most interesting approaches in Chap. 4. More specifically, we will discuss:

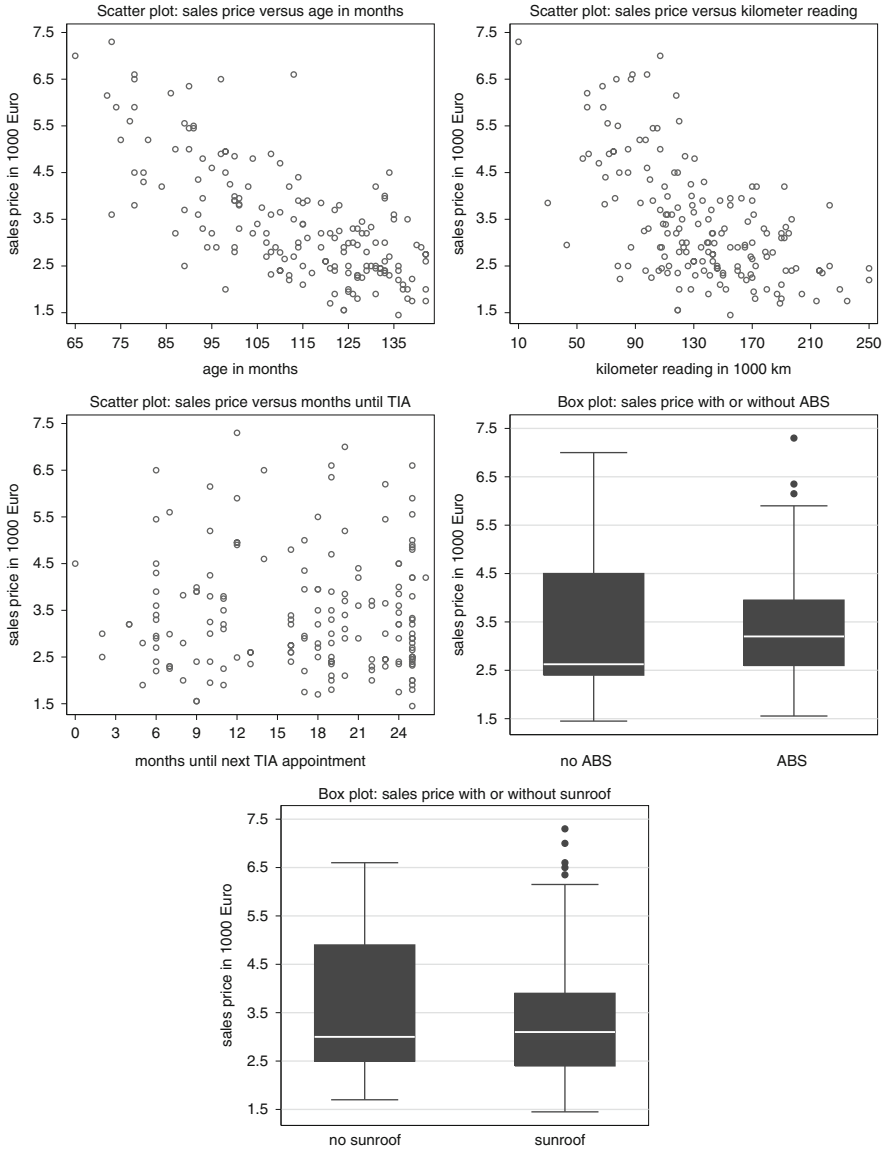


Fig. 3.20 Prices of used cars: scatter plots between sales prices and the continuous covariates *age*, *kilometer*, and *TIA*. Box plots of sales prices stratified according to the values of the binary variables *extras1* and *extras2*

- *Regularization techniques for the least squares estimator; in particular the LASSO:* Regularization of the least squares estimator allows to set some of the regression coefficients zero, so that the corresponding covariates are essentially removed from the model; see Sect. 4.2 for details.

Table 3.5 Prices of used cars: potential models

Model	<i>kilometer</i> degree 3	<i>age</i> degree 3	<i>extras1</i> linear	<i>extras2</i>	<i>TIA</i>	AIC
1	x	x				393.234 (1)
2	x	x	x			394.566 (2)
3	x	x		x		395.119 (4)
4	x	x			x	394.973 (3)
5	x	x	x	x		396.481 (6)
6	x	x	x		x	396.143 (5)
7	x	x		x	x	396.881 (7)
8	x	x	x	x	x	398.085 (8)

The values in brackets indicate the rank of the models according to AIC

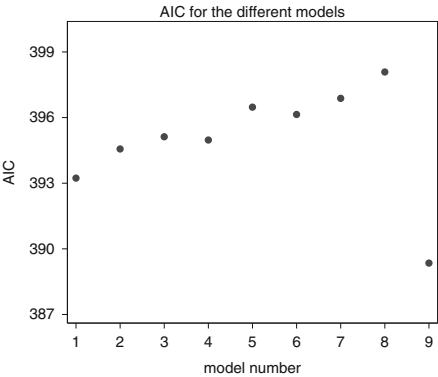


Fig. 3.21 Prices of used cars: AIC values for the potential models

Table 3.6 Prices of used cars: estimation results for the best model according to AIC

Variable	Coefficient	Standard error	t-value	p-value	95 % Confidence interval	
<i>intercept</i>	3.397	0.056	60.220	<0.001	3.285	3.508
<i>ageop1</i>	−0.705	0.061	−11.470	<0.001	−0.826	−0.584
<i>ageop2</i>	0.187	0.057	3.270	0.001	0.074	0.300
<i>kilometerop1</i>	−0.439	0.061	−7.170	<0.001	−0.560	−0.318
<i>kilometerop2</i>	0.141	0.057	2.460	0.015	0.028	0.254

- *Boosting*: Similar to the LASSO, boosting is able to simultaneously estimate the regression coefficients and select relevant variables; see Sect. 4.3.
- *Bayesian variable selection*: The last decades have seen a large number of practical Bayesian approaches for variable selection. We will present one approach with corresponding software in detail and give an overview over alternative approaches; see Sect. 4.4, in particular Sect. 4.4.3.

We will apply the alternative techniques for variable selection in our case study on prices of used cars and compare the results to those obtained by optimizing AIC in the previous example; refer to Examples 4.11 (LASSO), 4.12 (boosting), and 4.15, 4.16 (Bayesian variable selection).

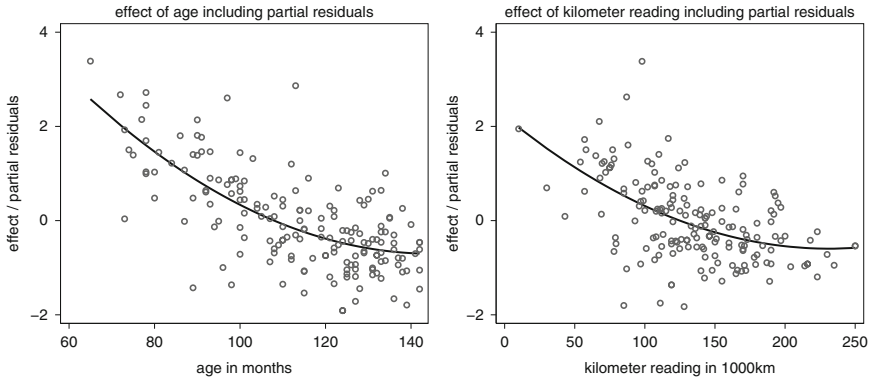


Fig. 3.22 Prices of used cars: model 9 based on all-subset selection, effects of age, and kilometer reading including partial residuals

3.4.4 Model Diagnosis

Having selected a working “best” model, the next step is to assess the validity of the model using diagnostic tools, in search of an improved model. In principle, model diagnosis pursues the following goals:

- *Examination of the model assumptions:* No model is correct. Nevertheless, the assumptions on which a model is based should be met at least approximatively. In the classical linear model, we have to evaluate the assumptions of homoscedastic, uncorrelated, and possibly normally distributed errors, as well as the linearity of the predictors.
- *Outlier detection:* In some situations, the results are heavily influenced by a few “unusual” observations. These observations should be identified and their effect on the estimates quantified.
- *Collinearity analysis:* Theoretical results have shown that highly correlated covariates can have a strong influence on the stability of the estimates; see pages 115ff. or Box 3.9 on p. 119. Thus it is important to examine the correlation structure of the covariates in the model.

In the following sections and Sect. 4.1, we describe a selection of tools useful for model diagnosis.

Examination of Model Assumptions

Homoscedasticity: The examination of the assumption of homoscedastic error variances will be outlined in Sect. 4.1.3 (p. 182). Important tools include residual plots and tests for heteroscedasticity. When heteroscedasticity occurs, transformation of the response variable or the use of weighted regression are possible remedies; see Sect. 4.1.3 in the next chapter.

Uncorrelated Errors: We can detect correlated errors through residual plots over time or through the use of statistical tests; see Sect. 4.1.4 (p. 191) of the

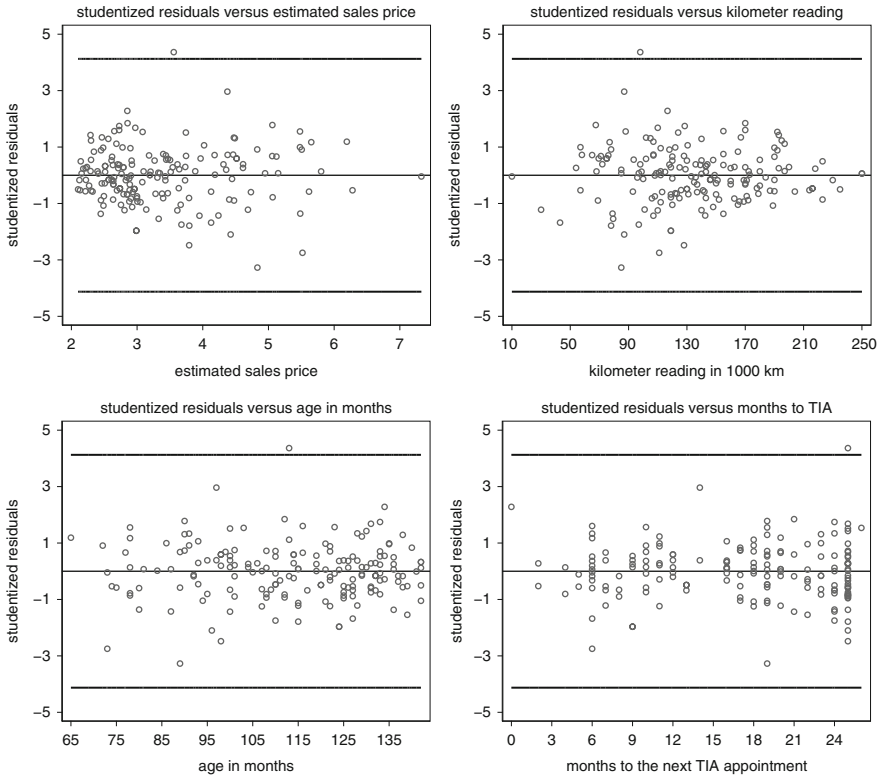


Fig. 3.23 Prices of used cars: studentized residuals against $\widehat{price}$ (top left) and against the continuous covariates. The horizontal lines mark the critical values of the test for outliers to be discussed in Sect. 3.4.4 (p. 160)

next chapter. Since correlated errors indicate misspecification (especially through omitted explanatory variables and unnoticed nonlinearity), we should first examine whether or not model specification can be improved. If not, we can use one of the estimation procedures discussed in Sect. 4.1.4 (p. 191).

Linearity: We can eventually discover undetected nonlinear effects of covariates with the help of scatter plots of (standardized or studentized) residuals as a function of estimated values. Moreover, plots of partial residuals $\hat{\varepsilon}_{x_j,i} = \hat{\varepsilon}_i + \hat{\beta}_j x_{ij}$ against covariate x_{ij} can be useful.

Assumption of Normality: We can test distributional assumptions with the quantile–quantile plot (Q–Q plot). In Q–Q plots, the empirical quantiles are compared to the quantiles of the theoretical distribution. If the data follows the distribution, the points should closely follow the 45° bisecting line.

Example 3.20 Price of Used Cars—Model Diagnosis

Figure 3.23 plots the studentized residuals versus the estimated price and the continuous covariates for the model estimated in Example 3.19. There is no evidence of any additional



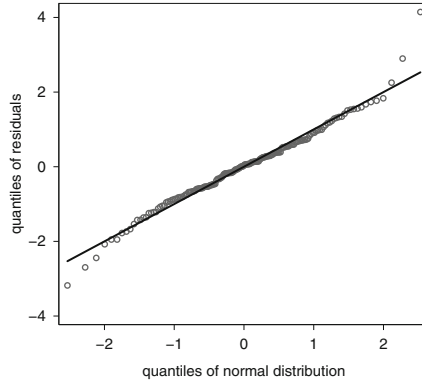


Fig. 3.24 Prices of used cars: Q–Q plot of standardized residuals

nonlinear effects. Nevertheless, evidence of (mild) heteroscedastic errors exists. The variability of studentized residuals decreases as kilometer reading increases. Also note the increasing variability of residuals for VW Golf models that have 18 months or more until their next TIA checkup. It appears that the variable TIA does not affect the conditional mean of sales prices but rather the conditional variance. Figure 3.24 shows the Q–Q plot for the standardized residuals. Significant deviations from the normal distribution are visible only in the extreme left and right tails.

△

Collinearity Analysis

We first illustrate the goals of collinearity analysis with the most extreme situation. Consider the model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon,$$

with covariates x_1 , x_2 , and x_3 . We first assume that there is perfect linear dependence $x_1 = a x_2$ between x_1 and x_2 . The variable x_1 could represent, for example, weight in kilogram and x_2 weight in tons, that is, x_1 and x_2 only differ in the measure unit. In this extreme case, we cannot uniquely estimate the regression coefficients β_1 and β_2 belonging to x_1 and x_2 , since

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon = \beta_0 + (\beta_1 a + \beta_2) x_2 + \beta_3 x_3 + \varepsilon.$$

In practice, the covariates typically do not show such an exact linear dependence, but they can be highly correlated, which is called (multi)collinearity. As we have seen in Sect. 3.2.3 on p. 116, highly correlated covariates cause imprecise estimation of the regression parameters, as again seen through the variance formula for $\hat{\beta}_j$,

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{(1 - R_j^2) \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}.$$

The greater the linear dependence of a covariate x_j with the other explanatory variables (measured through the coefficient of determination R_j^2), the greater is the variance $\text{Var}(\hat{\beta}_j)$. In the extreme case of $R_j^2 \rightarrow 1$, the variance explodes towards infinity. The variance formula also provides a diagnostic tool for measuring the degree of collinearity, i.e., the *variance inflation factor*:

$$\text{VIF}_j = \frac{1}{1 - R_j^2}.$$

The variance inflation factor quantifies the factor increase of the variance of $\hat{\beta}_j$ due to the linear dependence of x_j with the other regressors. The greater the correlation between x_j and the other covariates, the greater is R_j^2 , and thus the greater is VIF_j . As a general benchmark, we say that a serious collinearity problem exists when the variance inflation factor is greater than 10, i.e., $\text{VIF}_j > 10$.

Example 3.21 Prices of Used Cars—Collinearity



For the regression model of Example 3.19, the variance inflation factors range between 1.03 and 1.19. In other words, the variances of the estimated coefficients increase, at most, by factor 1.19, when compared to an “ideal” (orthogonal) setting. Thus, there is no evidence for a serious collinearity problem.

△

The literature offers a variety of proposals on how to deal with strong collinearity:

- *Omission of some affected covariates:* This method is probably both the most practiced and the most criticized in literature. However, in many cases there are different measures for the same purpose. Consider, for instance, the various measures for the nutritional status of the mother in our data example on malnutrition in Zambia. The original data set offers the body mass index, as well as the Rohrer Index as measures of nutritional status. Although constructed differently, these two variables are highly correlated, with a correlation coefficient greater than 0.95. In this case, as well as in other similar cases, we are forced to decide which index to use and which to exclude from further analysis.
- *Construct a single combined (easily interpretable) variable from the variables in question:* Of particular interest are linear combinations of the variables or simple summaries, such as the minimum or maximum of the variables. Recall the supermarket scanner data, which examines different marketing measures and relates them to the sale of a particular coffee brand (see Example 3.2 on p. 86). When modeling the sales of a coffee brand, in addition to its own price, the prices of other competitors are important covariates. For some competitors, several brands exist, which only differ from each other by package size. The prices for these brands often change at the same time and proportional to package size. Thus, strong collinearity results if we include all brands as separate effects into the model. We can avoid the collinearity problem by computing, for example, the average price (a specific linear combination) of the competitive brands, which is then included in the model in place of the individual brands. Another possibility is to use a simple summary, e.g., the lowest price (minimum).

- *Ridge Regression:* The ridge estimator is an alternative to the least squares estimator and is defined as

$$\hat{\beta} = (X'X + \lambda I)^{-1} X'y,$$

where $\lambda \geq 0$ is an appropriately chosen tuning-parameter. The ridge estimator, unlike the least squares estimator, is biased. The literature suggests the use of the ridge estimator, especially in the presence of collinearity problems. The addition of λI to $X'X$ leads to a regularized and invertible matrix, even if $X'X$ is near singular due to severe collinearity. Thus, the components of the ridge estimator often have a smaller MSE when compared to the corresponding components of the least squares estimator. The least squares estimator is only optimal in the class of linear and unbiased estimators. If we also allow biased estimators, it is possible that other estimators are better in terms of the MSE than the least squares estimators. Even though the ridge estimator has been highly researched and elaborated on in many publications, until recently it was rarely used in practice. One reason is that even when using the ridge estimator, separating the covariate effects is impossible with severe collinearity. However, ridge and related estimators have recently received increasing attention in order to cope with situations that have a huge number of possible regressors in combination with a relatively small sample size, i.e., large p , small n problems. Such problems occur frequently in biostatistics when analyzing gene expression and related data. More details on ridge and related estimators are given in Sect. 4.2.

Estimators with a similar structure to that of the ridge estimator are also widely used in nonparametric regression; see Chap. 8.

- *Principal component regression:* This variant does not directly include all available covariates in the model but rather appropriate linear combinations of covariates. Throughout, continuous covariates are assumed. The basis of principal component regression is the spectral analysis of the matrix $X'X$ (see Theorem A.25 on p. 633), i.e.,

$$X'X = P \text{diag}(\lambda_1, \dots, \lambda_p) P'.$$

The matrix P consists of the eigenvectors p_j or the principal components of $X'X$. The first principal component p_1 has the characteristic that the linear combination $z_1 = Xp_1$ has the largest variance among all linear combinations of the columns of X . The second principal component p_2 determines the linear combination $z_2 = Xp_2$, which has the second highest variance among all linear combinations that are also orthogonal to z_1 . Accordingly, the linear combination $z_j = Xp_j$ of the j th principal component has the highest variance relative to the linear combinations which are orthogonal to all of the first $j - 1$ principle components. Instead of the original regressor variables, we use the derived linear combinations $z_1, z_2, \dots$ as covariates in the principal component regression. Typically only the first $q \leq p$ of the p linear combinations are considered and sufficient for modeling. The remaining ones are neglected due to a lack of interpretability or due to low variance.



An application of principal component regression arises in the context of modeling determinants of malnutrition. Here, a number of variables describing the economical situation of the household is available. In most cases it does not make sense to include all economical covariates in the model. Based on a principal component analysis, easily interpretable linear combinations as measurements for the economical status are included instead. In almost all cases, only the linear combination with the highest variance based on the first principle component is used.



Outlier and Influence Analysis

In addition to checking model assumptions and collinearity analysis, another important aspect of model diagnosis is to investigate the influence of individual observations on the estimators. We will see that *outliers* are of particular importance, which we next examine in more detail.

Outliers

There is no exact definition for the term outlier. In this book, outliers are observations which do not adhere to the fitted model. We (informally) define the i th observation as an outlier if the (conditional) mean of y_i is not $E(y_i | x_i) = x_i' \beta$, as is postulated by the model, but rather shifted by the value d , i.e., $E(y_i | x_i) = x_i' \beta + d$.

We illustrate the outlier problem using the simulated data presented in Fig. 3.25. Panel (a) displays simulated observations (y_i, x_i) together with an estimated regression line (solid line). The solid point in the center of the range of values of x appears to be a severe outlier. If we estimate a new regression line, this time without considering the outlier, the resulting estimate (dotted line) only slightly changes. The estimated standard errors and, thus, also the confidence intervals however do change dramatically. Using all observations, we obtain 0.241 as estimated standard error for $\hat{\beta}_1$, whereas without the outlier, we obtain the value 0.069. Accordingly, we obtain the confidence interval [1.53, 2.50] for β_1 based on all observations and [1.84, 2.11] without considering the outlier. Even though in this case the outlier only slightly changes the estimated regression line, statistical inference is influenced considerably. In comparison to panel (a), panel (c) shows a slightly modified situation, where in this case, the outlier is not in the center but rather on the boundary of the range of values. As such, the outlier has now an effect on both the estimated standard errors *and* the estimated regression line. If multiple outliers appear as in panel (e), the impact is even greater.

Summarizing, we can state: Outliers can have a significant effect on estimation and inference. Further, the influence becomes more significant on the boundary of the covariate range and with multiple outliers. Thus, the following questions arise: First, how can we detect outliers? Secondly, what should we do when outliers are detected? The literature offers a multitude of possible answers to both questions. We first address the question how to detect outliers.

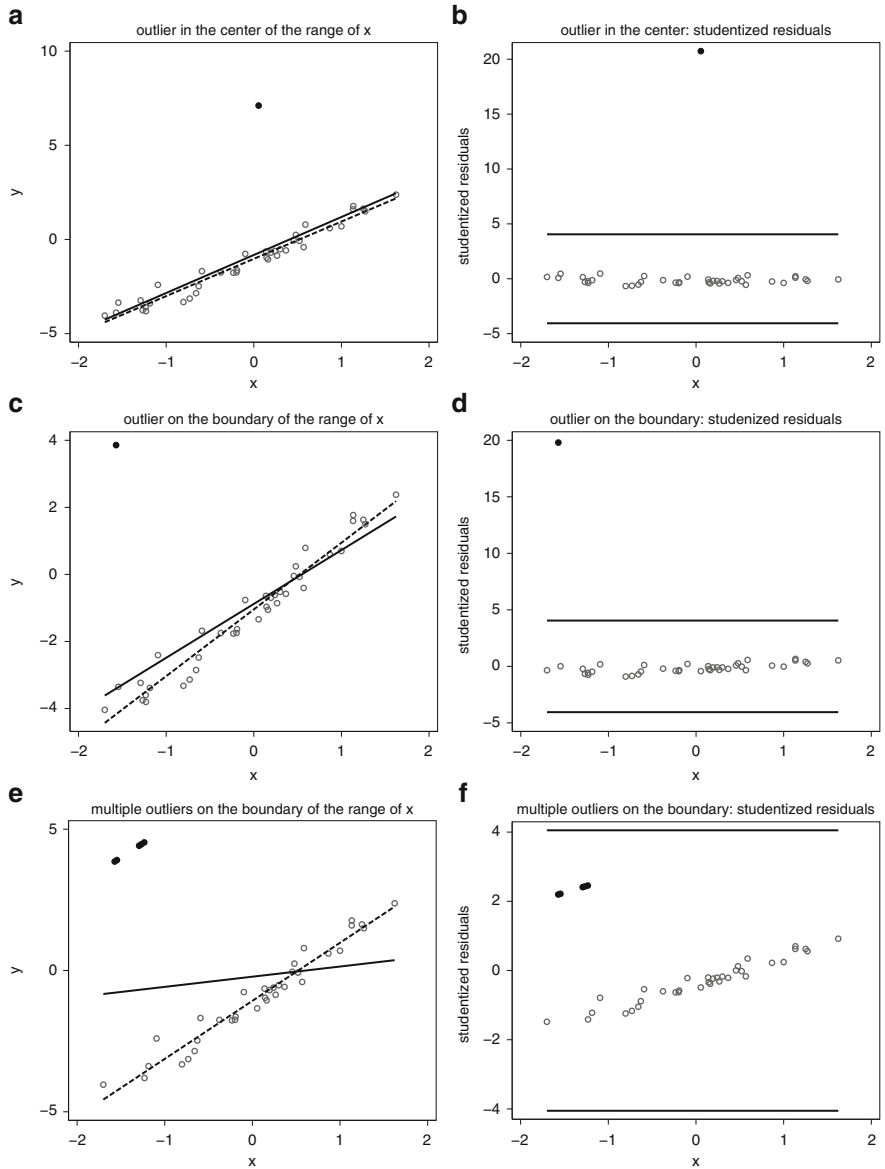


Fig. 3.25 Illustration of the outlier problem: The *left panels* show scatter plots between y and x together with the estimated regression line (*solid line*) and estimated regression line if the outliers are not considered (*dashed line*). The *right panels* display studentized residuals against x . Outliers are marked by *solid points*

A good starting point to detect outliers seems to be the search for large residuals. The discovery of outliers is, however, complicated by the fact that they can have a considerable effect on estimation. Note that estimates are “pulled” towards the outliers, such that the corresponding residuals can be relatively small, especially at the boundary of the covariate range. Therefore, if we wish to examine whether or not the i th observation is an outlier, it should be *excluded from estimation*.

In Sect. 3.10 on p. 123, we defined the “leave-one-out” residuals:

$$\hat{\varepsilon}_{(i)} = y_i - \hat{y}_{(i)} = y_i - \mathbf{x}'_i (\mathbf{X}'_{(i)} \mathbf{X}_{(i)})^{-1} \mathbf{X}'_{(i)} \mathbf{y}_{(i)}.$$

These residuals are based on estimates which do not consider the i th observation. With standardization, we obtained the studentized residuals:

$$r_i^* = \frac{\hat{\varepsilon}_{(i)}}{\hat{\sigma}_{(i)} (1 + \mathbf{x}'_i (\mathbf{X}'_{(i)} \mathbf{X}_{(i)})^{-1} \mathbf{x}_i)^{1/2}},$$

which follow a t-distribution with $n - p - 1$ degree of freedom when the model is correctly specified. Observations with large studentized residuals are thus potential outlier candidates. Since we know the distribution of studentized residuals in a correctly specified model, we can formally test whether or not the i th observation y_i is an outlier. In order to do so, we have to compare the i th studentized residual r_i^* with the $\alpha/2$ or the $1 - \alpha/2$ quantile of the t-distribution with $n - p - 1$ degree of freedom, for a given significance level α . If the residual is smaller than the $\alpha/2$ or larger than the $1 - \alpha/2$ quantile, then it is an outlier candidate.

In most cases, we do not want to test a single observation, but we want to test all observations *simultaneously*. In order to obtain a correct overall significance level α , we have to adjust the significance level for each observation. A straightforward approach is the Bonferroni correction, which uses the level α/n in each test. The Bonferroni correction is relatively conservative, which means that in general (especially with a large sample size) we tend not to discover all outliers even when they exist.

Figure 3.25b,d,f shows studentized residuals as a function of x for the simulated data sets. The plotted horizontal lines mark the 0.5 % and 99.5 % quantiles of the t-distribution with $n - p - 1 = 40 - 2 - 1 = 37$ degrees of freedom. Observations that are below or above the two lines are potential outlier candidates. It appears that the two individual outliers are identified, one in the center and the other on the boundary of the covariate range. The multiple outliers in panel (e) are, however, not identified. In general, it can be very difficult to discover multiple outliers. In this case, the visualization of the studentized residuals in panel (f) does offer us some help to notice the unusual observations.

Example 3.22 Prices of Used Cars—Outliers

Figure 3.23 shows the studentized residuals as a function of the fitted values and the covariates in the regression model derived from Example 3.19 (p. 152). The 0.5 % and 99.5 % quantiles of the t-distribution with $n - p - 1 = 172 - 6 - 1 = 165$ degrees of freedom (horizontal lines) are also plotted. Based on the overall significance level of $\alpha = 0.01$, we obtain one observation above the line with a studentized residual of $r_i^* = 4.36$.

△



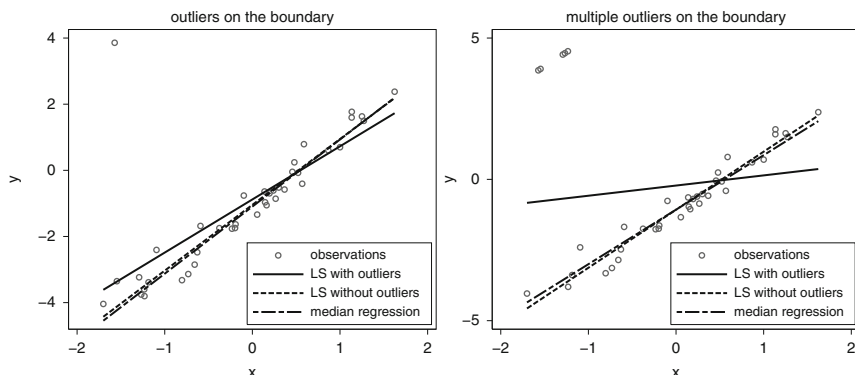


Fig. 3.26 Robust regression: estimated regression line based on various estimation techniques (least squares estimator based on all observations, least squares estimator without considering the outliers and median regression). *Left*: outlier on the border of the covariate range. *Right*: multiple outliers on the border of the covariate range

In practice, outlier candidates are often excluded from the analysis without further comment. This procedure is often not the best alternative. Instead, we recommend the following:

- *Exclusion of data errors*: Outliers can indicate data errors, for example, incorrect data coding. This source of error should first be eliminated.
- *Search for an explanation of outliers*: In some applications, the outliers of a model are the most interesting observations. A case in point is the satellite Nimbus 7 operated by the NASA to provide information about the atmosphere. In the year 1985, the satellite had been in use for several years and scientists observed an extremely large decrease of the ozone concentration in the atmosphere, the ozone hole was discovered. In this case, an explanation for the unusual observations existed.
- *Description of the differences in results*: Instead of excluding outliers from the analysis, it is possible to estimate the model with and without outliers. In this way, we can describe the emerged model differences and we do not lose the outlier information.
- *Robust regression*: Since the 1960s, robust methods have become increasingly more important in statistics. These methods are less sensitive to outliers, i.e., estimation results are not (or only slightly) influenced by outliers. The first robust method is median regression proposed by Boscovich already in 1760 (see Sect. 3.2.1). Median regression minimizes the sum of the absolute deviations instead of the sum of the squared deviations. Such an approach is much more robust against outliers than the least squares method; see Fig. 3.26, which compares several estimation methods in the presence of one or multiple outliers at the boundary of the covariate range. The least squares results are only insensitive to the outliers, in the cases that they are excluded from estimation. When using all observations, the estimated least squares regression lines are “pulled” towards the

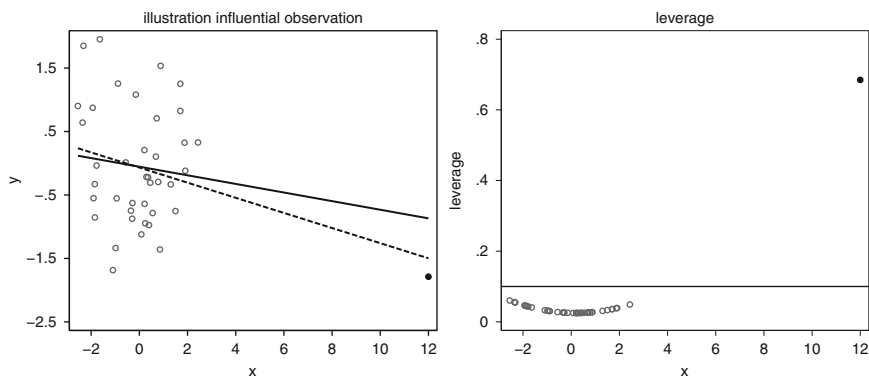


Fig. 3.27 The *left panel* shows a scatter plot between y and x including an estimated regression line (*dashed line*) and an estimated regression line if the influential observation is not considered (*solid line*). The *right panel* shows the leverage depending on x

outliers. However, outliers do not affect the median regression, even when they are not excluded from estimation. Median regression then has the advantage that we do not need to detect potential outliers, since estimation is unaffected. The use of median regression does become problematic when we have a relatively large amount of outliers. In such cases, we have to use other specialized methods, which, however, would go beyond the scope of this book; see, for example, [Rousseeuw and Leroy \(2003\)](#). Note that median regression is a special case of more general *quantile regression* which has recently become quite popular. More on quantile regression can be found in Chap. 10.

We may wonder why the least squares method is still used more often than the method of median regression, especially in light of the fact that median regression is much less sensitive to outliers. The answer is that the least squares estimator has much better estimation properties when no outliers exist. When the data follow a normal distribution, median regression needs about 50 % more data in order to obtain the same estimation precision compared to the least squares method. Nevertheless, within the scope of diagnostics, we also recommend to use robust methods (when available) to examine the sensitivity or robustness of the obtained least squares estimators.

Influence Analysis

Influence analysis focuses on measuring the impact of specific observations on estimation. The main objective is to find observations which have a large influence on $\hat{\beta}$ and, thus, $\hat{y}$. Estimation can drastically change, when such observations are removed. In many cases, influential observations are outliers, as informally defined above. In extreme cases, a single observation can drive the entire estimated relationship or the elimination of a certain observation destroys the relationship. Figure 3.27 (left panel) illustrates such a case using simulated data.

The literature offers numerous suggestions on how to detect highly influential observations including the following:

- *Leverage*: The diagonal elements h_{ii} of the hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ measure the leverage of the i th data point. The leverage ranges from $1/n$ to 1; see Box 3.5 on p. 108. A large leverage (close to 1) implies two things. First, $\text{Var}(\hat{\epsilon}_i) = \sigma^2(1 - h_{ii})$ is small, or even near zero when leverage is close to one. In the case of a single covariate, the regression line nearly passes through the point (y_i, x_i) , regardless of any other observation. Thus an observation with high leverage has a considerable influence on the estimation results. These considerations can easily be extended to observations with more covariates, where now, in the extreme case, the regression hyperplane nearly intersects the point $(y_i, \mathbf{x}_i)$.

Secondly, large leverage values indicate some unusual covariates values $\mathbf{x}_i$. This property is best seen in the case of only one covariate x_i as then the leverage is given by

$$h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_j (x_j - \bar{x})^2}.$$

The leverage h_{ii} increases, the further the observation x_i is from the average $\bar{x}$. In this sense, leverage values can also be understood as outliers in x -direction. In general, we have

$$h_{ii} = \frac{1}{n} + (\mathbf{x}_i - \bar{\mathbf{x}})'(\tilde{\mathbf{X}}'\tilde{\mathbf{X}})^{-1}(\mathbf{x}_i - \bar{\mathbf{x}})$$

using the design matrix of centered covariates

$$\tilde{\mathbf{X}} = \begin{pmatrix} (x_{11} - \bar{x}_1) & \cdots & (x_{1k} - \bar{x}_k) \\ (x_{21} - \bar{x}_1) & \cdots & (x_{2k} - \bar{x}_k) \\ \vdots & \cdots & \vdots \\ (x_{n1} - \bar{x}_1) & \cdots & (x_{nk} - \bar{x}_k) \end{pmatrix}.$$

Ideally, the leverages h_{ii} are uniformly distributed. As a rule of thumb, observations with $h_{ii} > 2p/n$ (twice the average) should be examined more closely. We should keep in mind, however, that large leverages do not necessarily lead to problems.

- *Cook's Distance*: Denote by $\hat{\mathbf{y}}_{(i)}$ the estimator for $E(\mathbf{y})$ that uses all observations with exception of the i th observation. An obvious measure for the difference between $\hat{\mathbf{y}}$ based on all observations and $\hat{\mathbf{y}}_{(i)}$ is the Euclidean distance between the estimators. More precisely, Cook's distance is defined by

$$D_i = \frac{(\hat{\mathbf{y}}_{(i)} - \hat{\mathbf{y}})'(\hat{\mathbf{y}}_{(i)} - \hat{\mathbf{y}})}{p \cdot \hat{\sigma}^2},$$

where the numerator is standardized by the estimated variance $\hat{\sigma}^2$. As a rule of thumb, observations with $D_i > 0.5$ are worthy of attention, and observations with $D_i > 1$ should always be examined.

For our simulated data, the right panel of Fig. 3.27 shows the leverages h_{ii} as a function of x_i . The horizontal line marks the rule of thumb $h_{ii} > 2p/n = 2 \cdot 2/40 = 0.1$ for influential observations. The most influential observation is with $x = 12$ having $h_{ii} = 0.6$ and is the only leverage larger than the rule of thumb. Cook's distance with $D_i = 2.55$ is extremely large for this observation, indicating extreme influence of the observation. It is also the only observation with $D_i > 0.5$.

Example 3.23 Prices of Used Cars—Influence Analysis



Cook's distances are all low for the regression model outlined in Example 3.19 (p. 152). The largest Cook's distance is 0.12, and no observation has a D larger than 0.5. Approximately ten observations have leverages larger than the cutoff $h_{ii} > 2p/n = 0.058$. For the most part, these are observations with a rather small (*kilometer* < 50) or large kilometer reading (*kilometer* > 230). The largest leverage is $h_{ii} = 0.23$. It is therefore reasonable to examine the effect on estimation by removing the observations associated with extremely low or large kilometer readings and refitting; see the next Example 3.24.

△

Alternative Modeling Approaches Due to Model Diagnosis

Model diagnosis often provides evidence for alternative modeling approaches to be investigated. When estimating alternative models, the main goal is, on the one hand, to find even more adequate models and, on the other hand, to evaluate model stability. In most cases, we do not find a single model that clearly outperforms all the others. On the contrary, we typically find a few competing models with almost equal fit. It is important to extract the common features of these models, as well as refer to their differences. Typically, the models will be different only with respect to some aspects which then cause uncertainty regarding our conclusions.

The following alternative modeling variants are possible:

- Residual plots can particularly provide useful information about *alternative modeling of the effect of one or more covariates*.
- We can *reestimate the model after removing conspicuous observations* having extremely large residuals, leverages, and/or Cook's distance.
- We can reestimate the model using *alternative approaches*, especially robust regression.
- Sometimes model diagnosis provides evidence for transforming the response variable. Especially if the response is strictly positive with a large range, then a logarithmic transformation of the response might be useful.
- If there is evidence of heteroscedastic errors, an alternative is weighted regression, as developed in Sect. 4.1.3 of the next chapter.

Example 3.24 Prices of Used Cars—Alternative Models



The regression model of Example 3.19 (p. 152) has undergone detailed regression diagnostics, as provided in the Examples 3.20–3.23. We conclude the following:

- There is no evidence of misspecification of continuous covariate effects; see Fig. 3.22 (p. 155).
- Figure 3.23 suggests moderately heteroscedastic errors.

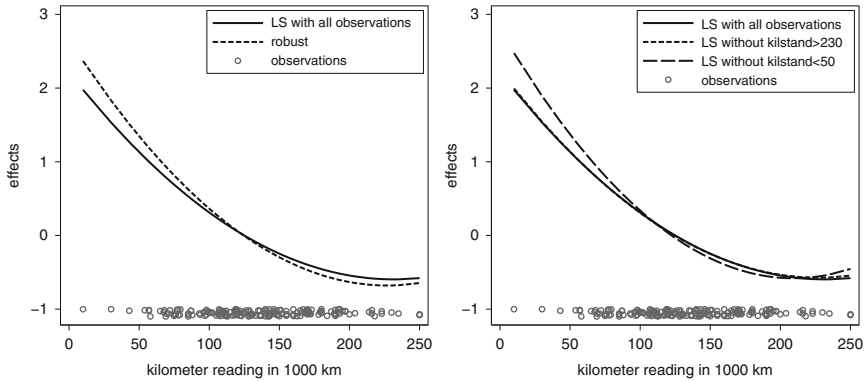


Fig. 3.28 Prices of used cars: effect of the kilometer reading for different modeling alternatives

- The influence analysis suggests unstable effects for Golf models with a kilometer reading smaller than 50 or larger than 230.

Hence, the following alternative model specifications were tested:

- *Weighted regression to cope with heteroscedastic errors*: Since heteroscedastic errors will be treated in Sect. 4.1.3 of Chap. 4, we defer the reader to Example 4.5 for details. For the moment, it suffices to know that the difference between the classical linear model and the heteroscedastic model is marginal. Example 4.5 elaborates this point further.
- *Logarithmic transformation of the response variable price*: Due to the positive and skewed response, we also estimated models with the log price. When using transformed responses, we have to keep in mind that model choice and model diagnoses have to be conducted again. For example, it is possible that outliers in the model with an untransformed response disappear in a model with a log-response, or vice versa. With the transformed response, we obtain a somewhat more parsimonious model, than with an untransformed one, since we can now model the effect of the kilometer reading linearly. Moreover, the model is more stable regarding outliers and influential observations. However, the main conclusions remain the same.
- *Model estimation after removing observations with kilometer < 50 or kilometer > 230, as well as robust regression*: Figure 3.28 shows the impact of different specifications on the effect of the kilometer reading. All models show a monotonically decreasing effect. We observe some uncertainty in the effect only in the domain *kilometer* < 50.

Based on our results, we can now draw the following final conclusions:

- We find weakly nonlinear and monotonically decreasing effects for age and the kilometer reading.
- The other explanatory variables do not have a significant effect (at most a very minor one) on sale's price.
- The models are relatively stable regarding outliers and influential observations. For the untransformed price response, the effect of the kilometer reading for *kilometer* < 50 is somewhat unstable.
- Any apparent heteroscedasticity can be neglected; see Example 4.5 for more details. $\triangle$

3.5 Bibliographic Notes and Proofs

3.5.1 Bibliographic Notes

Since linear models play a very prominent role in statistics, a vast amount of literature exists. [Faraway \(2004\)](#), [Rawlings, Pantula, and Dickey \(2001\)](#), and [Weisberg \(2005\)](#) are among some of the textbooks that we recommend.

The linear model also finds a strong representation in econometrics textbooks, often with a different focus. In particular, exploratory and graphical tools for model diagnosis and specification are rare. Rather, statistical tests for model diagnostics and specification do play an important role. A standard textbook is [Wooldridge \(2006\)](#). A somewhat more challenging presentation is the detailed overview provided by [Greene \(2000\)](#); its fifth edition appeared in 2008. We recommend the second edition (2000), since the subsequent editions appear to be somewhat confusing and excessive in coverage.

Some specific topics, which were only lightly touched upon in this chapter are outlined in monographs. We recommend [Rousseeuw and Leroy \(2003\)](#) for robust regression (but see also Chap. 10) and [Belsley, Kuh, and Welsch \(2003\)](#) for identification of outliers and influential observations. A standard book for variable selection in linear models is [Miller \(2002\)](#). We also did not examine models with a multivariate response. These models are described, for example, in [Anderson \(2003\)](#). Further elaboration on SUR models (seemingly unrelated regression) and simultaneous equation systems can be found in econometric textbooks, for example, [Greene \(2000\)](#).

3.5.2 Proofs

Derivation of the Expected Sum of Squared Residuals (p. 109)

Applying Theorem B.2 (p. 647) and using the properties of the matrix $I - H$ (see Box 3.5 on p. 108), we obtain the expected sum of squared residuals as follows:

$$\begin{aligned}
 E(\hat{\epsilon}'\hat{\epsilon}) &= E(y'(I - H)y) \\
 &= \text{tr}((I - H)\sigma^2 I) + \beta'X'(I - H)X\beta \\
 &= \sigma^2(n - p) + \beta'X'(I - X(X'X)^{-1}X')X\beta \\
 &= \sigma^2(n - p) + \beta'X'X\beta - \beta'X'X(X'X)^{-1}X'X\beta \\
 &= \sigma^2(n - p) + \beta'X'X\beta - \beta'X'X\beta \\
 &= \sigma^2(n - p). \quad \square
 \end{aligned}$$

Proof of Properties 3–5 in Box 3.7 on p. 112

3. Because the first column $\mathbf{x}^0$ of the design matrix is the vector $\mathbf{1} = (1, \dots, 1)'$, property 3 follows from the orthogonality property 2:

$$0 = (\mathbf{x}^0)' \hat{\boldsymbol{\varepsilon}} = \mathbf{1}' \hat{\boldsymbol{\varepsilon}} = \sum_{i=1}^n \hat{\varepsilon}_i.$$

4. Property 3 implies

$$\sum_{i=1}^n \hat{y}_i = \sum_{i=1}^n (y_i - \hat{\varepsilon}_i) = \sum_{i=1}^n y_i - \sum_{i=1}^n \hat{\varepsilon}_i = \sum_{i=1}^n y_i.$$

5. We have

$$\begin{aligned} \bar{y} &= \frac{1}{n} \sum_{i=1}^n y_i = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i + \hat{\varepsilon}_i) = \frac{1}{n} \sum_{i=1}^n (\hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \dots + \hat{\beta}_k x_{ik} + \hat{\varepsilon}_i) \\ &= \frac{1}{n} \sum_{i=1}^n (\hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \dots + \hat{\beta}_k x_{ik}) = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}_1 + \dots + \hat{\beta}_k \bar{x}_k. \end{aligned}$$

Note that the proof of these properties essentially relies on the fact that the model contains an intercept term. Therefore, these properties are only valid in models with an intercept. $\square$

Proof of the Variance Decomposition Formula (3.19) on p. 112

For the derivation of the formula for variance decomposition we make use of the special $n \times n$ -matrix

$$\mathbf{C} = \mathbf{I} - \frac{1}{n} \mathbf{1}\mathbf{1}'.$$

Clearly, $\mathbf{C}$ is symmetric and idempotent. The matrix has several remarkable properties. For example, multiplying $\mathbf{C}$ with a vector $\mathbf{a} \in \mathbb{R}^n$, we obtain

$$\mathbf{C}\mathbf{a} = \begin{pmatrix} a_1 - \bar{a} \\ \vdots \\ a_n - \bar{a} \end{pmatrix}. \quad (3.31)$$

Thus, multiplication with $\mathbf{C}$ centers the vector $\mathbf{a}$. For the quadratic form $\mathbf{a}'\mathbf{C}\mathbf{a}$, we obtain

$$\mathbf{a}'\mathbf{C}\mathbf{a} = \sum_{i=1}^n (a_i - \bar{a})^2. \quad (3.32)$$

Further properties of C are given in Definition A.12 (p. 625). For deriving the variance decomposition formula, however, we only need properties (3.31) and (3.32).

We first multiply the identity $y = \hat{y} + \hat{e}$ with C from the left and obtain

$$C y = C \hat{y} + C \hat{e}.$$

Using Eq. (3.31) and property 3 on p. 112 gives $C \hat{e} = \hat{e}$. This implies

$$C y = C \hat{y} + \hat{e} \quad \text{and} \quad y' C = \hat{y}' C + \hat{e}'.$$

Thus, we get

$$\begin{aligned} y' C C y &= (\hat{y}' C + \hat{e}') (C \hat{y} + \hat{e}) \\ &= \hat{y}' C C \hat{y} + \hat{y}' C \hat{e} + \hat{e}' C \hat{y} + \hat{e}' \hat{e} \\ &= \hat{y}' C \hat{y} + \hat{y}' \hat{e} + \hat{e}' \hat{y} + \hat{e}' \hat{e}. \end{aligned} \tag{3.33}$$

From Eq. (3.33), the left-hand side of Eq. (3.32) is $y' C C y = y' C y = \sum (y_i - \bar{y})^2$. Using $\hat{y} = \bar{y}$ (see property 4 on p. 112), it follows that $\hat{y}' C \hat{y} = \sum (\hat{y}_i - \bar{y})^2$. Property 1 on p. 112 implies $\hat{y}' \hat{e} = \hat{e}' \hat{y} = 0$ and we obtain

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^n \hat{e}_i^2. \quad \square$$

Proof of the Gauß–Markov Theorem on p. 119

We first consider conditions under which linear estimators are unbiased. For an unbiased linear estimator, the equation $E(\hat{\beta}^L) = a + A X \beta = \beta$ must hold for all $\beta \in \mathbb{R}^p$. Setting $\beta = 0$, we obtain $a = 0$ as a necessary condition for $\hat{\beta}^L$ being unbiased. Rearranging $A X \beta = \beta$ gives $(A X - I_p) \beta = 0$, implying $A X = I_p$ as a further condition for $\hat{\beta}^L$ being unbiased. From $\text{rk}(A X) = \min(\text{rk}(X), \text{rk}(A)) = \text{rk}(I_p) = p$, we also obtain $\text{rk}(A) = p$.

Now, without loss of generality, let A be of the form $A = (X' X)^{-1} X' + B$. Insertion into the condition $I_p = A X$ yields

$$I_p = A X = (X' X)^{-1} X' X + B X = I_p + B X$$

resulting in the condition

$$B X = 0.$$

This implies

$$\begin{aligned}
 \text{Cov}(\hat{\beta}^L) &= \sigma^2 \mathbf{A} \mathbf{A}' \\
 &= \sigma^2 \{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{B}\} \{ \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} + \mathbf{B}' \} \\
 &= \sigma^2 \{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{B}' \\
 &\quad + \mathbf{B}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} + \mathbf{B}\mathbf{B}'\} \\
 &= \sigma^2 (\mathbf{X}'\mathbf{X})^{-1} + \sigma^2 \mathbf{B}\mathbf{B}' \\
 &= \text{Cov}(\hat{\beta}) + \sigma^2 \mathbf{B}\mathbf{B}'
 \end{aligned}$$

for the covariance matrix of $\hat{\beta}^L$. From Theorem A.30 (p. 634), $\mathbf{B}\mathbf{B}'$ is nonnegative definite. Rearranging results in

$$\text{Cov}(\hat{\beta}^L) - \text{Cov}(\hat{\beta}) = \sigma^2 \mathbf{B}\mathbf{B}' \geq \mathbf{0}.$$

Based on this, we can now derive the optimality properties of the estimator. The variances of $\mathbf{b}'\hat{\beta}^L$ and $\mathbf{b}'\hat{\beta}$ are given by

$$\text{Var}(\mathbf{b}'\hat{\beta}^L) = \mathbf{b}'\text{Cov}(\hat{\beta}^L)\mathbf{b} \quad \text{and} \quad \text{Var}(\mathbf{b}'\hat{\beta}) = \mathbf{b}'\text{Cov}(\hat{\beta})\mathbf{b}.$$

Since the difference of covariance matrices is nonnegative definite, the difference of variances is nonnegative for any vector $\mathbf{b}$ (see Definition A.27 (p. 633) of definite matrices). This proves the proposition

$$\text{Var}(\mathbf{b}'\hat{\beta}^L) \geq \text{Var}(\mathbf{b}'\hat{\beta})$$

for vector $\mathbf{b}$. In particular, we can set $\mathbf{b} = (0, \dots, 1, \dots, 0)'$ with 1 at position $j + 1$ for $j = 0, \dots, k$. Then we obtain

$$\text{Var}(\hat{\beta}_j^L) = \text{Var}(\mathbf{b}'\hat{\beta}^L) \geq \text{Var}(\mathbf{b}'\hat{\beta}) = \text{Var}(\hat{\beta}_j). \quad \square$$

Proof of Propositions 5 and 6 on p. 123

5. The proof makes use of the matrix

$$\mathbf{Q} = \mathbf{I} - \mathbf{H} = \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}',$$

where $\mathbf{H}$ is the hat matrix from Sect. 3.2.1 (p. 107). The matrix $\mathbf{Q}$ is symmetric and idempotent with rank $\text{rk}(\mathbf{Q}) = n - p$. Multiplication with the design matrix $\mathbf{X}$ gives

$$\mathbf{Q}\mathbf{X} = \mathbf{X} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X} = \mathbf{0}.$$

From this, and because $\mathbf{Q}$ is symmetric and idempotent, we obtain

$$\hat{\boldsymbol{\varepsilon}}'\hat{\boldsymbol{\varepsilon}} = \mathbf{y}'\mathbf{Q}\mathbf{y} = (\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon})'\mathbf{Q}\mathbf{Q}(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = (\boldsymbol{\beta}'\mathbf{X}'\mathbf{Q} + \boldsymbol{\varepsilon}'\mathbf{Q})(\mathbf{Q}\mathbf{X}\boldsymbol{\beta} + \mathbf{Q}\boldsymbol{\varepsilon}) = \boldsymbol{\varepsilon}'\mathbf{Q}\boldsymbol{\varepsilon}.$$

The proposition now follows from $\boldsymbol{\varepsilon}/\sigma \sim N(\mathbf{0}, \mathbf{I})$ and from Theorem B.8 (2) on p. 651.

6. To prove the proposition, we show that $\frac{1}{\sigma}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$ and $\frac{1}{\sigma^2}\hat{\boldsymbol{\varepsilon}}'\hat{\boldsymbol{\varepsilon}}$ are independent. For this we apply Theorem B.8 (2) on p. 651 with $\mathbf{R} = \mathbf{Q}$ and $\mathbf{B} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$. The condition $\mathbf{B}\mathbf{R} = \mathbf{0}$ holds since

$$(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Q} = \mathbf{0}.$$

Since $\boldsymbol{\varepsilon}/\sigma \sim N(\mathbf{0}, \mathbf{I})$, the theorem implies that

$$\frac{\boldsymbol{\varepsilon}'}{\sigma}\mathbf{Q}\frac{\boldsymbol{\varepsilon}}{\sigma} = \frac{1}{\sigma^2}\hat{\boldsymbol{\varepsilon}}'\hat{\boldsymbol{\varepsilon}} \quad \text{and} \quad (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}/\sigma$$

are independent. The proposition finally follows from

$$\begin{aligned} \frac{1}{\sigma}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) &= \frac{1}{\sigma} \{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} - \boldsymbol{\beta}\} \\ &= \frac{1}{\sigma} \{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) - \boldsymbol{\beta}\} \\ &= \frac{1}{\sigma}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon} \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\frac{\boldsymbol{\varepsilon}}{\sigma}. \end{aligned} \quad \square$$

Distribution of the Test Statistic Under Linear Null Hypothesis H_0

(p. 128)

Computation of the Least Squares Estimator Under H_0

To minimize the residual sum of squares under the null hypothesis $\mathbf{C}\boldsymbol{\beta} = \mathbf{d}$, we apply the *Lagrange approach*

$$\begin{aligned} \text{LSR}(\boldsymbol{\beta}; \boldsymbol{\lambda}) &= \text{LS}(\boldsymbol{\beta}) - 2\boldsymbol{\lambda}'(\mathbf{C}\boldsymbol{\beta} - \mathbf{d}) \\ &= \mathbf{y}'\mathbf{y} - 2\mathbf{y}'\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta} - 2\boldsymbol{\lambda}'\mathbf{C}\boldsymbol{\beta} + 2\boldsymbol{\lambda}'\mathbf{d}, \end{aligned}$$

where $\boldsymbol{\lambda}$ is a column vector of Lagrangian multipliers of dimension r . Using Theorems A.33.1 and A.33.3 (p. 636) on differentiation of matrix functions, we obtain

$$\begin{aligned}\frac{\partial \text{LSR}(\beta; \lambda)}{\partial \beta} &= -2 X' y + 2 X' X \beta - 2 C' \lambda \\ \frac{\partial \text{LSR}(\beta; \lambda)}{\partial \lambda} &= -2 C \beta + 2 d.\end{aligned}$$

Setting these derivatives to zero gives

$$\begin{aligned}X' X \beta - X' y &= C' \lambda \\ C \beta &= d.\end{aligned}$$

We solve these equations first for λ and then for β . Multiplying the first equation from the left with $(X' X)^{-1}$ gives

$$\beta - \hat{\beta} = (X' X)^{-1} C' \lambda.$$

Multiplying this equation from the left with the matrix C yields

$$C \beta - C \hat{\beta} = C (X' X)^{-1} C' \lambda.$$

Inserting the second equation $C \beta = d$, we obtain

$$d - C \hat{\beta} = C (X' X)^{-1} C' \lambda,$$

and

$$\lambda = (C (X' X)^{-1} C')^{-1} (d - C \hat{\beta}).$$

In the last step, we have used the fact that $C (X' X)^{-1} C'$ is positive definite [Theorem A.29.2 (p. 634)] and, therefore, invertible.

Inserting λ into the first equation gives

$$X' X \beta - X' y = C' (C (X' X)^{-1} C')^{-1} (d - C \hat{\beta}),$$

and finally, the (restricted) least squares estimator

$$\hat{\beta}^R = \hat{\beta} - (X' X)^{-1} C' (C (X' X)^{-1} C')^{-1} (C \hat{\beta} - d).$$

Derivation of the Difference of Residual Sum of Squares

We first denote the restricted least squares estimator $\hat{\beta}^R$ as

$$\hat{\beta}^R = \hat{\beta} - \Delta_{H_0},$$

where Δ_{H_0} is defined through

$$\Delta_{H_0} = (X' X)^{-1} C' (C (X' X)^{-1} C')^{-1} (C \hat{\beta} - d).$$

For the values $\hat{\mathbf{y}}_{H_0}$ fitted under the restriction of the null hypothesis we then obtain

$$\hat{\mathbf{y}}_{H_0} = \mathbf{X} \hat{\boldsymbol{\beta}}^R = \mathbf{X} (\hat{\boldsymbol{\beta}} - \Delta_{H_0}) = \mathbf{X} \hat{\boldsymbol{\beta}} - \mathbf{X} \Delta_{H_0} = \hat{\mathbf{y}} - \mathbf{X} \Delta_{H_0}$$

and

$$\hat{\boldsymbol{\varepsilon}}_{H_0} = \mathbf{y} - \hat{\mathbf{y}}_{H_0} = \mathbf{y} - \hat{\mathbf{y}} + \mathbf{X} \Delta_{H_0} = \hat{\boldsymbol{\varepsilon}} + \mathbf{X} \Delta_{H_0}$$

for the residuals under the null hypothesis. Then the residual sum of squares SSE_{H_0} under H_0 can be written in the form

$$\begin{aligned} \text{SSE}_{H_0} &= \hat{\boldsymbol{\varepsilon}}'_{H_0} \hat{\boldsymbol{\varepsilon}}_{H_0} \\ &= (\hat{\boldsymbol{\varepsilon}} + \mathbf{X} \Delta_{H_0})' (\hat{\boldsymbol{\varepsilon}} + \mathbf{X} \Delta_{H_0}) \\ &= \hat{\boldsymbol{\varepsilon}}' \hat{\boldsymbol{\varepsilon}} + \hat{\boldsymbol{\varepsilon}}' \mathbf{X} \Delta_{H_0} + \Delta'_{H_0} \mathbf{X}' \hat{\boldsymbol{\varepsilon}} + \Delta'_{H_0} \mathbf{X}' \mathbf{X} \Delta_{H_0} \\ &= \hat{\boldsymbol{\varepsilon}}' \hat{\boldsymbol{\varepsilon}} + \Delta'_{H_0} \mathbf{X}' \mathbf{X} \Delta_{H_0}. \end{aligned}$$

The last equality holds because the residuals of the full model and the columns of the design matrix are orthogonal, i.e., $\hat{\boldsymbol{\varepsilon}}' \mathbf{X} = \mathbf{0}$ (see Box 3.7 on p. 112).

The matrix $\mathbf{X}' \mathbf{X}$ is positive definite, implying that $\Delta'_{H_0} \mathbf{X}' \mathbf{X} \Delta_{H_0}$ is nonnegative. Therefore, the residual sum of squares SSE_{H_0} under H_0 is always greater than or equal to the residual sum of squares SSE in the unrestricted model; see p. 128.

For the difference ΔSSE of residual sum of squares we finally obtain

$$\begin{aligned} \Delta \text{SSE} &= \hat{\boldsymbol{\varepsilon}}' \hat{\boldsymbol{\varepsilon}} + \Delta'_{H_0} \mathbf{X}' \mathbf{X} \Delta_{H_0} - \hat{\boldsymbol{\varepsilon}}' \hat{\boldsymbol{\varepsilon}} \\ &= \left\{ (\mathbf{X}' \mathbf{X})^{-1} \mathbf{C}' (\mathbf{C} (\mathbf{X}' \mathbf{X})^{-1} \mathbf{C}')^{-1} (\mathbf{C} \hat{\boldsymbol{\beta}} - \mathbf{d}) \right\}' \mathbf{X}' \mathbf{X} \\ &\quad \cdot \left\{ (\mathbf{X}' \mathbf{X})^{-1} \mathbf{C}' (\mathbf{C} (\mathbf{X}' \mathbf{X})^{-1} \mathbf{C}')^{-1} (\mathbf{C} \hat{\boldsymbol{\beta}} - \mathbf{d}) \right\} \\ &= (\mathbf{C} \hat{\boldsymbol{\beta}} - \mathbf{d})' (\mathbf{C} (\mathbf{X}' \mathbf{X})^{-1} \mathbf{C}')^{-1} \\ &\quad \cdot \mathbf{C} (\mathbf{X}' \mathbf{X})^{-1} \mathbf{C}' (\mathbf{C} (\mathbf{X}' \mathbf{X})^{-1} \mathbf{C}')^{-1} (\mathbf{C} \hat{\boldsymbol{\beta}} - \mathbf{d}) \\ &= (\mathbf{C} \hat{\boldsymbol{\beta}} - \mathbf{d})' (\mathbf{C} (\mathbf{X}' \mathbf{X})^{-1} \mathbf{C}')^{-1} (\mathbf{C} \hat{\boldsymbol{\beta}} - \mathbf{d}). \end{aligned}$$

Stochastic Properties of the Difference of Residual Sum of Squares

1. To prove this property, we apply Theorem B.2.8 (p. 647) on the expectation of quadratic forms. We have

$$\mathbf{E}(\mathbf{C} \hat{\boldsymbol{\beta}} - \mathbf{d}) = \mathbf{C} \boldsymbol{\beta} - \mathbf{d}$$

and

$$\text{Cov}(\mathbf{C} \hat{\boldsymbol{\beta}} - \mathbf{d}) = \sigma^2 \mathbf{C} (\mathbf{X}' \mathbf{X})^{-1} \mathbf{C}'.$$

Setting $\mathbf{Z} = \mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d}$ and $\mathbf{A} = (\mathbf{C}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{C}')^{-1}$ in Theorem B.2.8, we obtain

$$\begin{aligned} E(\Delta\text{SSE}) &= E\left\{(\mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d})' (\mathbf{C}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{C}')^{-1} (\mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d})\right\} \\ &= \text{tr}\left\{\sigma^2 (\mathbf{C}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{C}')^{-1} \mathbf{C}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{C}'\right\} \\ &\quad + (\mathbf{C}\boldsymbol{\beta} - \mathbf{d})' (\mathbf{C}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{C}')^{-1} (\mathbf{C}\boldsymbol{\beta} - \mathbf{d}) \\ &= \text{tr}(\sigma^2 \mathbf{I}_r) + (\mathbf{C}\boldsymbol{\beta} - \mathbf{d})' (\mathbf{C}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{C}')^{-1} (\mathbf{C}\boldsymbol{\beta} - \mathbf{d}) \\ &= r\sigma^2 + (\mathbf{C}\boldsymbol{\beta} - \mathbf{d})' (\mathbf{C}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{C}')^{-1} (\mathbf{C}\boldsymbol{\beta} - \mathbf{d}). \end{aligned}$$

2. The proposition follows from Theorem B.8.1 (p. 651). Defining the random vector $\mathbf{Z} = \mathbf{C}\hat{\boldsymbol{\beta}}$, we obtain

$$E(\mathbf{Z}) = \mathbf{C}\boldsymbol{\beta} = \mathbf{d}$$

and

$$\text{Cov}(\mathbf{Z}) = \sigma^2 \mathbf{C}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{C}'$$

under H_0 . Since $\hat{\boldsymbol{\beta}}$ is normally distributed, it follows that

$$\mathbf{Z} \sim N(\mathbf{d}, \sigma^2 \mathbf{C}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{C}').$$

The proposition follows directly by applying Theorem B.8.1 to the random vector $\mathbf{Z}$.

3. The difference of residual sum of squares ΔSSE only depends on the LS estimator $\hat{\boldsymbol{\beta}}$. The proposition now follows directly from the independence of $\hat{\boldsymbol{\varepsilon}}'\hat{\boldsymbol{\varepsilon}}$ and $\hat{\boldsymbol{\beta}}$; see Box 3.11 on p. 123. $\square$